Design & Optimization of a 3D-Printed Trichotomous Virtual Impactor
for Fukushima Daiichi decommissioning

Pierre Roger - 5A EMME MFE - Nuclear Engineering Track
The University of Tokyo in collaboration with JAEA
02/03/2026 - 31/08/2026

ABSTRACT

Aerosol inhalation can pose serious threats to human health, and real complications when they contain radioactive particles. In the context of the Fukushima Daiichi nuclear power plant decommissioning process, operators can face this kind of situation. It is essential to prevent them from any hazard. This internship in the Department of Nuclear Engineering and Management of the University of Tokyo, in collaboration with the Japan Atomic Energy Agency (JAEA), explored the development of µSPLIT, a two-stage 3D-printed virtual impactor created by the CLADS (JAEA). A parametric study has been conducted with the creation of a fully automated CFD pipeline to explore 250 sets of the separation region parameters allowing the optimization of the size separation and the reduction of related issues. In this automated Python pipeline, Latin Hypercube Sampling and multiple OpenFOAM solvers are implemented to observe reliable trends between parameters and performance indicators. An optimal configuration emerged from this study.

Keywords: Aerosol Assessment, CFD, Fukushima Daiichi, Latin Hypercube Sampling, OpenFOAM, Virtual Impactors.

ACKNOWLEDGEMENTS

I would like to express my deepest gratitude to my supervisors at the University of Tokyo and JAEA, Ryo Yokoyama and Hugo Laffolley, for their invaluable guidance and profound scientific insights throughout the entirety of this internship. Their expertise and support were essential in shaping the direction and quality of this work.

I am also sincerely grateful to my supervisor at Polytech Nancy, Abdelhamid Kheiri, for his dedication to the entire process of this internship, and for his contribution to the creation of Nuclear Engineering Track in Lorraine-INP which led me to this internship.

I would also like to extend my sincere gratitude to Shuichiro Miwa, Prof. Associate of the Department of Nuclear Engineering and Management of the University of Tokyo, for allowing me to work in the Nuclear Professional School laboratory, with the Nuclear Reactor Safety Laboratory team and utilize its facilities. Additionally, I am grateful to Youichi Tsubota, director of the µSPLIT project from CLADS (JAEA) for enabling my collaboration with the agency and for granting me access to JAEA's experimental facilities.

Merci !

Thank you !

ありがとうございました

HOST INSTITUTION

The internship is conducted in the Nuclear Professional School (NPS) in Tōkai, Ibaraki, JAPAN. This is a laboratory associated to the doctoral school of the Department of Nuclear Engineering and Management of the University of Tokyo. It aims to train nuclear power technicians to the security, design, construction, operation and management of nuclear power stations and nuclear power related facilities.

The project involves a collaboration with the Collaborative Laboratories for Advanced Decommissioning Science (CLADS), a core division of the Japan Atomic Energy Agency created in response of the Fukushima Daiichi nuclear accident. It coordinates research and development for decommissioning, radioactive waste management, and fuel debris retrieval.

CLADS develops a two-stage virtual impactor named µSPLIT to replace industrial ones by low cost 3D-printed ones. µSPLIT must separate the aerosol in three class of size : > 10 µm, $\in 1 - 10$ µm, < 1 µm.

(a) (b)

Figure 1: (a) µSPLIT (b) Fluid domain of the µSPLIT

The objective of my internship is to optimize the 10 µm separation stage (large separation region in the Figure 1b) that separates the aerosol in > 10 µm and < 10 µm distributions. Those optimization patterns may also be useful for future work on the 1 µm separation stage, as it follows the same principles, but on a different scale.

CONTENTS

NOTATIONS

Acronyms

CAD
Computer-Aided Design

CFD
Computational Fluid Dynamics

CFL
Courant-Friedrichs-Lewy

CLADS
Collaborative Laboratories for Advanced Decommissioning Science

DNS
Direct Numerical Simulation

FDM
Finite Difference Method

FDNPP
Fukushima Daiichi Nuclear Power Plant

FEM
Finite Element Method

FVM
Finite Volume Method

HEPA
High Efficiency Particulate Air

HPC
High Performance Computer

JAEA
Japan Atomic Energy Agency

KPIs
Key Performance Indicators

LES
Large Eddy Simulation

LHS
Latin Hypercube Sampling

MCCI
Molten Core-Concrete Interaction

NPS
Nuclear Professional School

PIV
Particle Image Velocimetry

PM
Particulate Matters

RANS
Reynolds-Averaged Navier-Stokes

SDGs
Sustainable Development Goals

SST
Shear Stress Transport

STL
Stereolithography

TEPCO
Tokyo Electric Power Company

UTokyo
Department of Nuclear Engineering and Management of The University of Tokyo

VI
Virtual Impactor

Nomenclature

Roman Symbols

Symbol	Description	Unit	S.I. Unit
a_{s}	Specific activity of an isotope	$\mathrm{Bq} \cdot \mathrm{g}^{-1}$	$\mathrm{kg}^{-1} \cdot \mathrm{s}^{-1}$
A	Activity of an isotope	Bq	s^{-1}
c_{air}	Speed of the sound in the air	$\mathrm{m} \cdot \mathrm{s}^{-1}$	$\mathrm{m} \cdot \mathrm{s}^{-1}$
C_{c}	Cunningham slip correction factor	$\varnothing$	$\varnothing$
C_{μ}	turbulent viscosity constant	$\varnothing$	$\varnothing$
d_{ae}	Aerodynamic diameter	µm	m
d_{e}	Volume equivalent diameter	µm	m
d_{p}	Particle diameter	µm	m
d_{50}	Cutoff diameter	µm	m
D_{eff}	Effective dose resulting from the intake of an isotope's activity	mSv	$\mathrm{m}^2 \cdot \mathrm{s}^{-2}$
D_{f}	Characteristic diameter of the flow	µm	m
D_{h}	Hydraulic equivalent diameter	µm	m
e	Coefficient of restitution	$\varnothing$	$\varnothing$
$e(50)$	Dose coefficient for an adult	$\mathrm{Sv} \cdot \mathrm{Bq}^{-1}$	$\mathrm{m}^2 \cdot \mathrm{s}^{-1}$
E_{VI}	Young modulus of the VI	MPa	$\mathrm{kg} \cdot \mathrm{m}^{-1} \cdot \mathrm{s}^{-2}$
E^{*}	Reduced Young modulus	MPa	$\mathrm{kg} \cdot \mathrm{m}^{-1} \cdot \mathrm{s}^{-2}$
F_{B}	Buoyancy Force	N	$\mathrm{kg} \cdot \mathrm{m} \cdot \mathrm{s}^{-2}$
F_{D}	Drag force	N	$\mathrm{kg} \cdot \mathrm{m} \cdot \mathrm{s}^{-2}$
F_{L}	Lift force	N	$\mathrm{kg} \cdot \mathrm{m} \cdot \mathrm{s}^{-2}$
F_{PG}	Pressure gradient force	N	$\mathrm{kg} \cdot \mathrm{m} \cdot \mathrm{s}^{-2}$
F_{VM}	Virtual mass force	N	$\mathrm{kg} \cdot \mathrm{m} \cdot \mathrm{s}^{-2}$
$\mathbf{g}$	Gravitational acceleration vector	$\mathrm{m} \cdot \mathrm{s}^{-2}$	$\mathrm{m} \cdot \mathrm{s}^{-2}$
k	Turbulent kinetic energy	$\mathrm{m}^2 \cdot \mathrm{s}^{-2}$	$\mathrm{m}^2 \cdot \mathrm{s}^{-2}$
L_N	Length of the acceleration Nozzle	mm	m
m	Number of LHS sample points	$\varnothing$	$\varnothing$
m_{air}	Mass of the air in the VI	µg	kg
m_{i}	Mass of the isotope collected on the VI Walls	µg	kg
m_{p}	Mass of a particle	µg	kg
M_{air}	Molar mass of the air	$\mathrm{g} \cdot \mathrm{mol}^{-1}$	$\mathrm{kg} \cdot \mathrm{mol}^{-1}$
n	Number of LHS input parameters	$\varnothing$	$\varnothing$
n_{IG}	Amount of substance in the ideal gas	mol	mol
N_{x}	Number of grid point	$\varnothing$	$\varnothing$
p	Static pressure	Pa	$\mathrm{kg} \cdot \mathrm{m}^{-1} \cdot \mathrm{s}^{-2}$
P	Pressure of the ideal gas	Pa	$\mathrm{kg} \cdot \mathrm{m}^{-1} \cdot \mathrm{s}^{-2}$
P_{air}	Mean pressure of air in the VI	Pa	$\mathrm{kg} \cdot \mathrm{m}^{-1} \cdot \mathrm{s}^{-2}$
q_{m}	Ratio between mass flow rates of minor outlet and inlet	%	%
q_{Major}	Mass flow rate in the major outlet	$\mathrm{kg} \cdot \mathrm{s}^{-1}$	$\mathrm{kg} \cdot \mathrm{s}^{-1}$
q_{minor}	Mass flow rate in the minor outlet	$\mathrm{kg} \cdot \mathrm{s}^{-1}$	$\mathrm{kg} \cdot \mathrm{s}^{-1}$
R	Universal gas constant	$\mathrm{J} \cdot \mathrm{mol}^{-1} \cdot \mathrm{K}^{-1}$	$\mathrm{kg} \cdot \mathrm{m}^2 \cdot \mathrm{s}^{-2} \cdot \mathrm{mol}^{-1} \cdot \mathrm{K}^{-1}$

Symbol	Description	Unit	S.I. Unit
R_1	Radius of the fillet between the inlet and the Major outlet	mm	m
R_2	Radius of the fillet between the Major outlet and the minor outlet	mm	m
$\mathscr{R}_{\text{air}}$	Specific constant of air	$\text{J} \cdot \text{kg}^{-1} \cdot \text{K}^{-1}$	$\text{m}^2 \cdot \text{s}^{-2} \cdot \text{K}^{-1}$
$seed$	LHS Randomness	$\varnothing$	$\varnothing$
Stk_{50}	Cutoff Stokes number	$\varnothing$	$\varnothing$
T	Thickness of the extrusion	mm	m
T_{air}	Temperature of the air in the VI	°C	K
T_{IG}	Temperature of the ideal gas	°C	K
$\mathbf{u}_{\text{f}}$	Fluid velocity vector	$\text{m} \cdot \text{s}^{-1}$	$\text{m} \cdot \text{s}^{-1}$
$\mathbf{u}_{\text{p}}$	Particle velocity vector	$\text{m} \cdot \text{s}^{-1}$	$\text{m} \cdot \text{s}^{-1}$
u_τ	Friction velocity	$\text{m} \cdot \text{s}^{-1}$	$\text{m} \cdot \text{s}^{-1}$
U_{f}	Mean velocity of the fluid	$\text{m} \cdot \text{s}^{-1}$	$\text{m} \cdot \text{s}^{-1}$
v_{s}	Critical sticking velocity	$\text{m} \cdot \text{s}^{-1}$	$\text{m} \cdot \text{s}^{-1}$
v_{i}^{n}	Incident velocity in the normal direction to the surface	$\text{m} \cdot \text{s}^{-1}$	$\text{m} \cdot \text{s}^{-1}$
v_{r}^{n}	Rebound velocity in the normal direction to the surface	$\text{m} \cdot \text{s}^{-1}$	$\text{m} \cdot \text{s}^{-1}$
V	Volume of the ideal gas	m^3	m^3
V_{air}	Volume of the VI filled with air	L	m^3
W_m	Width of the minor flow outlet	mm	m
W_M	Width of the Major flow outlet	mm	m
W_N	Width of the acceleration Nozzle	mm	m
X_{p}	Particle position	m	m
y_{w}	Distance from the wall	mm	m

Greek Symbols

Symbol	Description	Unit	S.I. Unit
Γ	Interface energy	$J \cdot m^{-2}$	$kg \cdot s^{-2}$
δ_{ij}	Kronecker delta	$\varnothing$	$\varnothing$
Δt	Time step	s	s
Δx	Cell size	m	m
ε	Dissipation rate	$m^2 \cdot s^{-3}$	$m^2 \cdot s^{-3}$
λ_{air}	Mean free path of the air	µm	m
λ_{f}	Mean free path of the fluid	µm	m
μ	Mean particle size of the distribution	µm	m
μ_{air}	Dynamic viscosity of the air	$Pa \cdot s$	$kg \cdot m^{-1} \cdot s^{-1}$
μ_{f}	Dynamic viscosity of the fluid	$Pa \cdot s$	$kg \cdot m^{-1} \cdot s^{-1}$
μ_{t}	Turbulent viscosity	$Pa \cdot s$	$kg \cdot m^{-1} \cdot s^{-1}$
ν_{air}	Kinematic viscosity of the air	$m^2 \cdot s^{-1}$	$m^2 \cdot s^{-1}$
ν_{f}	Kinematic viscosity of the fluid	$m^2 \cdot s^{-1}$	$m^2 \cdot s^{-1}$
ν_{VI}	Poisson's ratio of the VI	$\varnothing$	$\varnothing$
$\tilde{\nu}$	Modified turbulent viscosity	$m^2 \cdot s^{-1}$	$m^2 \cdot s^{-1}$
ρ_{air}	Density of the air	$kg \cdot m^{-3}$	$kg \cdot m^{-3}$
ρ_{f}	Fluid density	$kg \cdot m^{-3}$	$kg \cdot m^{-3}$
ρ_{p}	Particle density	$g \cdot cm^{-3}$	$kg \cdot m^{-3}$
ρ_0	Aerodynamic reference density	$g \cdot cm^{-3}$	$kg \cdot m^{-3}$
σ	Geometric standard deviation of the distribution	$\varnothing$	$\varnothing$
σ_{VI}	Sharpness of the virtual impactor	$\varnothing$	$\varnothing$
$\boldsymbol{\tau}$	Viscous stress tensor	Pa	$kg \cdot m^{-1} \cdot s^{-2}$
τ_{w}	Wall shear stress	$kg \cdot m^{-1} \cdot s^{-2}$	$kg \cdot m^{-1} \cdot s^{-2}$
Φ_{f}	Proportion of the fluid phase in the total volume	$\varnothing$	$\varnothing$
Φ_{p}	Proportion of the particulate phase in the total volume	$\varnothing$	$\varnothing$
χ	dynamic shape factor	$\varnothing$	$\varnothing$
ω	Specific dissipation rate	s^{-1}	s^{-1}

Dimensionless Numbers

Symbol	Description
$\mathcal{C}$	Courant Number
Ma	Mach Number
Re	Reynolds Number
Re_{ℓ}	Large Eddy Reynolds number
Stk	Stokes Number
y^+	Dimensionless wall distance

1

INTRODUCTION

1.1 Context

On March 11, 2011, a 9.0-magnitude earthquake off the coast of Japan resulted in a tsunami that overwhelmed the Fukushima Daiichi Nuclear Power Plant (FDNPP)'s seawalls and emergency power systems, leading to the loss of reactor core cooling in Units 1, 2, and 3. This was the worst nuclear incident since the Chernobyl disaster.
In the context of the FDNPP decommissioning, supervised by the Tokyo Electric Power Company (TEPCO), the operator of this power plant, in collaboration with the Department of Nuclear Engineering and Management of the University of Tokyo (UTokyo) and the Japan Atomic Energy Agency (JAEA), nuclear aerosol released during the decommissioning process is an important matter and has to be monitored. It is critical to determine the composition of aerosols and the hazards posed by cutting or cleaning operations in the Fukushima Daiichi area.

1.2 Problematic

As the FDNPP decommissioning takes places, technical operators need to be present on site. Aerosols can enter their respiratory system and risk their health.
In aerosol science, the virtual impactor is an instrument often used to classify particles according to their physical characteristics, mainly their size. Since Bill Conner created the first one in 1960 (Virgil A Marple et al., 2011), the industry produces them at a high price. To increase their durability and precision they are made of metal. However they are not practical in nuclear decommissioning activities : Laffolley and Tsubota (2026) explain that their own decontamination would expose operators to unnecessary radioactive emissions, and their excessive prices prevent their single use.

1.3 Research Question

This report aims to present my contribution in the µSPLIT project conducted in collaboration with JAEA's Collaborative Laboratories for Advanced Decommissioning Science (CLADS) on the development of a low cost 3D-printable virtual impactor for nuclear aerosol characterization. The device aims to separate particles into three size fractions based on their aerodynamic diameter. This leads us to the following research question :

What are the key geometric and flow parameters influencing the separation efficiency of a 3D-printable trichotomous virtual impactor, and how can they be optimized for nuclear decommissioning applications ?

2

STATE OF THE ART

2.1 Aerosols in Nuclear Decommissioning

Aerosols form the core of our investigation. The classification and sampling of aerosol relies heavily on our knowledge of their behavior in flows, their physical and chemical properties, and their transport mechanisms. The virtual impactor optimization for nuclear decommissioning issue depends on the understanding of what aerosols are and how they are formed in nuclear severe accident scenarios or in the decommissioning process, and why they pose a hazard.

2.1.1 Definitions and Physical Properties

Definitions

Colbeck et al. (2014) defines aerosols as a suspension of liquid or solid particles present in a gaseous medium. It is a two-phase system composed of a continuous phase (gas) and of a discrete phase (particulate matter).

They can originate from nature (pollen, fungal spores, bacteria, viruses, volcanic ash and gases) or from human activities (Industrial emissions (coal-fired power plants, incinerators, oil and gas drilling platforms), automotive traffic, airplane contrails, ship smokestack plumes, spraying of pesticides and fertilizers in agriculture).

Particulate Matter (PM) are the solid or liquid particles present in the gaseous medium that forms the aerosol. In recent years, PM have received increasing attention from the scientific community due to its impact on human health: they alters air quality and atmospheric conditions. It is necessary to know the structure, origin, transport and provenance of PM to understand the impact of aerosols.

Particle Properties

Shape Particle shapes can be divided into three general classes (Colbeck et al., 2014):

- Isometric: The particle's three dimensions are roughly equal (spherical particles).
- Platelets: The particle has two long dimensions and a third small one (paper sheet).
- Fibres: The particle has one long dimension and two much smaller ones (needles).

Composition An aerosol is called homogeneous when all particles are of the same nature (size, shape, chemical composition), which is very rare in the nature. Otherwise they are described as heterogeneous. It is difficult to define a generic composition for nuclear aerosols, it varies a lot between different scenarios (Chernobyl, Fukushima, Three Miles Island, URASOL project, etc...).

Density Particles density is strongly related to their composition, however it is often assumed to be evenly distributed inside a same particle (NEA, 2009).

Transport behaviour

Hari et al. (2006) explain that particles do not behave in the same way as gas molecules when dispersed in air. They deposit under gravity, impact on bends due to particle inertia, are deposited on internal surfaces by molecular and turbulent diffusion and are affected by thermal, electrostatic and acoustic forces. Particle motion in the flow field under consideration is governed by Newton's second law:

$$\begin{aligned} m_{\mathrm{p}}\frac{\mathrm{d}\mathbf{u}_{\mathrm{p}}}{\mathrm{d}t} &= F_D + F_B + F_{PG} + F_{VM} + F_L \ , \\ \frac{\mathrm{d}X_{\mathrm{p}}}{\mathrm{d}t} &= \mathbf{u}_{\mathrm{p}} \ , \end{aligned} \tag{2.1}$$

where m_{p} is the mass of a particle, $\mathbf{u}_{\mathrm{p}}$ is the particle velocity vector (u_{p}, v_{p}, w_{p}), F_{D} is the drag force, F_{B} is the buoyancy force, F_{PG} is the pressure gradient force, F_{VM} is the virtual mass force, F_{L} is the lift force, X_{p} is the particle position (x_{p}, y_{p}, z_{p}).

Crowe et al. (1998) assumes that the dispersed phase can be considered negligible compared to the continuous phase, and consequently, other forces besides drag are insignificant. Particle motion equation becomes:

$$m_{\mathrm{p}}\frac{\mathrm{d}\mathbf{u}_{\mathrm{p}}}{\mathrm{d}t} = F_D \ . \tag{2.2}$$

The drag force is exerted on the particle by the continuous phase in the opposite direction of the flow. Several drag models exist: Stokes, Newton, Schiller-Naumann, etc... However in aerosol science, the flow around a moving particle is usually in the Stokes regime, and consequently, the Stokes drag model is mainly considered. It is associated with a Cunningham slip correction factor that adjusts the Stokes drag model for the slip flow regime encountered by ultrafine particles. This drag force is calculated as follows (Hinds et al., 1999):

$$F_D = \frac{3\pi\mu_{\mathrm{f}}\mathbf{u}_{\mathrm{p}}d_{\mathrm{p}}}{C_c\left(d_{\mathrm{p}}\right)} \ , \tag{2.3}$$

where μ_{f} is the dynamic viscosity of the fluid, d_p is the particle diameter, $C_c(d_p)$ is the Cunningham slip correction factor for the particle having a diameter d_p defined as:

$$C_c\left(d_{\mathrm{p}}\right) = 1 + \frac{\lambda_{\mathrm{f}}}{d_{\mathrm{p}}}\left[2.34 + 1.05\exp\left(-0.39\frac{d_{\mathrm{p}}}{\lambda_{\mathrm{f}}}\right)\right] \ , \tag{2.4}$$

where λ_{f} is the mean free path of the fluid.

Aerodynamic diameter Spherical particles are easier to describe in a flow than fibers or platelets. However, most particles are not spherical. In aerosol science this problem is being bypassed using the equivalent aerodynamic diameter d_{ae}: it allows the consideration of any particle as a spherical particle. It is defined as the diameter of a sphere of $\rho_0 = 1\ \mathrm{g\cdot cm^{-3}}$ density and the same settling velocity as the particle under study.

It is defined with the volume equivalent diameter d_e (diameter of a sphere having the same volume of the particle) via (Colbeck et al., 2014):

$$d_\text{ae} = d_\text{e} \sqrt{\frac{\rho_\text{p}}{\chi \rho_0} \times \frac{C_c(d_\text{e})}{C_c(d_\text{ae})}}\ , \tag{2.5}$$

where ρ_p is the density of the particle, χ is the dynamic shape factor ($\chi = 1$ for spherical particles, $\chi \in]1, 2]$ for others (Hinds et al., 1999)) and ρ_0 is the aerodynamic reference density.

Classification by size

Colbeck et al. (2014) characterizes PM on the behalf of their aerodynamic diameter referred as coarse (< 10 µm), fine (< 2.5 µm) and ultrafine (< 1 µm):

- PM_{10} are inhalable coarse particle. Their diameter is less than 10 µm, which leads them to fall quickly on the ground, they principally come from road or dirty industries
- $PM_{2.5}$ are fine particle. Their diameter is less than 2.5 µm, they suspend in the air for long distances they are emitted from power plants, industries and automobiles. Moreover, $PM_{2.5}$ is considered as the main challenge for air pollution activities.
- PM_1 are ultrafine particles of less than 1 µm diameter. They consist primarily of inorganic ions, hydrocarbons and metals.

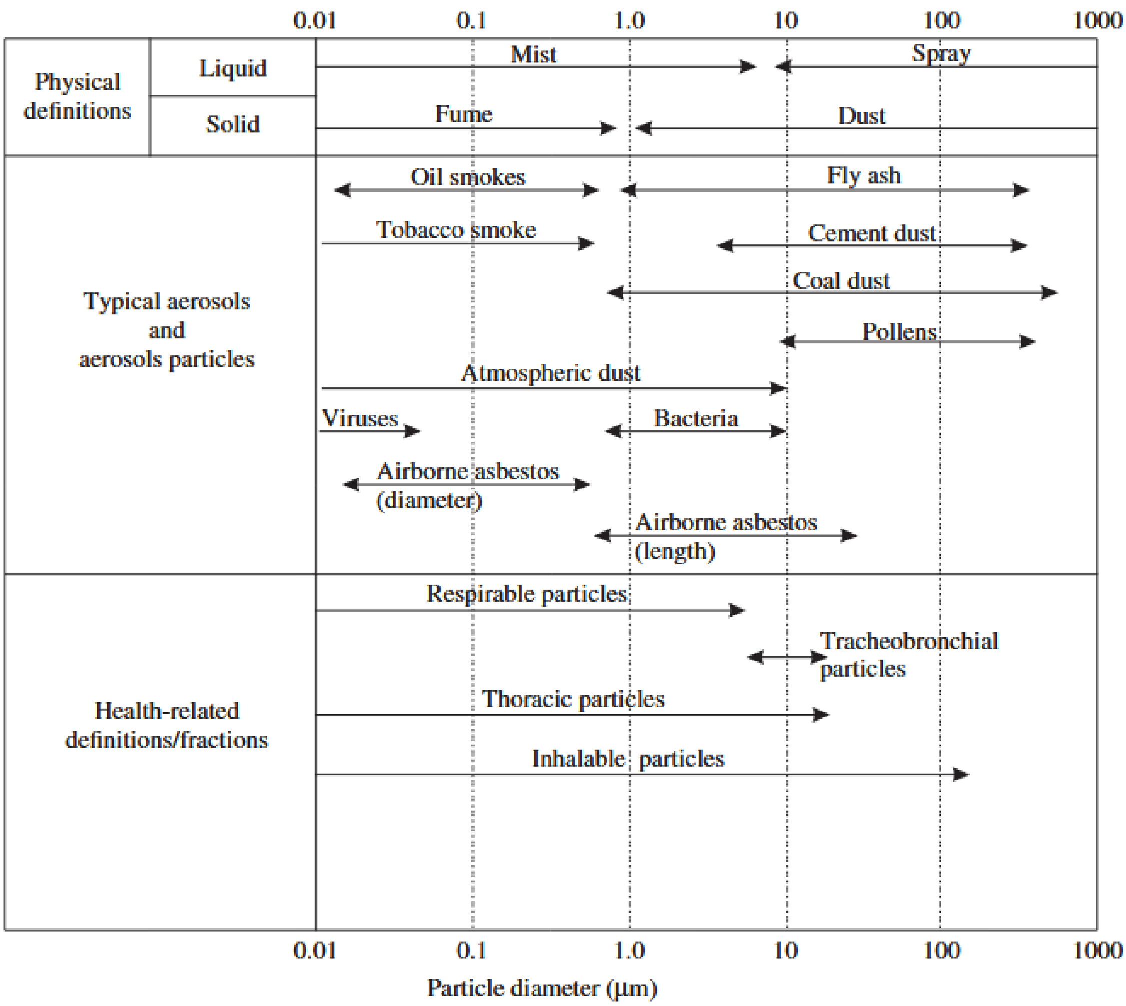


Figure 2.1: Particle size range for atmospheric aerosols (Colbeck et al., 2014).

Aerosols can cause health problems when deposited on the skin, but generally, the most sensitive routes of entry into the body are through the respiratory system and ingestion. Knowledge of the deposition of particulate matter in the human respiratory system is important for dose assessment and the risk analysis of airborne pollutants. The deposition process is controlled by the physical characteristics of the inhaled particles and by the physiological factors of the individuals involved. Among the physical factors, particle size and size distribution are the most important. The same physical properties that govern aerosols in the atmosphere apply within the lungs (Colbeck et al., 2014). In nuclear applications, particle size characterization is essential: since radioactive emissions scale with particle volume $\left(\propto d^3\right)$. A PM_{10} emits ≈ 1000 times more radiation than a PM_1. Size classification becomes essential for hazard assessment.

Size distribution

A monodisperse sample represent a group of particles having the same size, which is extremely rare in nature. Generally, particles vary in size in a sample, and this is called polydisperse.

For nuclear aerosol, it is common to report aerosol size data in terms of the parameters of the lognormal size distribution (μ and σ). The probability density function for particles having sizes in the interval d_p to $d_\mathrm{p} + \mathrm{d}d_\mathrm{p}$ is (NEA, 2009):

$$\mathrm{pdf}(d_\mathrm{p}) = \frac{1}{\sqrt{2\pi}\ln\sigma} \exp\left[-\left(\frac{\ln\left(d_\mathrm{p}/\mu\right)}{\sqrt{2}\ln\sigma}\right)^2\right] \mathrm{d}\ln d_\mathrm{p} \ , \tag{2.6}$$

where σ is the geometric standard deviation of the distribution and μ is the mean particle size of the distribution.

Zhuang et al. (2026) found diverse μ and σ for the characterization of nuclear aerosol from Fukushima Daiichi.

2.1.2 Aerosols at Fukushima Daiichi

Historical Aerosols: Formation and Dispersion

On March 11, 2011, an earthquake occurred in the Pacific ocean, followed an hour after by a tsunami that flooded the FDNPP. Nominal and emergency power source were lost, resulting in the loss of coolant for unit 1 to 3. In unit 4 the core had been unloaded to the spent fuel pool. Units 5 and 6 were not under operation. Cooling restriction caused core degradation (corium) which led to the production of large amount of radioactive gas and radioactive aerosols, and caused the release of radioactivity on site and into the environment (Högberg, 2013).

Formation Aerosols originating from corium arise from diverse processes, here are two detailed examples:

The first mechanism for aerosol generation involves the interaction between corium and concrete (Molten Core-Concrete Interaction (MCCI)). During this interaction, the extremely hot melted corium reacts with concrete, producing gases that bubble through the molten mixture. These bubbles carry tiny solid particles from the corium, concrete, and fission products, which then form fine aerosols as they cool. Among these, cesium microparticles ($CsMPs$) are particularly notable. These particles, typically 0.1 to 2 µm in size, have a silicate structure and contain, in addition to cesium, elements such as Silicon and Oxygen as the major elements, along with the presence of elements such as Sodium, Magnesium, Aluminium, Potassium, Calcium, Iron (Laffolley, Journeau, et al., 2024).

The second source of aerosol is the chemical reaction between the fuel cladding and the (supposed to) cooling water:

$$Zr + 2H_2O \rightarrow ZrO_2 + 2H_2 + \mathrm{heat} \qquad (\Delta H < 0) \ , \tag{2.7}$$

IRSN (2012) explains that at temperatures above 830 °C the cladding made of Zircaloy (an alloy made of Zirconium, used for nuclear power plant combustible coating) is deteriorating, and the reaction between the zirconium and water becomes rapid. At 1200 °C the cladding is broken leading to hydrogen gas formation and as the reaction is exothermic, this results in explosions. During those explosions, multiple fissile products are mixed with vapor created and forms aerosols by nucleation.

In the Fukushima Daiichi accident, the top part of the combustible assemblage was not cooled. It led to this reaction between zirconium and water resulting in multiple explosions that burst confinement vessels. The radioactive aerosols made of fission products exit the confinement vessels and spread in the environment.

Dispersion

During the first days after the accident, the atmospheric dispersion of radioactive aerosols was influenced by meteorological conditions and release dynamics throughout the world. Figure 2.2 shows that from March 12 to 17, the simulated plume initially moved toward the ocean. After the hydrogen explosion at Unit 1, a high concentration plume traveled north-northwest from FDNPP.

Dispersion models and measures around the globe showed that the radioactive plume in the atmosphere was initially captured by an extratropical cyclone over the western pacific ocean during the first few days of the release period, then moved on every continents and returned over Japan within 20 days. And after the first month of the dispersion, 16 % of the atmospheric ^{131}I activity was over tropical latitudes and 20 % in northern polar regions (Mészáros et al., 2016).

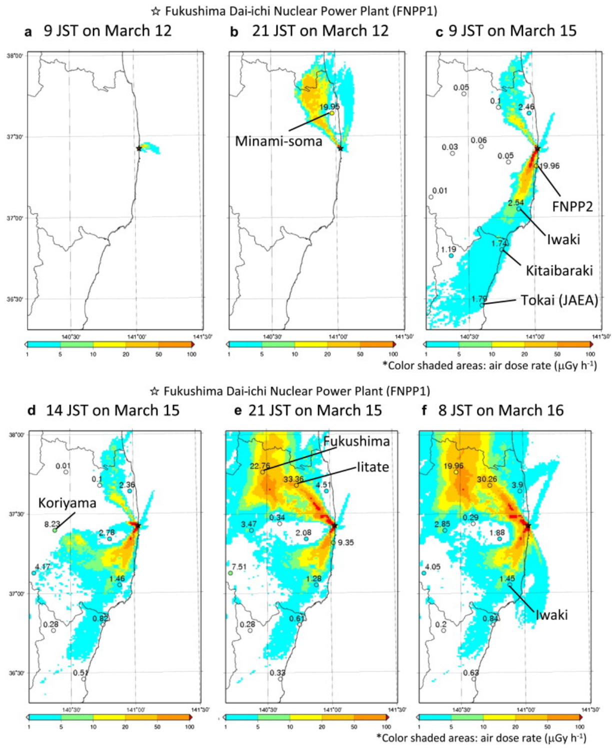


Figure 2.2: Simulated spatial distributions of air dose rate from March 12 to 16, 2011 (Katata et al., 2012).

Newly Generated Aerosols during Decommissioning

Since the accident in 2011, the decommissioning process is underway. Aerosol assessment is especially useful for major decommissioning activities near the damaged cores.

More than 10 years after the accident, it is realistic to believe that all the particles suspended in aerosols has fallen on the existing surfaces (such as ground, walls or structures). However, human activities through decommissioning can generate radioactive aerosols. Some activities are simply resuspending historical aerosols (ex: drone exploration (TEPCO, 2026)), but others create new aerosols (ex: corium retrieval, concrete structure cutting, etc... (Porcheron et al., 2021)).

2.1.3 Health Risks and Need for Size Classification

The FDNPP accident released an estimated 520 PBq of radionuclides into the environment, approximately an order of magnitude less than Chernobyl, with volatile radionuclides like ^{131}I, ^{137}Cs. The majority of atmospheric releases were transported offshore by the wind, leading to significant marine contamination of ^{131}I and ^{137}Cs near the plant, and ^{134}Cs and ^{137}Cs could be detected in California's Pacific ocean side. On land, ^{137}Cs deposition exceeded 185 $\text{kBq} \cdot \text{m}^{-2}$ across $1,700\ \text{km}^2$, with soil concentrations reaching 1.79 $\text{MBq} \cdot \text{kg}^{-1}$ near the site, far below Chernobyl's levels. Food contamination was effectively managed through strict regulatory limits and rapid monitoring, resulting in any excess contamination by the marketed food. The evacuation zone covered 600 km^2 (vs. $4,300\ \text{km}^2$ for Chernobyl), with $160,000$ people relocated (Steinhauser et al., 2014).

At a global scale authorities managed to secure environment and population from the radioactivity, Li et al. (2022) shows that the accident resulted in no observed acute radiation syndrome among workers or the public. Internal contamination from ^{134}Cs and ^{137}Cs was minimal (below 1 mSv for nearly all examined individuals). The Fukushima Health Management Survey initiated long-term thyroid monitoring for 360,000 children, though no radiation-attributed health effects have been detected to date.

However Długosz-Lisiecka et al. (2025) explains that significant impact on human body can exist, by deep interaction and distribution of isotope in most morphological parts, only in case of presence of high concentrations of radionuclides attached on the particles, especially those with smaller nanometer sizes. If large amounts of radionuclides are injected into the atmosphere and attach to submicrometric particles, they pose a significant health risk, particularly through the inhalation of contaminated air over extended periods. And that different radioactive fractions of aerosols will cause different health effects on the human body upon inhalation.

2.2 Virtual Impactors

As Długosz-Lisiecka et al. (2025) showed previously, nuclear aerosols can pose hazards to the human body. Assuming that nuclear aerosol particles are spherical and of homogeneous composition, the magnitude of the impact through radioactive emission is directly correlated to their mass, thus their aerodynamic diameter: a ten-micrometer difference in diameter results roughly in a one-thousand-becquerel change in emissions. That is a reason why aerosol sampling has to be mainly size-dependent. Another reason is that size influence their transport behavior, thus their depth penetration in the human body.

2.2.1 Principle

Inertial Impaction

Virgil A Marple et al. (2011) explains with the Figure 2.3 that the human lung acts as a natural size-selective particle classifier, relying on these deposition mechanisms: inertial impaction, gravitational settling, Brownian diffusion, deposition by interception (generally negligible except for elongated particles), electrostatic forces (important only for highly charged particles), and turbulent deposition (occurs mainly in upper airways).

Each mechanism is associated with certain characteristics of the particles and the fluid flow. According to Darquenne (2020), the inertial impaction is the primary mechanism of deposition for the particle > 2 µm, where sedimentation mechanism is more important for $1 - 8$ µm diameter particles and Brownian diffusion needs the particles diameter to be close to the mean free path (λ_f) of the gas molecules so that the particle no longer moves as a continuum in the gas, but as a particle among discrete gas molecules.

Inertial impaction occurs when there is a sudden change in the direction of the flow, which causes coarse particles to deviate from the air streamlines as the inertia of the particles keeps them on their initial trajectories. As a result, coarse particles may impact on airway walls and be removed from the flow and fine particles keep following the streamlines. The cut-off diameter is mainly influenced by the velocity magnitude in the acceleration nozzle.

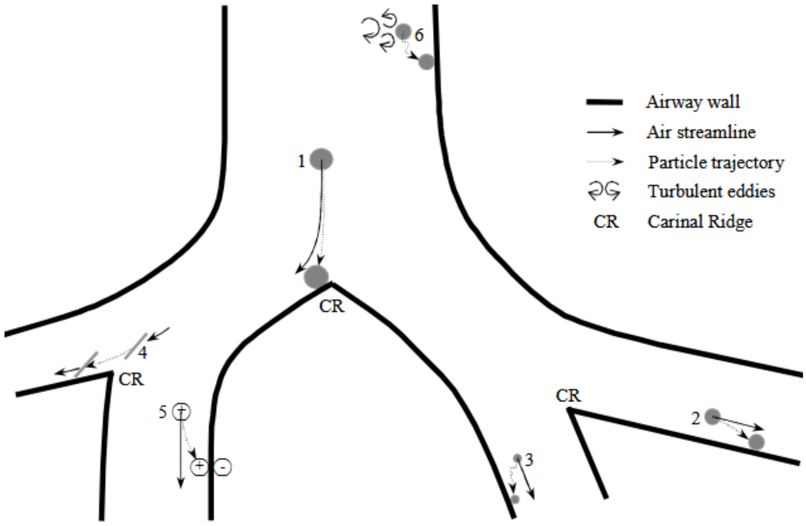


Figure 2.3: Mechanisms of particle deposition in the respiratory tract: (1) inertial impaction, (2) gravitational settling, (3) Brownian motion, (4) interception, (5) electrostatic forces, (6) turbulent deposition (Virgil A Marple et al., 2011).

Conventional vs Virtual Impactor

A conventional impactor is a device that uses this principle of inertial impaction, with an impaction plate for coarse particles while fine particles are deviated Figure 2.4 (A). It is mainly called a physical impactor or a plate impactor.

On the other hand, the virtual impactor (VI) does not need this impaction plate and rely exclusively on airflow principle. The carrying fluid is sucked by the outlets with a ratio between the flow channels. In the Figure 2.4 (B), this ratio is between the bottom channel and the sides channels. The bottom channel is called "Minor Flow Channel" and the sides channels are called "Major Flow Channel" due to the ratio of the flow considered in these channels, their names represent their identity in the fluid flow repartition: the minor flow must be $5 - 20\%$ of the total flow or the major flow must be $80 - 95\%$ of the total flow (Hinds et al., 2022). As the major flow deviates from the acceleration nozzle, fine particles are deviated into the major flow channel, coarse particles go in the minor flow channel (represented by the collection probe).

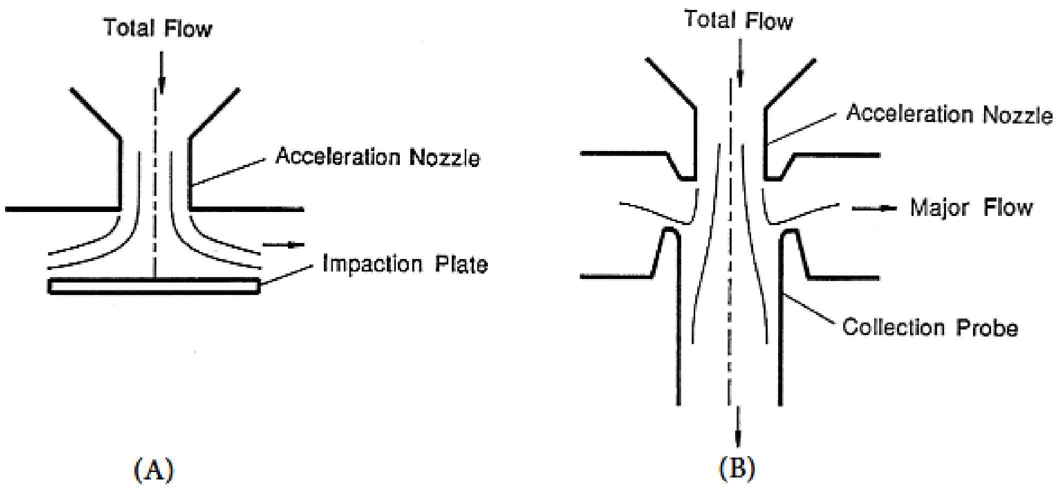


Figure 2.4: Representation of (A) a conventional impactor, (B) a virtual impactor (Virgil A Marple et al., 2011).

In the context of nuclear decommissioning, it is preferable to use VI than conventional impactors because:

- The impaction plate creates rebounds of coarse particles, it degrades with utilization time and would need decontamination or stocking as a radioactive trash because of saturation.
- Both flows (major and minor flow) of the VI allows particles to stay in motion which reduces operator danger with the manipulation operations.
- VI combines classification and direct analysis, which is really difficult to implement in plate impactors.

The objective of VI is to separate coarse and fine particles around a reference size. It is possible to specify this separation using the following parameters.

2.2.2 Physics of the Virtual Impactor

Reynolds Number

The airflow regime in the VI is governed by the Reynolds Number Re:

$$Re = \frac{\text{viscous diffusion time}}{\text{transport time}} = \frac{\rho_{\mathrm{f}} D_{\mathrm{f}}}{\mu_{\mathrm{f}}} \times U_{\mathrm{f}} \ , \tag{2.8}$$

where ρ_{f} is the density of the fluid, D_{f} is the characteristic diameter of the flow, U_{f} is the mean velocity of the fluid.
It characterizes the behavior of the fluid:

→ $Re < 2 \times 10^3$: Laminar flow regime (clean streamlines).

→ $Re \in \left]2 \times 10^3 - \ 1 \times 10^4\right[$: Transition from Laminar to Turbulent flow.

→ $Re > 1 \times 10^4$: Turbulent flow regime (random perturbations).

Stokes Number

The Stokes Number Stk represent the probability that a particle will diverge from the streamlines (Colbeck et al., 2014):

$$Stk = \frac{\text{particle response time}}{\text{fluid time scale}} = \frac{\rho_{\mathrm{p}} d_{\mathrm{e}}^2 C_{\mathrm{c}}\left(d_{\mathrm{e}}\right)}{18 \mu_{\mathrm{f}} \chi} \times \frac{U_{\mathrm{f}}}{D_{\mathrm{f}}} \ , \tag{2.9}$$

or, to refer as aerodynamic diameter:

$$Stk = \frac{\rho_0 d_{\mathrm{ae}}^2 C_{\mathrm{c}}\left(d_{\mathrm{ae}}\right)}{18 \mu_{\mathrm{f}}} \times \frac{U_{\mathrm{f}}}{D_{\mathrm{f}}} \ . \tag{2.10}$$

→ A particle with $Stk \ll 1$ follows the stream lines.

→ A particle with $Stk \gg 1$ can't follow the stream lines due to its inertia, and then follows a straight trajectory.

2.2.3 Performance of the Virtual Impactor

The performance of VI is measured through multiple indicators:

Cutoff diameter

The Cutoff diameter d_{50} represents the diameter of the particle that has 50 % chances of going into the minor flow. This is the reference size used to separate coarse ($> d_{50}$) and fine ($< d_{50}$) particles.

$$d_{50} = \sqrt{\frac{18\mu_\mathrm{f} D_\mathrm{f}}{\rho_\mathrm{p} C_\mathrm{c}\left(d_{50}\right) U_\mathrm{f}}} \times \sqrt{Stk_{50}}\ , \tag{2.11}$$

where Stk_{50} is the cutoff Stokes number.

Sharpness

The sharpness σ_{VI} represents the deviation of the distribution of particles collected through the VI to d_{50}, Le et al. (2021) proposes one definition:

$$\sigma_{\mathrm{VI}} = \sqrt{\frac{d_{84}}{d_{16}}}\ , \tag{2.12}$$

where d_{84} is the diameter of the particle that has 84 % chances of going into the minor flow, d_{16} is the diameter of the particle that has 16 % chances of going into the minor flow.

According to the Figure 2.5, which represents a collection efficiency diagram of ideal and realistic VI, $\sigma_{\mathrm{VI}} \geq 1$. $\sigma_{\mathrm{VI}} = 1$ corresponds to the ideal separation in which all coarse particles are in the minor flow channel and all fine particles are in the major flow channel.

However, particles that have a high stokes number will deviate from their streamlines, increasing σ_{VI} and possibly modifying d_{50}.

- If fine particles deviate to the minor flow channel, it is called **Minor flow contamination**.
- If coarse particles deviate to the major flow channel, it is called **Major flow contamination**.
- If particles deviate on the wall of any channel, it is called **Wall loss** (Figure 2.6).

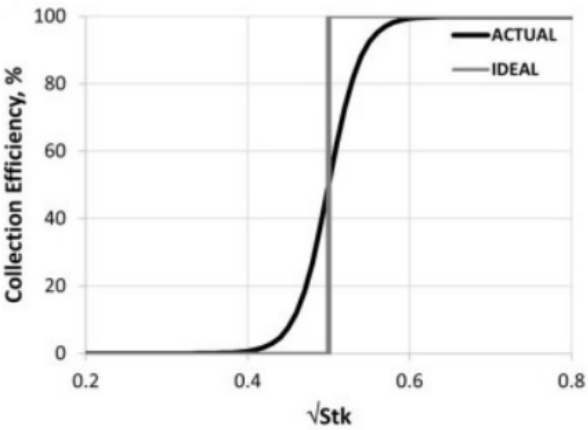


Figure 2.5: Ideal vs. Actual virtual impactor (Romay et al., 2023)

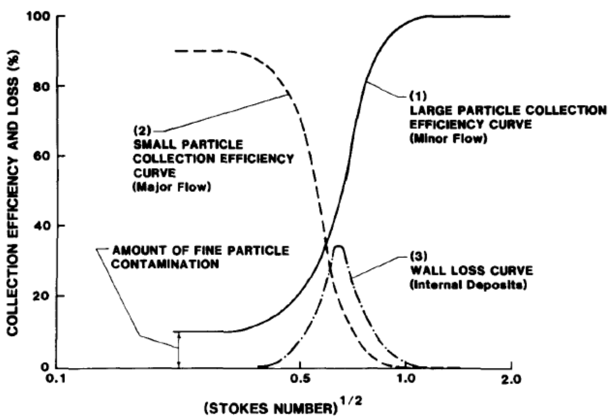


Figure 2.6: Collection efficiency diagram (Chen et al., 1987).

2.2.4 Optimization Strategies

VI with precise d_{50} and σ_{VI} relies mainly on the adjustment of geometric and fluid separation region parameters, focusing on the elimination of contamination and losses.

Geometric Parameters

VI are 3D geometries, often a thickened or revolved 2D geometry. The Figure 2.7 is a representation of a 2D geometry containing every parameter discussed below and some others that can be thought of.

Hinds et al. (2022) show that W_2 should be about 30-50 % larger than W_1, R_2 should be about 30 % of W_1, W_3 should be about $1 - 1.8$ times W_1, and the axes of the nozzle should be aligned closely to that of the collection probe. Le et al. (2021) explains the value of W_2/W_1 do not have an important effect on d_{50}.

Virgil A. Marple et al. (1980) and Hari et al. (2006) dictates that particle loss or contamination is affected by W_2/W_1 and nozzles shapes characterized by θ_1, θ_2, θ_3, θ_3', L_1, L_3 and the alignment between the symmetry axis and the major flow channel characterized by θ. High losses or contamination occur for particles having a diameter close to d_{50}. Certain values of L_1, R_1, and W_2 might reduce particle losses and contamination. Rectangular nozzle shape can increase the flow rate without changing d_{50} (Masuda et al., 1988). Hari et al. (2006) explains that increasing θ_1 up to $45°$ reduces wall losses from 24 % to 9 % which improves σ_{VI} and the diminution of R_2 (with a limit of $R_2 = 0.16W_1$) reduces wall losses, that the variation of L_2 and L_3' does not show any incidence.

Flow Parameters

Virgil A. Marple et al. (1980) and McFarland et al. (1978) explains that d_{50} increases as the ratio Q_{minor}/Q_{major} decreases and a reduction of wall loss and contamination is observed when the ratio $Q_{minor}/Q_{inlet} \in 5 - 20$ %, respectively $Q_{major}/Q_{inlet} \in 80 - 95$ %. Hari et al. (2006) explains that the reduction of the ratio Q_{minor}/Q_{inlet} from 10 % to 30 % decreases $\sqrt{Stk_{50}}$ from 0.94 to 0.64, and wall losses decreases, more over Hari et al. (2006) found that increasing Re improves σ_{VI}, but decreases d_{50} and that if $Re > 500$ recirculation zones appears and wall losses increase by 20 %. Also, Zhao et al. (2026) shows that low temperature CO_2 (instead of air in standard conditions) as the carrier fluid reduces d_{50}.

Multistage Virtual Impactors

The classical VI is a dichotomous separator. Literature doesn't mention polytomous VI relying exclusively on inertia principle. The multi-stage VI allows to collect particles in different size ranges simultaneously by the superposition of classical single-stage VI (Novick, 1987; V. Marple et al., 2013; Virgil et al., 2014).

Others optimizations

d_{50} and σ_{VI} precision can also be improved using other physics than inertia:

- Zahir et al. (2021) developed a VI with electromagnetic field as shown in the Figure 2.8.
- P. Wang et al. (2025) placed a small triangle structure in the inlet as shown in the Figure 2.9.

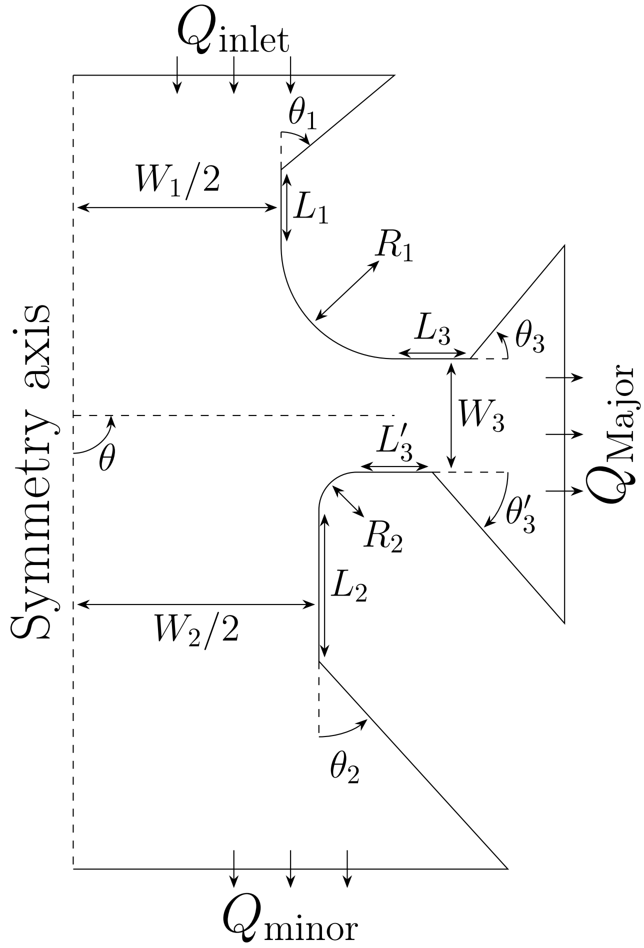


Figure 2.7: 2D geometry of the separation region of a virtual impactor.

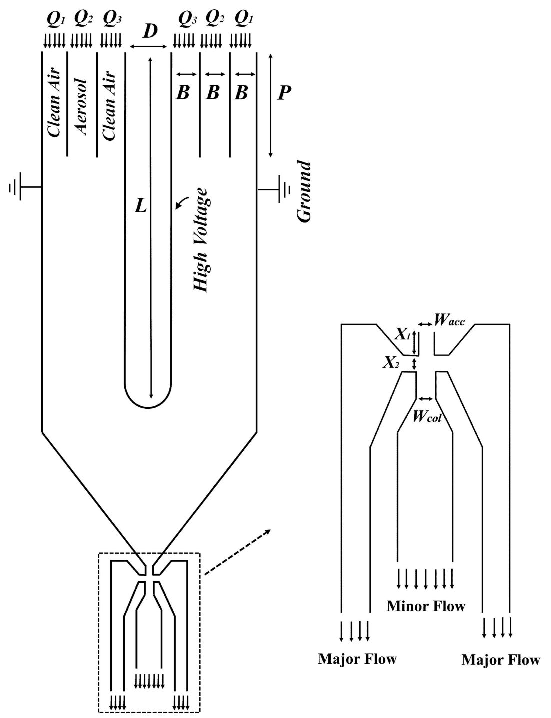


Figure 2.8: Virtual impactor with electromagnetic field in the acceleration nozzle (Zahir et al., 2021).

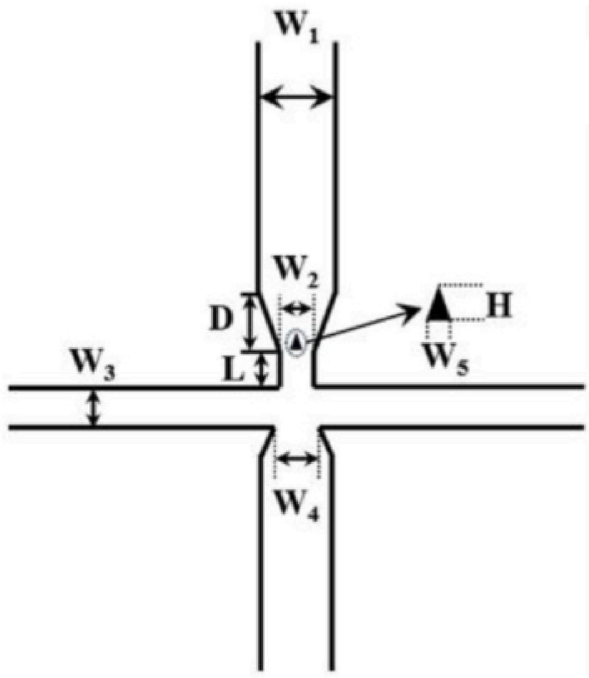


Figure 2.9: Virtual impactor with triangle shape in the acceleration nozzle (P. Wang et al., 2025).

2.3 Numerical Modeling of Particle-Laden Flows

The simulation of aerosol transport in virtual impactors requires a comprehension of the equations governing both continuous and discrete phases, interactions between these phases and discretization of those equations.

2.3.1 Continuous Phase Governing Equations

Continuity Equation

The continuity equation expresses the conservation of mass for the carrier fluid. In the general compressible form, the continuity equation is written as:

$$\frac{\partial \rho_\mathrm{f}}{\partial t} + \nabla \cdot (\rho_\mathrm{f} \mathbf{u}_\mathrm{f}) = 0 \ , \tag{2.13}$$

where $\mathbf{u}_\mathrm{f}$ is the fluid velocity vector.

For an incompressible flow, where the fluid density ρ_f is assumed constant, the equation reduces to:

$$\nabla \cdot \mathbf{u}_\mathrm{f} = 0 \ . \tag{2.14}$$

Navier-Stokes Equations

The motion of the carrier fluid is governed by the Navier-Stokes equations, which express the conservation of momentum. In their most general form, for a compressible viscous fluid, they are written as:

$$\frac{\partial (\rho_\mathrm{f} \mathbf{u}_\mathrm{f})}{\partial t} + \nabla \cdot (\rho_\mathrm{f} \mathbf{u}_\mathrm{f} \otimes \mathbf{u}_\mathrm{f}) = -\nabla p + \nabla \cdot \boldsymbol{\tau} + \rho_\mathrm{f} \, \mathbf{g} \ , \tag{2.15}$$

where p is the static pressure, $\boldsymbol{\tau}$ is the viscous stress tensor, and $\mathbf{g}$ is the gravitational acceleration vector.

For a Newtonian fluid, the viscous stress tensor is related to the strain rate tensor by:

$$\boldsymbol{\tau} = \mu_\mathrm{f} \left(\nabla \mathbf{u}_\mathrm{f} + (\nabla \mathbf{u}_\mathrm{f})^\top - \frac{2}{3} (\nabla \cdot \mathbf{u}_\mathrm{f}) \mathbf{I} \right) \ , \tag{2.16}$$

where and $\mathbf{I}$ is the identity tensor.

Under the incompressible assumption, the equation simplifies to:

$$\rho_\mathrm{f} \left(\frac{\partial \mathbf{u}_\mathrm{f}}{\partial t} + (\mathbf{u}_\mathrm{f} \cdot \nabla) \mathbf{u}_\mathrm{f} \right) = -\nabla p + \mu_\mathrm{f} \nabla^2 \mathbf{u}_\mathrm{f} + \rho_\mathrm{f} \mathbf{g} \ . \tag{2.17}$$

The system of equations that combines (2.14) and (2.17) fully describes the dynamics of the carrier phase (Ferziger et al., 2020).

2.3.2 Multiphase Flow Modeling

Coupling Regimes

For a multiphasic flow, Cao et al. (2023) mentioned there are four types of considerations on how to couple fluids to particles and characterize their interaction:

- One-way coupling: Effect of fluids on particles.
- Two-way coupling: Effect of fluids on particles and vice versa.
- Four-way coupling: Effect of fluids on particles and vice versa, Interparticle reactions.

Figure 2.10 describes conditions for choosing coupling regime based on Φ_p , with $\Phi_\mathrm{p} + \Phi_\mathrm{f} = 1$, where Φ_p is the proportion of the particulate phase in the total volume and Φ_f is the proportion of the fluid phase in the total volume.

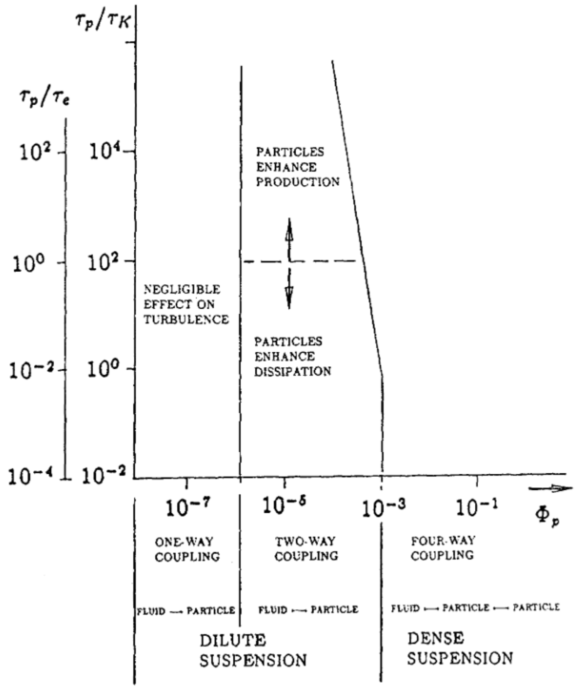


Figure 2.10: Map of regimes of interaction between particles and turbulence (Elghobashi, 1994).

Eulerian vs Lagrangian Approaches

For Elghobashi (1994), two main approaches exist for modeling particle-laden flows. The Euler-Euler (two-fluid) method treats both phases as interpenetrating continua, it is well-suited for dense suspensions ($\Phi_\mathrm{p} > 10^{-3}$) where particle-particle interactions dominate. The Euler-Lagrange method solves the continuous phase on a fixed Eulerian mesh using the continuity (2.14) and Navier-Stokes equations (2.17) while particles trajectories are tracked individually using their motion equations (2.1).

Particle-Wall Interaction Models

The interaction between aerosol particles and VI surfaces impacts the efficiency and selectivity of particle classification. When a particle impacts a wall, several outcomes are possible: elastic rebound (full energy recovery), inelastic rebound (energy loss), or adhesion to the surface. The transition between these regimes is governed by impact energy and surface properties, and is characterized by two key parameters: the coefficient of restitution and the critical sticking velocity.

Coefficient of Restitution The coefficient of restitution e quantifies energy loss upon impact. It varies between 0 (adhesion) and 1 (elastic rebound) and is defined as the ratio of the rebound velocity v_r^n to the incident velocity v_i^n in the normal direction of the surface (Thornton et al., 1998):

$$e = \frac{v_\mathrm{r}^\mathrm{n}}{v_\mathrm{i}^\mathrm{n}} \ . \tag{2.18}$$

Critical Sticking Velocity and Thornton-Ning Model Thornton et al. (1998) defined a critical sticking velocity v_s corresponding to the maximum impact velocity below which a particle adheres to the surface upon impact. Above v_s, particles rebound.

For a sphere of radius R impacting a flat surface, the critical sticking velocity is expressed as (Thornton et al., 1998):

$$v_\mathrm{s} = 1.84 \left[\frac{(\Gamma/R)^5}{\rho_\mathrm{p}^3 E^{*2}} \right]^{1/6} , \tag{2.19}$$

where Γ is the interface energy, $E^* = E_\mathrm{VI}/(1-\nu_\mathrm{VI}^2)$ is the reduced modulus with E_VI Young's modulus of the VI and ν_VI Poisson's ratio of the VI.

For incident velocities below or equal v_s, adhesion dominates and $e = 0$ (critical sticking condition). For $v_\mathrm{i}^\mathrm{n} > v_\mathrm{s}$, the particle rebounds with $e > 0$. The complete model provides an explicit formulation for e as a function of impact velocity:

$$e = \left[1 - \left(\frac{v_\mathrm{s}}{v_\mathrm{i}^\mathrm{n}} \right)^2 \right]^{1/2} . \tag{2.20}$$

2.3.3 Turbulence Modeling

Choice of Turbulence Model

In contrast to a laminar flow (where fluid layers slide in parallel without disruption), a turbulent flow is characterized by chaotic fluctuations in pressure and velocity. Most engineered and natural flows are turbulent, and in aerosol flows, turbulence governs particle dispersion, mixing, and wall deposition (Shukla et al., 2022).

Turbulence arises when fluid kinetic energy overcomes viscous damping, creating unsteady vortices across multiple scales. Because resolving all these scales is computationally prohibitive, turbulence must be modeled.

Three principal modeling approaches exist, distinguished by the range of scales they resolve:

Direct Numerical Simulation (DNS) DNS solves the instantaneous three-dimensional Navier-Stokes equations without turbulence modeling, resolving all scales of motion from the integral scale to the Kolmogorov microscale. This approach yields the most accurate representation of turbulent flow physics but at prohibitive computational cost: the required number of grid points scales as $N_\mathrm{x} \sim Re_\ell^{9/4}$, where Re_ℓ is the large-eddy Reynolds number (Moin et al., 1998).

Large Eddy Simulation (LES) LES filters the Navier-Stokes equations to separate large-scale (resolved) and small-scale (subgrid) motions (Sagaut, 2006). The large structures, which carry most of the turbulent kinetic energy and are geometry-dependent, are resolved explicitly. The small, universal scales are modeled using subgrid-scale models, such as the Smagorinsky model or dynamic procedures (Germano et al., 1991).

Reynolds Averaged Navier-Stokes (RANS) RANS decomposes flow variables into time-averaged and fluctuating components, leading to equations for the mean flow that include unknown Reynolds stresses (Reynolds, 1895). These stresses representing the effect of turbulent fluctuations on the mean flow must be modeled. RANS approaches are the most computationally economical, requiring modest grid resolution, and have achieved wide industrial acceptance due to their robustness and relatively low computational demand. The trade-off is that all scales of turbulence are averaged out and modeled: RANS can not capture instantaneous turbulent fluctuations or the temporal evolution of large structures.

In RANS, velocity is decomposed as $u_{\mathrm{f},i} = \bar{u}_{\mathrm{f},i} + u'_{\mathrm{f},i}$, the momentum equation becomes:

$$\frac{\partial(\rho_\mathrm{f}\,\bar{u}_{\mathrm{f},i})}{\partial t} + \frac{\partial(\rho_\mathrm{f}\,\bar{u}_{\mathrm{f},i}\bar{u}_{\mathrm{f},j})}{\partial x_j} = -\frac{\partial \bar{p}}{\partial x_i} + \frac{\partial}{\partial x_j}\left(\mu_\mathrm{f}\,\frac{\partial \bar{u}_{\mathrm{f},i}}{\partial x_j}\right) - \frac{\partial}{\partial x_j}(\rho_\mathrm{f}\,\overline{u'_{\mathrm{f},i}u'_{\mathrm{f},j}}) \ , \tag{2.21}$$

where the term $-\rho_\mathrm{f}\overline{u'_{\mathrm{f},i}u'_{\mathrm{f},j}}$ represents the Reynolds stress tensor, which bears the effects of turbulent fluctuations.

The Boussinesq viscosity hypothesis assumes:

$$-\rho_\mathrm{f}\overline{u'_{\mathrm{f},i}u'_{\mathrm{f},j}} = \mu_\mathrm{t}\left(\frac{\partial \bar{u}_{\mathrm{f},i}}{\partial x_j} + \frac{\partial \bar{u}_{\mathrm{f},j}}{\partial x_i}\right) - \frac{2}{3}\rho_\mathrm{f} k \delta_{i\,j} \ , \tag{2.22}$$

where μ_t is the turbulent viscosity, k is the turbulent kinetic energy and $\delta_{i\,j}$ is the Kronecker delta.

Different RANS models distinguish themselves by how they model the turbulent viscosity μ_t, principally defined as (Versteeg, 2007):

$$\mu_\mathrm{t} = \rho_\mathrm{f} C_\mu \times f(\phi_1, \phi_2), \quad \phi_1, \phi_2 \in \{k, \varepsilon, \omega\} \ , \tag{2.23}$$

where C_μ is the turbulent viscosity constant, ε is the dissipation rate and ω is the specific dissipation rate.

$k - \varepsilon$ Model

The standard $k - \varepsilon$ model solves transport equations for k and ε with:

$$\mu_\mathrm{t} = \rho_\mathrm{f} C_\mu \times \frac{k^2}{\varepsilon} \ . \tag{2.24}$$

It is economical and robust, particularly in free shear layers and bulk flows with high Reynolds number. However, it struggles with adverse pressure gradients and boundary layer separation, and its behavior near walls requires special treatment (wall functions or near-wall resolution) (Launder et al., 1983).

$k - \omega$ Model

The $k - \omega$ model solves transport equations for k and ω with:

$$\mu_\mathrm{t} = \rho_\mathrm{f} \times \frac{k}{\omega} \ . \tag{2.25}$$

The advantage of the $k - \omega$ formulation is that ω can be prescribed directly at the wall (as a boundary condition), enabling the model to resolve the viscous layer without explicit wall functions (Wilcox, 1988).

$k - \omega$ SST Model

The $k - \omega$ SST (Shear Stress Transport) model, developed by Menter (1994), uses the standard $k - \omega$ model near walls (where it handles boundary layers well) and transitions to a $k - \varepsilon$ model behavior in the outer flow (where it exhibits better stability). $k - \omega$ SST also includes additional cross-diffusion and viscosity-limiting terms to improve prediction of separated flows and adverse pressure gradients. It has become the industry standard for engineering CFD.

However, one-equation RANS closure models also exist, such as the Spalart-Allmaras model, which transports a single variable, the modified turbulent viscosity $\tilde{\nu}$, and performs well in attached boundary layers (Spalart et al., 1992).

Among those, there is no universal model: the choice depends on flow characteristics and computational budget.

Wall Functions and Near-Wall Treatment

The treatment of turbulence near solid walls is critical for accurate prediction of particle deposition and wall shear stresses in aerosol impactors. The region immediately adjacent to a wall exhibits strong gradients in velocity and turbulent quantities, RANS models require special treatment in these near-wall regions, two strategies exist: wall functions and near-wall resolution. The choice depends on the dimensionless wall distance y^+, defined as:

$$y^+ = \frac{u_\tau y_\mathrm{w}}{\nu_\mathrm{f}} \ , \tag{2.26}$$

where u_τ is the friction velocity ($u_\tau = \sqrt{\tau_\mathrm{w}/\rho_\mathrm{f}}$), τ_w being the wall shear stress), y_w is the distance from the wall and ν_f is the kinematic viscosity of the fluid.

Wall Functions (High-Reynolds Treatment) Wall functions uses empirical relationships to relate mean velocity and turbulent quantities at a point outside the viscous sublayer (typically $y^+ \approx 30 - 300$) to wall shear stress, without explicitly resolving the viscous layer. This approach reduces mesh resolution requirements near walls and is economical (Versteeg, 2007).

Near-Wall Resolution (Low-Reynolds Treatment) Alternatively, the grid can be refined to resolve down into the viscous sublayer, typically with $y^+ < 1$. This requires many more grid points near walls but captures the full velocity and turbulence profiles without empirical assumptions.

2.3.4 Numerical Schemes and Discretization

Numerical resolution of the governing equations presented above rely on the spatial and temporal discretization of the continuous flow domain.

Spatial Discretization

Mainly three spatial discretization methods exist in continuum mechanics: Finite Difference Method (FDM), Finite Element Method (FEM), Finite Volume Method (FVM).

FDM covers the fluid domain with a grid and approximates the differential equation in each point by polynomial fitting of the functions. It doesn't automatically ensure conservation and is restricted to simple geometries. FEM decomposes the domain into an unstructured set of elements (triangles/quadrilaterals in 2D, tetrahedra/hexahedra in 3D), then the equations are multiplied by a weight function before they are integrated over the entire domain. And FVM subdivides the domain into a finite number of control volumes, and the conservation equations are applied to each control volumes. At the centroid of each control volumes lies a computational node at which the variable values are to be calculated. Interpolation is used to express variable values at the control volumes surface in terms of the control volumes center values. Surface and volume integrals are approximated using quadrature formulas. FVM can accommodate any type of grid, so it is suitable for complex geometries. It is the most common method for fluid dynamics numerical resolution (Ferziger et al., 2020).

Temporal Discretization

A flow is said to be steady-state when its properties (velocity, pressure, concentration) no longer change with time and doesn't require temporal discretization. In contrast, a transient flow evolves over time and requires time-accurate integration of the governing equations, which implies discretizing the time domain into finite time steps.

For physics such as aerosol transport, where particle trajectories evolve from the inlet to the outlet of the impactor, transient formulations are necessary. Two main approaches exists for advancing the solution in time: explicit and implicit methods.

Explicit Formulations In explicit schemes, the solution at the next time step is computed directly from the current known state. They are straightforward to implement but are subject to stability constraints. The Courant-Friedrichs-Lewy (CFL) condition imposes a maximum allowable time step (Ferziger et al., 2020):

$$\mathcal{C} = \frac{\mathbf{u}_{\mathrm{f}} \Delta t}{\Delta x} \leq 1, \tag{2.27}$$

where $\mathcal{C}$ is the Courant number, Δt is the time step and Δx is the cell size.

Implicit Formulations Implicit schemes solve a system of equations coupling all cells simultaneously at each time step, which relaxes the CFL stability constraint and allows larger time steps. It is more stable than explicit method (Ferziger et al., 2020).

Mesh Topology and Quality

The spatial discretization of the flow domain into a computational mesh directly conditions the accuracy and computational cost of the simulation. Structured meshes which consist of hexahedral elements arranged in a regular, logically ordered grid. They offer high numerical accuracy and computational efficiency due to their regular connectivity. While Unstructured meshes use elements of arbitrary shape (tetrahedra, prisms, polyhedra) that can conform to complex geometries automatically. They are easier to generate for intricate geometries but generally require more elements to achieve the same accuracy as structured meshes.
Several quality metrics must be monitored to ensure reliable simulation results:

- **Orthogonality** measures the alignment between cell face normals and the vector connecting adjacent cell centers. Maximum acceptable orthogonality is 85° otherwise, convergence is compromised.

- **Skewness** measures the deviation between the actual cell volume and the volume of the corresponding ideal regular shape. A skewness of 0 indicates a perfectly regular cell, while a skewness of 1 corresponds to a fully degenerate cell with zero volume.

- **Aspect ratio** is the ratio of the longest to the shortest dimension of a cell. High aspect ratios are acceptable in boundary layers where gradients are predominantly unidirectional, but degrade accuracy in regions of complex three-dimensional flow.

Despite their wide adoption, RANS-based CFD simulations remain computationally expensive, as achieving accurate results requires high mesh resolution and iterative convergence. This cost becomes a significant constraint when parametric studies are needed to optimize impactor geometry, as each configuration requires a full simulation cycle.

A promising perspective to address this limitation is the use of machine learning-based reduced order models. As demonstrated by Pant et al. (2021), a deep learning reduced order model studies many pre-calculated fluid flow examples (from full Navier-Stokes simulations) and learns to recognize patterns in how the flow changes over time. Once trained, it can quickly predict new flow behaviors without solving the complex equations again, instead of calculating every possible move from scratch. So, it doesn't copy the equations, it learns their effects from examples and mimics them efficiently. This technology is quite recent and functional only on simple fluid resolutions. It is a powerful tool, but can induce irregularities in more complex situations.

2.4 3D Printing for Aerosol Instrumentation

The fabrication of a low-cost, single-use virtual impactor relies heavily on 3D printing: Waldbaur et al. (2011) explains that material selection and manufacturing techniques allow to ensure reproducibility, precision, and performance.

2.4.1 Materials Selection

The range of materials used in 3D printing is wide, but not all of them are suitable for aerosol devices applications.

Metals are robust and chemically stable, but they are rarely used for microfluidic chips due high costs (gold, platinum) or rough surfaces. They remain relevant for high-temperature applications (plasma, flames) but are not ideal for aerosol monitoring or cell culture (Waldbaur et al., 2011).

Silicon was historically dominant in microfluidics, this material offer high bio-compatibility and chemical resistance. However, its fabrication via etching is a two-step process, and silicon manufactured items lack gas permeability, making them unsuitable for aerosol applications (Waldbaur et al., 2011).

Polymers, in contrary to these two, have become the material of choice for microfluidics, particularly for disposable, low-cost devices. They share the main properties of metals and silicones such as chemical resistance, stiffness, and surface tension, but they are mainly optical transparent. They can be classified by their chemistry (acrylates, epoxy resins, fluorinated) or by their physical behavior (thermoplastics, thermosets, elastomers) which ensures different properties (Waldbaur et al., 2011).

2.4.2 Manufacturing Techniques

3D-printed structures can be manufactured using two main techniques: Material removal or Material deposition.

Material removal techniques are often used for high-precision components or materials that are difficult to deposit, they generate material waste. Waldbaur et al. (2011) details some of them:

→ Electrical Discharge Machining: It uses electrical spark discharges to erode material from conductive substrates, allowing the creation of complex micro-structures and deep channels with high precision.

→ Laser Machining: It uses a focused laser beam to melt, burn, or vaporize material. It is suitable for precise cutting and patterning on polymer substrates.

However, material removal techniques may have difficulties to efficiently "dig" complex channels in a volume, which could be useful for VI manufacturing. On the other hand, material deposition techniques enable the creation of complex geometries for aerosols applications (Waldbaur et al., 2011), here are two techniques that can be considered for aerosol devices printing:

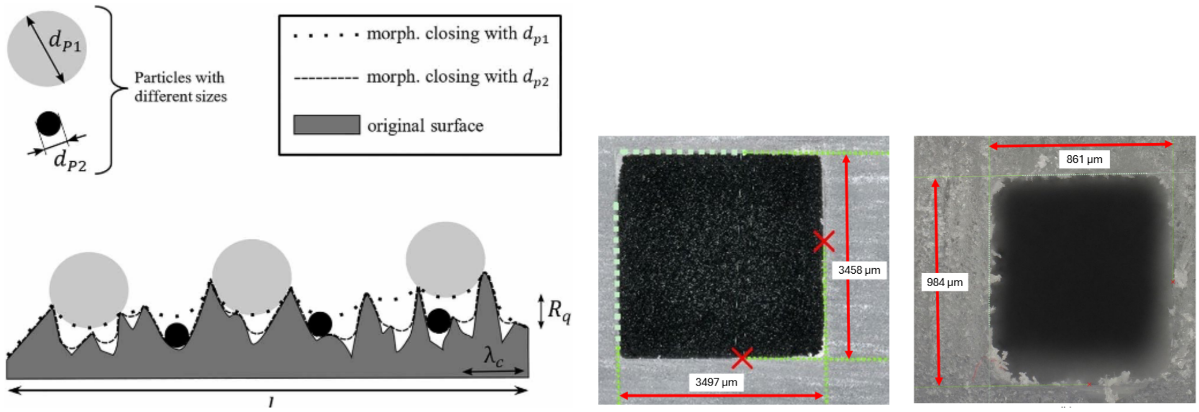
→ Stereolithography (STL): It uses UV light to polymerize liquid photoresists. It is high-resolution but limited by layer thickness and resin compatibility.

→ Two-Photon Polymerization: It allows sub-micron precision by using femtosecond laser pulses to polymerize resin at the focal point. However, it is slow and requires specialized equipment, making it less suitable for large-scale production.

2.4.3 Constraints and Opportunities for Nuclear Applications

Fluid and particles interaction on printed surfaces

Y. Wang et al. (2024) and Bazan et al. (2025) shows that printing technique and material used can create roughness on the internal surface of VI. Altmeppen et al. (2020) and Altmeppen et al. (2022) developed a theoretical and experimental spread model that characterizes the effect of target surface roughness on rebound spread. They explain that the effect of surface roughness is significant for a particle's individual rebound behavior and should not be neglected in precise numerical simulations: particles kinetic energy loss can attain 50 %. Depending on the plate angle and the surrounding fluid flow, particles are decelerated on their path to the surface or accelerated after the rebound, and they can be captured in particular cases, as shown in Figure 2.11 (a). Laffolley, Tsubota, et al. (2026) also showed (Figure 2.11 (b)) that for VI, the channel walls can be subject to these printing defaults, and therefore be able to capture particles passing by.



(a) (b)

Figure 2.11: (a) Morphological filter with respect to the particle size (Altmeppen et al., 2022) , (b) Digital microscope images of slices of µSPLIT, perpendicular to the flow channels.

Broader Deployment Perspectives

For nuclear applications, it would be interesting to develop a VI that can be fabricated at a reduced cost and disposed of easily after usage by incineration, without generating any hazardous human exposure during cleaning and decontamination activities, and free from hardly disposable metallic wastes (Laffolley, Tsubota, et al., 2026). The "low cost" characteristic of µSPLIT induces big quantities production: they can either be useful for frequent replacement single use for radioprotection matters, but these big quantities can also be used as a massive network for pollutant contamination assessment.

It could also be interesting for universities that can not afford to buy industrial VI due to their high price (TEC AG-E3 Aerosol Generator - IC-AG-E3: ≈ 12 000 € / SIOUTAS Cascade Impactor 9 L/min PK/1: ≈ 900 €) to use 3D-printed low cost VI. They could teach about aerosol management on air quality monitoring, pollination studies, or forest fire soot: for example Lilienfeld et al. (1990) uses a one stage VI to collect the coarse particles of asbestos in asbestos removal.

3

METHODS

3.1 Introduction

This chapter presents a complete methodology to answer efficiently the research question stated in the introduction chapter. The state of the art serves as a guide, and the methodological choices will be justified against existing approaches.

A parametric study testing dozens of simulations is conducted. Its objective is to identify an optimal set of parameters (fluid and geometry) to improve the performance of the current design of the µSPLIT and optimize particle discrimination.

3.1.1 Modeling Assumptions and Hypotheses

In order to conduct a focused study for the optimization of the 3D-printed VI, it is necessary to adopt assumptions and hypotheses. Below are listed the different choices that have been made for the frame of the study :

Flow Modeling

Hypothesis 1: Incompressible flow
The equation of state of the ideal gas is:

$$PV = n_{\mathrm{IG}} R T_{\mathrm{IG}} \tag{3.1}$$

where P is the pressure of the ideal gas, V is the volume of the ideal gas, n_{IG} is the amount of substance of the ideal gas, R is the universal gas constant $R = 8.314\ \mathrm{J} \cdot \mathrm{mol}^{-1} \cdot \mathrm{K}^{-1}$ and T_{IG} is the temperature of the ideal gas.

VI uses mainly air as the carrier fluid, which can be considered as ideal gas. Air is assumed to be at standard atmospheric pressure and ambient temperature inside the virtual impactor.

The density of air can be expressed as:

$$\rho_{\mathrm{air}} = \frac{m_{\mathrm{air}}}{V_{\mathrm{air}}} = \frac{n_{\mathrm{air}} M_{\mathrm{air}}}{V_{\mathrm{air}}} = \frac{P_{\mathrm{air}} M_{\mathrm{air}}}{R T_{\mathrm{air}}} = \frac{P_{\mathrm{air}}}{\mathscr{R}_{\mathrm{air}} T_{\mathrm{air}}} \tag{3.2}$$

where ρ_{air} is the density of air, m_{air} is the mass of air in the VI, V_{air} is the volume of the VI filled with air, M_{air} is the molar mass of air, P_{air} is the mean pressure of air in the VI, T_{air} is the temperature of the air in the VI and $\mathscr{R}_{\mathrm{air}}$ is the specific constant of the air.

Property	Value	Unit
ρ_{air}	1.184	$\text{kg} \cdot \text{m}^{-3}$
μ_{air}	1.85×10^{-5}	$\text{kg} \cdot \text{m}^{-1} \cdot \text{s}^{-1}$
ν_{air}	1.56×10^{-5}	$\text{m}^2 \cdot \text{s}^{-1}$
c_{air}	346.3	$\text{m} \cdot \text{s}^{-1}$
λ_{air}	6.0×10^{-8}	m
M_{air}	28.97	$\text{g} \cdot \text{mol}^{-1}$
$\mathscr{R}_{\text{air}}$	287	$\text{J} \cdot \text{kg}^{-1} \cdot \text{K}^{-1}$

Table 3.1: Air properties at 25 °C, 1 atm (Blevins, 1984).

The pressure and temperature variations inside the VI are assumed to be negligible, meaning that density variations can be ignored. This allows the flow to be considered incompressible, which is represented by:

$$Ma < 0.3 \tag{3.3}$$

where Ma is the Mach number.

Hypothesis 2: Steady state
The fluid flow is considered to be in steady state ($\partial/\partial t = 0$) to simulate the continuous, real-time operation of the device.

Hypothesis 3: No gravity
The gravity force is neglected due to uncertain orientation of the VI during operation, moreover it is assumed to be negligible to the other forces acting on the fluid and on the particles.

Particle Dynamics

Hypothesis 4: Rigid spherical particles
The particle diameter is assumed to be the aerodynamical diameter, which allows the simplification of numerical model of particles' geometries: each particle is assumed to be spherical with a density of $\rho_0 = 1000 \text{ kg} \cdot \text{m}^{-3}$.

Hypothesis 5: One-way coupling
Particle concentration is assumed low enough that interactions between fluid and particles and interactions between particles themselves are assumed to be negligible. Each particle is tracked independently. As defined in Multiphase Flow Modeling subsection, one-way coupling is assumed.

Hypothesis 6: No electrostatic effects
Particles are assumed neutral (uncharged). Electrostatic forces are neglected.

Geometry and Fabrication

Hypothesis 7: 3D-printable design
The prototypes are 3D printed using a FORMLABS FORM 4.

- Layer thickness/vertical resolution : $25 - 300\ \mu m$
- Printing material : Clear Resin V5 (FORM 4)

The technical specifications of this 3D printer and material used are specified in Appendix B.2.1.

Hypothesis 8: Planar half-geometry
The whole geometry has a symmetry plane. To reduce simulation costs, it is assumed that only half of the whole geometry is modeled. The symmetry plane is modeled as a symmetry boundary condition and not as a wall boundary condition.

Boundary Conditions

Hypothesis 9: Pressure and Velocity assumptions
The two outlets suck the fluid with a constant mass flow rate and the inlet is considered at atmospheric pressure.

Hypothesis 10: No-slip at walls
The condition of no-slip is applied on walls. This implies the velocity of the fluid on the wall to be zero.

Simulation Setup

Hypothesis 11: Converged flow field before particle injection
The fluid phase simulations are integrated until full convergence of the flow field is achieved. The particles are injected and tracked only in this steady state velocity field.

3.2 Automated Parametric Study Workflow

The parametric study aims to find optimal configuration for the separation stage ($d_{50} = 10$ µm). Optimal configuration is expected to maximize compliance with the JAEA's requirements, based on Key Performance Indicators (KPIs) : d_{50}, σ_{VI}, Wall loss , Minor contamination, Major contamination.

In order to have an efficient, reliable, and reproducible parametric study, the entire workflow has been automated with Python. It is structured as follows in Figure 3.1.

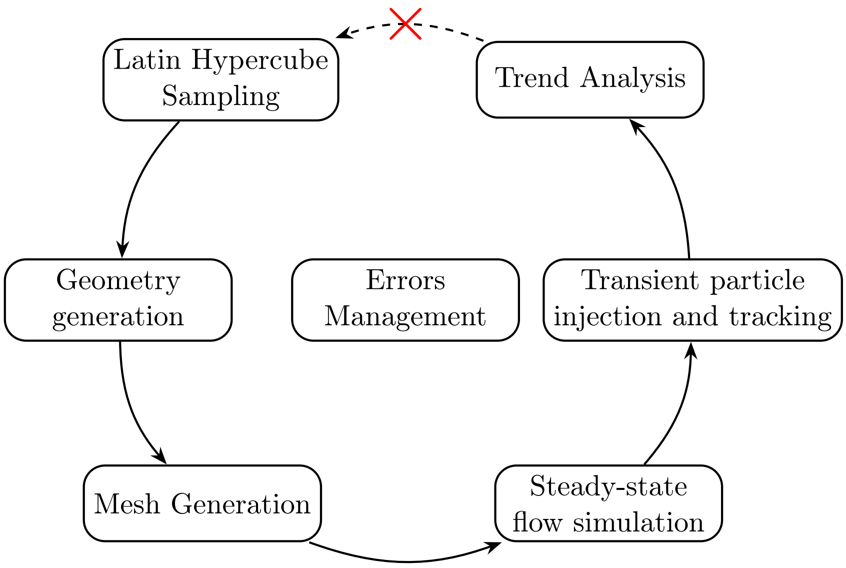


Figure 3.1: Parametric study automation structure

This automation creates multiple geometric configurations that will be exploited through simulation. The execution sequence for one geometry proceeds as follow, each step is detailed further. First, Latin Hypercube Sampling (LHS) selects values of geometric and fluid parameters that will be used in the process. Then, the geometry is created with geometric parameters values issued from LHS selection. Next, the geometry is meshed, the boundary conditions, issued from LHS, are applied on the corresponding surfaces of the meshed geometry. Then, the airflow simulation runs until convergence of the steady state. When this state of convergence is attained, the distribution of particles is injected by the inlet and the simulation runs until no particle is left in the geometry, each particle is tracked at the outlets and walls of the geometry. When all the configurations have been simulated, the overall performance and the influence of LHS parameters on KPIs is plotted.

Furthermore, an automated error management system is running continuously in the background of the workflow. The primary objective of this module is to monitor the simulations in real-time and automatically terminate any runs that shows unphysical behavior or poor convergence behavior (unclosed geometry, diverging residuals, or trapped particles removal). This ensures a reduction of computational time waste.

The red cross between the last step and the first step represents a possible interpretation of the results by the automaton, which would adjust the LHS boundaries itself. This approach considers the utilization of an adaptive sampling that relies on machine learning methods such as Bayesian Optimization, which would need a full trained surrogate model to be efficient.

The following paragraphs explore, step by step, the automation workflow functioning.

Please note that the general structure of an OpenFOAM case simulation and every function referred in those paragraphs explanations are available in the Appendix A.1.2

3.2.1 Latin Hypercube Sampling (LHS)

LHS is a stratified statistical method for generating combinations of parameter values to cover the design space efficiently. In the LHS method, the range for each parameter is divided into m bins with equal probability, where m is the number of required sample points (the number of simulations that will be conducted). Then, a sample is randomly selected from each bin for each uncertain variable. Next, the samples of different uncertain variables are randomly matched, resulting in an $n \times m$ sample matrix, where n is the number of input parameters. The initial binning enables LHS to cover the space of each uncertain parameter better than random sampling (when $m < 500$). LHS is not a sequential sampling method because the binning of the parameters changes as the value of m is modified (Mohammadi et al., 2022).

In order to grant reproducibility across the parametric study, a *seed* determines the randomness, it is natively implemented in the LHS algorithm. Consequently, each 3-tuple (n, m, *seed*) provide unique (thus reproducible) combination of parameters. This deterministic mapping ensures that any specific geometric configuration can be reconstructed identically at any time.

Configurations of the parameters (n)

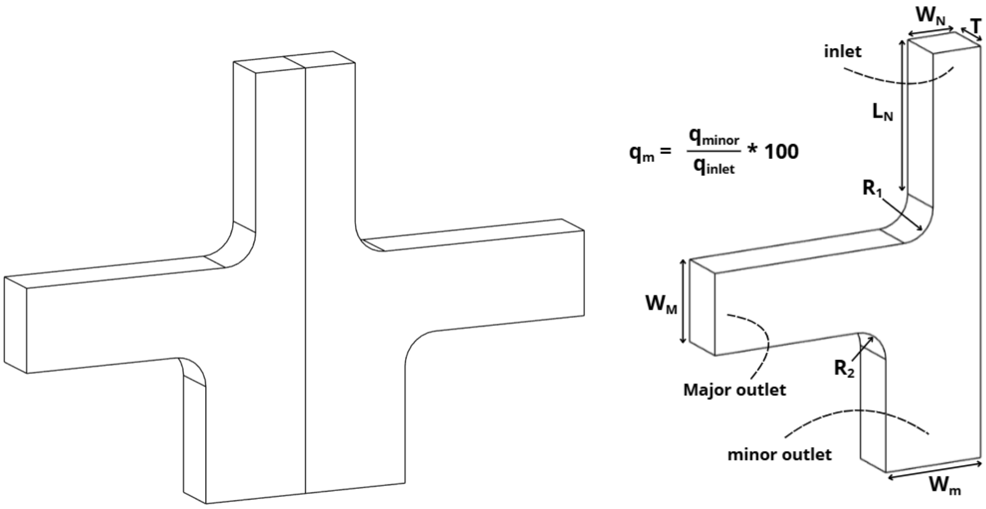
The general shape of the geometry that will be used for each simulation is represented in Figure 3.2.

As assumed in the Hypothesis 8, instead of working with the complete geometry of the 10 µm stage (3.2a), the study will focus on the optimization of half of a stage (3.2b).

On Figure 3.2b, the different parameters that will be modified for the optimization are represented. They are listed below with their significance:

- W_N : Width of the acceleration Nozzle (also called inlet),
- W_M : Width of the Major flow outlet,
- W_m : Width of the minor flow outlet,
- L_N : Length of the acceleration Nozzle (also called inlet),

- R_1 : Radius of the fillet between the inlet and the Major outlet,
- R_2 : Radius of the fillet between the Major outlet and the minor outlet,
- T : Thickness of the extrusion,
- q_m : Ratio between mass flow rates of minor outlet and inlet.



(a) (b)

Figure 3.2: (a) Geometry of the 10 µm stage (b) Half geometry referencing the parameters

Due to the Hypothesis 1 and Hypothesis 7, 2 constraints restrain the creation of the parameters bounds :

- Constraint 1 : Incompressible flow

$$\begin{aligned}\text{Incompressible air flow} &\iff Ma < 0.3\\ &\iff \frac{v_{air}}{c_{air}} < 0.3\\ &\iff \frac{Q}{WT} < 0.3 \times c_{air}\\ &\iff W > \frac{Q}{0.3 \times c_{air}T}\\ &\iff W_{min} = \frac{Q_{max}}{0.3 \times c_{air}T_{min}}\end{aligned} \tag{3.4}$$

where v_{air} represents the mean velocity of the air inside any section of the virtual impactor, c_{air} represents the speed of the sound in air, Q represents the flow rate in any section of the VI and W is the width of any section of the VI, Q_{max}, T_{min} and W_{min} represents, respectively, the maximum flow rate (which is Q_{inlet} by conservation of the flow rate) and the minimum thickness and width possible in every section of the VI.

- Constraint 2 : 3D-printable design The FORMLAB Forms 4 has a minimal resolution of 300 µm. Every geometric parameter must be superior to this value.

$$W_{min},\ T_{min}\ \&\ L_{min} \geq 300\ \mu\text{m} \tag{3.5}$$

where L_{min} represents the minimal length of any channel of the VI.

The determination of LHS parameters relies on the resolution of this system of equations:

$$\begin{cases} W_{min} = \dfrac{Q_{max}}{0.3 \times c_{air} T_{min}} & \text{Constraint 1} \\ W_{min},\ T_{min}\ \&\ L_{min} \geq 300\ \mu m & \text{Constraint 2} \end{cases} \tag{3.6}$$

The Table 3.2 presents the selected boundary values for LHS parametric fields.

Parameter	Min	Max	Unit	Justification
W_N	2.67	10.0	mm	Lower bound is the result of the Constraint 1, Upper bound is issued from previous µSPLIT analysis
W_M	2.67	18.0	mm	Lower bound is the result of the Constraint 1, Upper bound is issued from previous µSPLIT analysis. Additionally Hinds et al. (2022) explain that W_M/W_N should be comprised in $1 - 1.8$, the lower bound is associated to the value 1 and the upper bound with the value 1.8 to explore all possibilities.
W_m	2.67	10.0	mm	Lower bound is the result of the Constraint 1, Upper bound is issued from previous µSPLIT analysis
L_N	5	15	mm	Lower and upper bound are issued from previous µSPLIT analysis.
R_1	1.6	5.0	mm	Lower bound is issued from Hari et al. (2006) explaining R_1 must be superior to $0.16 \times W_N$, the recommendation is applied to the upper bound of W_N to ensure the 0.16 minimal factor to be always respected and upper bound is an arbitrary choice
R_2	1.6	5.0	mm	Lower bound is issued from Hari et al. (2006) explaining R_2 must be superior to $0.16 \times W_N$, the recommendation is applied to the upper bound of W_N to ensure the 0.16 minimal factor to be always respected and upper bound is an arbitrary choice
T	0.3	5.0	mm	Lower bound is the result of the Constraint 2 and upper bound is issued from previous µSPLIT analysis.
q_m	5	20	%	Lower and upper bounds are issued from Virgil A. Marple et al. (1980) and McFarland et al. (1978) recommendations.

Table 3.2: LHS parameters bounds selection

Configuration of the sample points (m)

The choice of the number of simulations m has to represent a good statistical exploration while considering the available computing budget. Loeppky et al. (2009) explains that the empirically based recommendation of $m = 10 \times n$ runs will provide reasonable and sufficient prediction accuracy. And that, unless a huge sample size is available, more than $m = 10 \times n$ runs would be wasteful.

To implement this, the pipeline utilizes `qmc.LatinHypercube`, from the SciPy library, which ensures that the m sample points are distributed with optimal uniformity across the n-dimensional hypercube, guaranteeing that no region remains unmapped. As the LHS explores $n = 8$

parameters, it might be a good approach to conduct ≈ 80 simulations for the optimization of each stage. However, due to a consideration that will be explained in the Mesh Generation section, a significantly higher number of simulations is performed to ensure the robustness of the parametric space exploration.

Configuration of the randomness (*seed*)

To ensure the reproducibility of the study, we will proceed as follows:

- **Seed 0**: Utilized for pipeline development.
- **Seed 1**: Utilized for the optimization of the 10 µm stage.
- **Seed 2**: Utilized for the validation of the optimal set of parameters.

Please note that the numerical value of the seed is just an index. The statistical distance and lack of correlation between Seed 0, 1, and 2 are equivalent to any other arbitrary set of integers.

Sampling Uniformity Study

To ensure the reliability of the parametric exploration, it is essential that the sampling distribution provides good coverage of the 8-dimensional space, through an uniform distance between every sample point.

A Sampling uniformity study has been conducted, it is detailed in Appendix A.3.1.

3.2.2 Geometry Generation

CAD softwares such as Fusion 360 or SALOME are not well suited for automated workflows with Python. The CadQuery library is employed to create a 2D sketch using different LHS geometric parameters for the design of straight lines and fillets, then extruded by T. To gather boundary conditions to the corresponding surfaces, different named selections are defined using a correlation between their normal direction of the surface and position. The named selections are referred to as boundaries in the following.

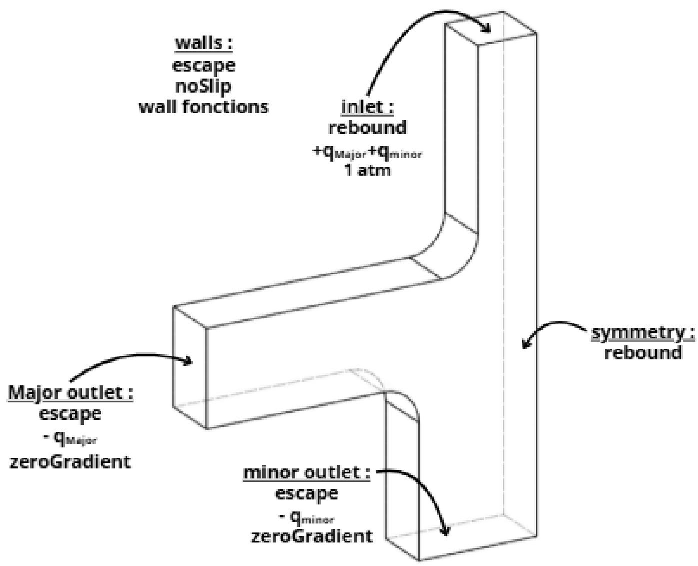


Figure 3.3: Named selection with their respective boundary conditions.

Those boundary conditions will be explained in further sections : *noSlip*/$\pm q_{major}$/$\pm q_{minor}$ & 1 *atm*/*zeroGradient* in Steady-state flow simulation section, *escape*/*rebound* in Transient particle injection and tracking section.

The geometry is then exported in .stl as a closed surface consisting in the gathering of boundary surfaces.

3.2.3 Mesh Generation

OpenFOAM offers two native mesh tools: *blockMesh* and *SnappyHexMesh*. *blockMesh* is the basic mesh tool of OpenFOAM, that works for simple geometries such as a cube, and it is not difficult to configure, on the other hand, *snappyHexMesh* is an extremely efficient meshing tool suitable for 99% of the geometries, but is time-consuming to configure, and more complex to automate with Python. *cfMesh* represents a good compromise between those two mesh tools: it is suitable for most geometries but it can be limited on some complex ones. More over, it is quite simple to integrate in a Python automation.

In the OpenFOAM framework, the dictionary *meshDict* (Appendix A.1.2) allows the configuration of *cfMesh*. The *meshDict* has been configured as an adaptive mesh to fit the m different geometries that are created with LHS parameters, so the maximum and minimum size of the cells are influenced by the parameters issued from the LHS. A refinement of the mesh near the *walls* region has been implemented to capture boundary layer effects.

Mesh Quality

Unfortunately, the fillets created with R_1 and R_2 bring the virtual impactor geometries considered into the complex geometries poorly managed by *cfMesh*. It creates negative cell volumes (reversed cells) that increase the maximal skewness up to 1.66907 on the configuration of $seed = 0$ while a skewness of < 0.85 is recommended. This problem results in a simulation completion rate of 36 % on the configuration of $seed = 0$. As Loeppky et al. (2009) recommends having a ratio of $m = 10 \times n$ simulations completed, the number of simulations required increases to a minimum of $m = \lceil 10 \times n/0.36 \rceil = 223$ for each stage, and to ensure reliability, 250 simulations will be conducted, which correspond to a safety margin of $\approx 10\%$. To achieve this large run, the UTokyo's High Performance Computer (HPC) will be used and linked to the Python automation.

The principle of HPC is detailed in Appendix A.2.

Mesh independence study

The mesh independence study ensures that the numerical results are not influenced by the discretization of the computational domain. To achieve this, multiple meshes with progressively refined cell sizes are generated, typically using a constant refinement ratio are submitted to simulations with identical boundary conditions and numerical settings for each mesh. The relative error of fluid velocities, measured in certain points of the geometry, between successive mesh resolutions is calculated, and the independence is assumed when the difference is under a predefined threshold (commonly 2%).

A mesh independence study has been conducted for this study, it is detailed in Appendix A.3.2. It is assumed that the mesh density ratio validated in the independence study remains sufficient to capture flow features across the range of geometries generated by the LHS.

3.2.4 Steady-State Flow Simulation

The particle tracking simulation needs to imitate a functioning VI. To do so, the air carrier flow has to be in a steady state (Hypothesis 1). As the airflow is modified by the geometry, it is necessary to have a complete explanation of how to run the carrier phase until convergence.

The *simpleFoam* solver is employed to model the steady-state incompressible airflow, imposing a mass flow rate at the *outlet_maj* and *outlet_min* boundaries Hypothesis 1. The JAEA requirements impose a cumulated flow rate at the inlet of the 10 µm stage at 10 $L \cdot min^{-1}$ for the entire geometry. Since we consider only half of the geometry (Hypothesis 8), this cumulated flow rate has to be 5 $L \cdot min^{-1}$ in the half inlet, which corresponds to a mass flow rate of 9.87×10^{-5} $kg \cdot s^{-1}$. This mass flow rate is split in two for *outlet_maj* and *outlet_min* boundaries, by a ratio of q_m defined by the LHS. Those mass flow rates are implemented with `MASSFLOWRATE_MIN` and `MASSFLOWRATE_MAJ` in the U dictionary contained in the **0** directory.

To ensure that the conservation of the flow rate is respected, the mass flow rate has been set to 0 $kg \cdot s^{-1}$ on the *inlet* boundary in the initial time step but it will update over time, and the *walls* boundary is considered with a condition of noSlip ($U = 0$ $m \cdot s^{-1}$ (Hypothesis 10)). Concerning the pressure, the *inlet* boundary is at atmospheric relative pressure (Hypothesis 9), and a Neumann boundary condition ($\partial\square/\partial x_i = 0$) is applied to the *walls*, *outlet_min* and *outlet_maj* boundaries ($\square$ represents any quantity). The dynamic viscosity is displayed on the whole internal field with Neumann condition on every boundaries, where the turbulent viscosity is set to 0 $m^2 \cdot s^{-1}$ on the *inlet* and *walls* boundaries, and with a Neumann condition on the *outlet_maj* and *outlet_min* boundaries with values issued from Hypothesis 1. The Table 3.3 is a summary of the initialized flow field quantities in the first time step corresponding to directory 0 in OpenFOAM with respect to the region of application.

Region	internalField	inlet	outlet_maj
`U`	$0\ m \cdot s^{-1}$	$0\ m \cdot s^{-1}$	MASSFLOWRATE_MAJ*
`p`	$0\ m^2 \cdot s^{-2}$	$0\ m^2 \cdot s^{-2}$*	zeroGradient*
`mu`	$1.85 \times 10^{-5}\ kg \cdot m^{-1} \cdot s^{-1}$	zeroGradient*	zeroGradient*
`nut`	$1 \times 10^{-5}\ m^2 \cdot s^{-1}$	$0\ m^2 \cdot s^{-1}$	zeroGradient*
`k`	$198\ m^2 \cdot s^{-2}$	$198\ m^2 \cdot s^{-2}$*	zeroGradient*
`omega`	$28000\ s^{-1}$	$28000\ s^{-1}$*	zeroGradient*

Region	outlet_min	walls	sym
`U`	MASSFLOWRATE_MIN*	$0\ m \cdot s^{-1}$*	symmetry
`p`	zeroGradient*	zeroGradient*	symmetry
`mu`	zeroGradient*	zeroGradient*	symmetry
`nut`	zeroGradient*	$0\ m^2 \cdot s^{-1}$*	symmetry
`k`	zeroGradient*	$198\ m^2 \cdot s^{-2}$*	symmetry
`omega`	zeroGradient*	$28000\ s^{-1}$*	symmetry

Table 3.3: Initialization of the flow field with respect to the region of application. **zeroGradient** corresponds to a Neumann condition, **MASSFLOWRATE_MAJ** and **MASSFLOWRATE_MIN** are considered as mass flow by OpenFOAM with a dedicated function, **symmetry** is acting as a symmetry surface, * indicates which are constant, p is a relative pressure.

The turbulence model chosen is *RANS* $k - \omega$ *SST* for considerations that will be explained in Transient particles injection and tracking section. Those parameters could not easily be automated and can be source of errors. The values of k and ω are issued from the JAEA's template.

Parallel computing

To significantly reduce computational time, the global domain of simulation is automatically partitioned via the *decomposeParDict* dictionary contained in **system** directory utility using the `scotch` method, which is simpler to automate. The solver is then deployed across 10 processors, and the fields are automatically merged back.

A detailed explanation of parallel computing is given in Appendix A.1.3.

3.2.5 Transient Particle Injection and Tracking

The particle transport in the simulation domain is governed by the *icoUncoupledKinematicParcelFoam* solver. It considers the kinematic of parcels (a parcel accounts for N homogeneous particles) under the influence of an incompressible fluid. The parcels have no effect on the behavior of the fluid and a *noCollision* model between parcels has been added: that creates a one-way coupling (Hypothesis 5). 10^5 parcels (containing each 10^4 particles) are injected for 0.1 seconds in the inlet nozzle with the initial velocity of the converged airflow. Every parcels are assumed to have escaped the VI within 1 second, other parcels are considered as trapped and the error management function handles the removal of trapped parcels at the end of the simulation.

The sample of particles simulated has to be representative of nuclear aerosols of Fukushima Daiichi, thus a log-normal distribution ($\sigma = 2.5$, $\mu = 6.4$) is considered for the size distribution (Zhuang et al., 2026).

This distribution is injected in the inlet nozzle of the 10 µm stage separation. The major flow outlet of the 10 µm stage harvests the distribution of the 1 µm stage separation.

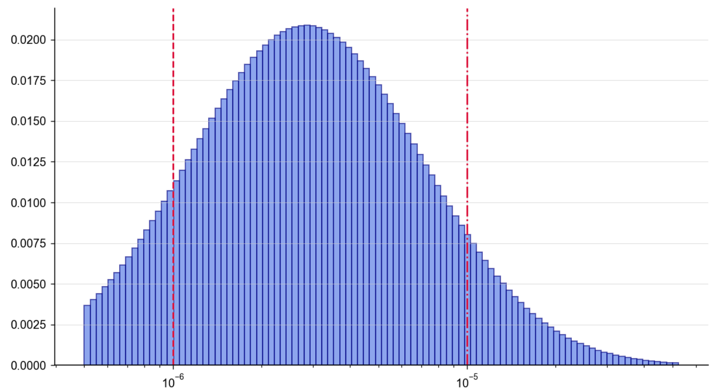


Figure 3.4: Lognormal distribution - $\sigma = 2.5$, $\mu = 6.4$

Particle rebound model

If a particle deviates from its streamlines, it might have an interaction with a boundary of the simulation. For the boundaries such as *inlet*, *symmetry*, the interaction is considered to be a full rebound (no loss of energy), to ensure these particles to stay in the simulation domain. For fluid outlets boundaries (*major_outlet*, *minor_outlet*), the particles have to escape from the domain, so they are virtually sticked to these boundaries. And for the *walls* boundaries, the rebound model of Thornton et al. (1998) has been considered.

According to Thornton et al. (1998), Figure 3.5 shows that, in simulations, fine particles tend to stick and coarse particles tend to have a elastic or near full rebound.
As the implementation of the Thornton et al. (1998) rebound function in OpenFOAM is beyond the scope of the study, a choice between adhesion and full rebound has to be made. Previous work of JAEA used a full rebound model on *walls*. However, in an ideal VI, *Stk* governs alone the particles trajectories. Coarse and fine particles are separated by inertia and follow streamlines,

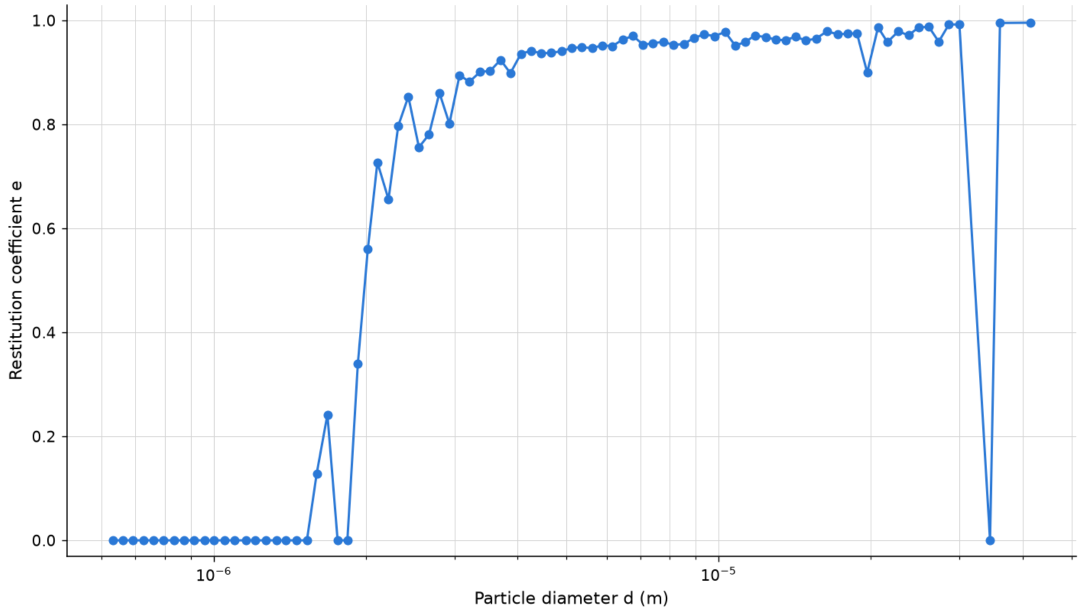


Figure 3.5: Coefficient of restitution of the kinetic energy of the particle in function of its diameter

with no impact on walls. Particles that hit walls show the imperfection of the VI's separation region. In this project, particles that deviate from their streamlines and impact walls will be identified using a sticking model, which allows better tracking than an unidentified rebound.

Numerically, OpenFOAM considers two types of adhesion models: stick and escape. The main difference resides in the removal or not of the particle after hitting the boundary: stick model conserves the particle in the simulation domain, and escape model does not. Since the particles are pairwise independent (one-way coupling), once a particle sticks to $major\_outlet/minor\_outlet/walls$ boundaries, it has no influence on the rest of the simulation. The escape model is then considered for these boundary regions. Moreover, the removal of the particles frees memory which improves the simulation.

Due to this adhesion model (escape) applied at the walls, the near-wall accuracy offered by the $RANS\ k-\omega\ SST$ model is no longer required for the particle-wall interaction. It could be tempting to use a $RANS\ k-\varepsilon$ model which is computationally cheaper. However, the $RANS\ k-\varepsilon$ model badly predicts flow separation and recirculation under adverse pressure gradients, which are present in the current flow and modify the particle trajectories. The computational gain offered by $RANS\ k-\varepsilon$ would not compensate for the loss of accuracy in the recirculation region, so the turbulence model considered is the $RANS\ k-\omega\ SST$ model.

Statistical Convergence Study

To ensure the reliability of the injection, it is essential that the number of parcels/particles does not influence the results. *A statistical convergence study has been conducted, it is detailed in the Appendix A.3.3.*

3.2.6 Pipeline Certification

To ensure that the numerical model can run simulations that are realistic, pipeline certification functions are created in the automation, which certify the functioning of every step: **LHS** (Constraints for geometry generation); **Geometry** (Closed surface and boundary regions well placed); **Mesh** (Definitions of metrics boundaries in checkMesh); **Fluid phase simulation** (Convergence and mass flow rate conservation); **Particles injection simulation** (Convergence and mass conservation).

4

RESULTS

4.1 Parametric Study Simulation

4.1.1 Performance Plot

The parametric study simulation are gathered in the Figure 4.1. It is a matrix of 8 columns, each one representing a LHS parameter, and 5 lines, each one representing a KPI. Each component of the matrix is a scatter plot, where x-axis describe the range of variation of the LHS parameter considered, y-axis describe the deviation of the KPI from ideal KPI and red curve is a LOESS (LOcally Estimated Scatterplot Smoothing) regression. Trend can be observed on which parameter has more or less influence on which KPI. Every scatter plot allows the selection of optimal LHS parameter value, but a blind modification can impact other KPIs in a bad way. For the modification of a parameter, a whole column must be considered, and a weighting of KPI as well.

µSPLIT objective is to separate the particles around $d_{50} = 10$ µm, and in the context of decommissioning operators' safety, coarse particles must to be eliminated from major flow channel and it have to be easily decontaminated as a trash to end its life cycle. The KPIs weighting proceeds a follow: d_{50} > Major Contamination > Wall losses > Minor Contamination > Sharpness. Contamination and wall losses are the root source of sharpness deviation, this is why it is placed in the end.

4.1.2 Performance Analysis

→ d_{50} : W_N variation has a major impact on this deviation, where every other parameter variation is less significant. Data points are vertically dispersed around the regression curve, only W_N data points corresponds to the regression curve.

→ σ_{VI} : No clear trend appears with any parameter. The regression curves remain flat across the sampled range, except for the upper bound of Q_m.

→ Major flow contamination : W_N and W_M have a big influence on the Major flow contamination, where other parameter are less significant. Data points are really dispersed, only W_N, W_M, W_m data points have a similar shape to the regression curve.

→ Minor flow contamination : W_N and Q_m seems to have an influence on the minor flow contamination where every other parameters don't. However, every parameters' data points have a weak vertical dispersion around the regression line, and W_N et Q_m ones are quite precise.

→ Wall losses : W_N, W_M, W_m, T have a great influence on the wall losses, L_N has a small influence, and other doesn't. Vertical dispersion around the regression curve seems to be weak, and the W_N one is quite precise.

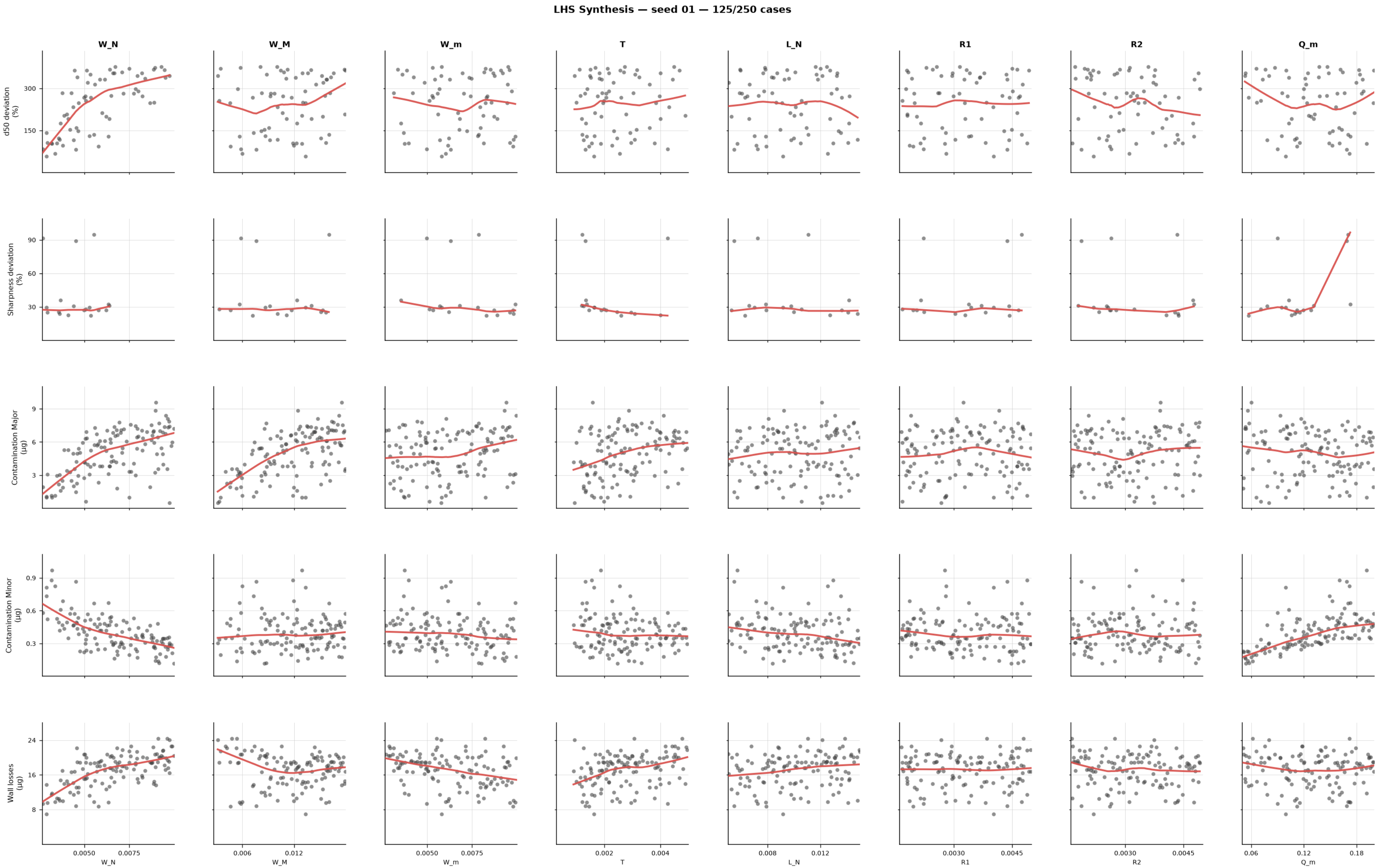


Figure 4.1: Performances of the valid simulations issued from parametric exploration of 250 simulations.

4.1.3 Interpretation

The dominant influence of W_N and W_M on d_{50}, Major flow contamination and Wall losses can be explained on behalf of the Stokes Number:

$$\frac{U_\mathrm{f}}{D_\mathrm{f}} = \frac{Q}{D_\mathrm{f} \times WT} = \frac{Q}{D_\mathrm{h} \times WT} = \frac{Q(W+T)}{4 \times (WT)^2} \ , \tag{4.1}$$

$$Stk = \frac{\rho_0 d_\mathrm{ae}^2 C_\mathrm{c}\left(d_\mathrm{ae}\right)}{18\mu_\mathrm{f}} \times \frac{U_\mathrm{f}}{D_\mathrm{f}} = \frac{\rho_0 d_\mathrm{ae}^2 C_\mathrm{c}\left(d_\mathrm{ae}\right)}{18\mu_\mathrm{f}} \times \frac{Q(W+T)}{4(WT)^2} \ , \tag{4.2}$$

where D_h is the hydraulic equivalent diameter.

With this formulation of Stk and the facts that flow rate conservation ensure the inlet flow rate constancy no matter the value of Q_m and as W_N mainly dominates T, the term $(W_N + T)/W_N T$ approaches $1/W_N$, it is understandable that for a polydisperse sample in the acceleration nozzle, the diminution of Stk results primarily on the increase of W_N.

This diminution of Stk results in less deviation of the fine and coarse particles driven by the major flow, which increase the quantities of major flow contamination, and wall losses in the major channel. Increasing W_M decreases the fluid velocity in the major flow channel, and particles that are already following the streamlines are decelerated, and will end up in the major flow boundary, not on the walls because they will follow the streamlines and almost never deviate from it, this is why the augmentation of W_M increases the Major flow contamination and decreases the Wall losses.

More over, Figure 4.2 shows that some streamlines going in the minor flow channel can end their course in the major flow channel so the fine particle can't contaminate the minor flow channel unless they are near the symmetry plane. This bifurcation might be dependent on the ratio Q_m, because if Q_m is increased, the quantity of streamlines having a bifurcation in the minor channel decreases, and those streamlines ends in the minor flow boundary, and this increases minor contamination of fine particles that could not escape their streamlines.

In contrary, R_1 and R_2 show a poor influence on the contamination and Wall losses, it could be thought that the trajectory of the particles is already decided by the acceleration nozzle Stokes number and that tiny modification near the separation region have no effect on the particles trajectory.

The dispersion of the data points is more important around the regression of d_{50} deviation and Major contamination than around Minor flow and wall losses regressions. The choice of $d_{50} = 10$ µm and the lognormal distribution are mainly responsible for this : particles > 10 µm, which are mainly responsible of the major contamination and d_{50} by definition, represent a small minority (12 %) of the injected particle sample compared to to the particles < 10 µm (88 %). With fewer particles representing d_{50} and major contamination KPIs, each data points among the 125 is less precise.

The sharpness is not a good indicator in the frame of the study. It is defined as $\sigma_\mathrm{VI} = \sqrt{d_{84}/d_{16}}$ but on most collection efficiency diagram that can be plotted during this campaign, d_{84} or d_{16} does not exist, those values are non existent due to the consideration of wall losses. The literature does mainly not consider wall losses, there is no reliable expression of σ_VI to express the sharpness.

4.1.4 Simulation Analysis

The campaign initially needed 250 simulations to ensure 80 valid cases that are necessary to confirm the good parametric study of 8 parameters. 125 simulations are valid, which leads in a success rate of 50 %. However, it is also important to know why other simulations have been dismissed. The Figure 4.3 show the status of the simulation correlated with the range of each parameter. Three status exists : valid (complete simulation), Airflow divergence (for *simpleFoam* divergence or non convergence on the airflow simulation) and Particles injection crash (for particles that were trapped early in the simulation and that caused further problems

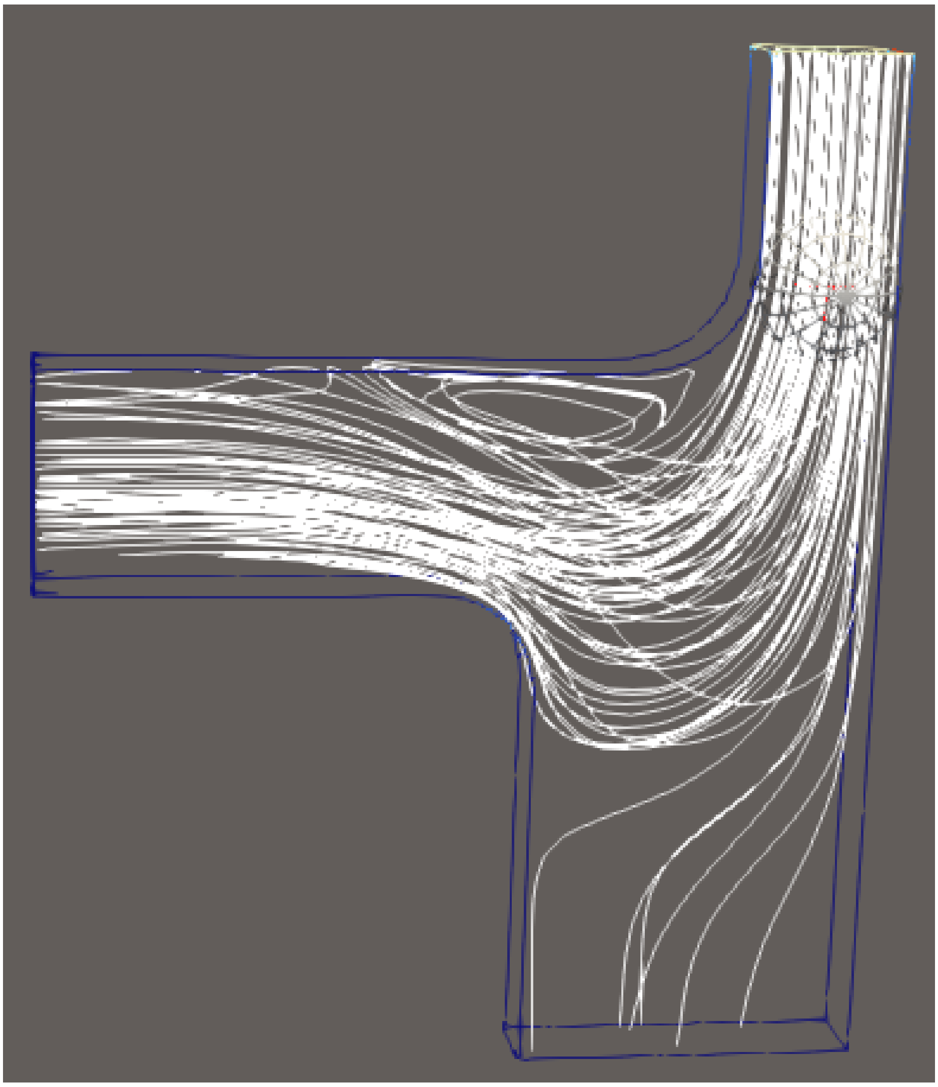


Figure 4.2: Bifurcation of the streamlines entering the minor flow channel

that stopped the simulation). The goal of this section is to observe if certain ranges of parameters have an influence on the good functioning of the simulation.

In every graphic, a simulation is represented by a single line. As LHS distributes the n parameters across the m simulations evenly, the concentrations of lines represent simply a similar behavior. The main convergence failure factor seems to be the thickness T of the channel near the lower bound, yet the upper bound values creates particles injection crash. It might be caused by a poor mesh near to the walls and by the turbulence model that could not be automated. The closer the value of W_N, W_N, L_N and Q_m is from the upper bound, the better the simulation validity is. The fluid seems to establish easier when it has space to do so, and that in the three dimensions. However, another main failure factor concerns the curvature of the particles around R_1: if W_N or W_M are close to the lower bound, and that R_1 has a intermediate value, the particle injection simulations crashes. Which corresponds to the identified problem of trapped particles in the negative volumes created by *cfMesh*. And when R_1 is close the upper or lower bound, the number of valid simulation is big. Regarding R_2, values in its lower bound creates weak proportion of failure. And if Q_m is close to its lower bound, there are lots of particles crash, that night correspond to the bifurcation of streamlines represented in Figure 4.2.

The major proportion of failure is due to particles injection simulation crashes rather than the research of convergence of the carrier airflow.

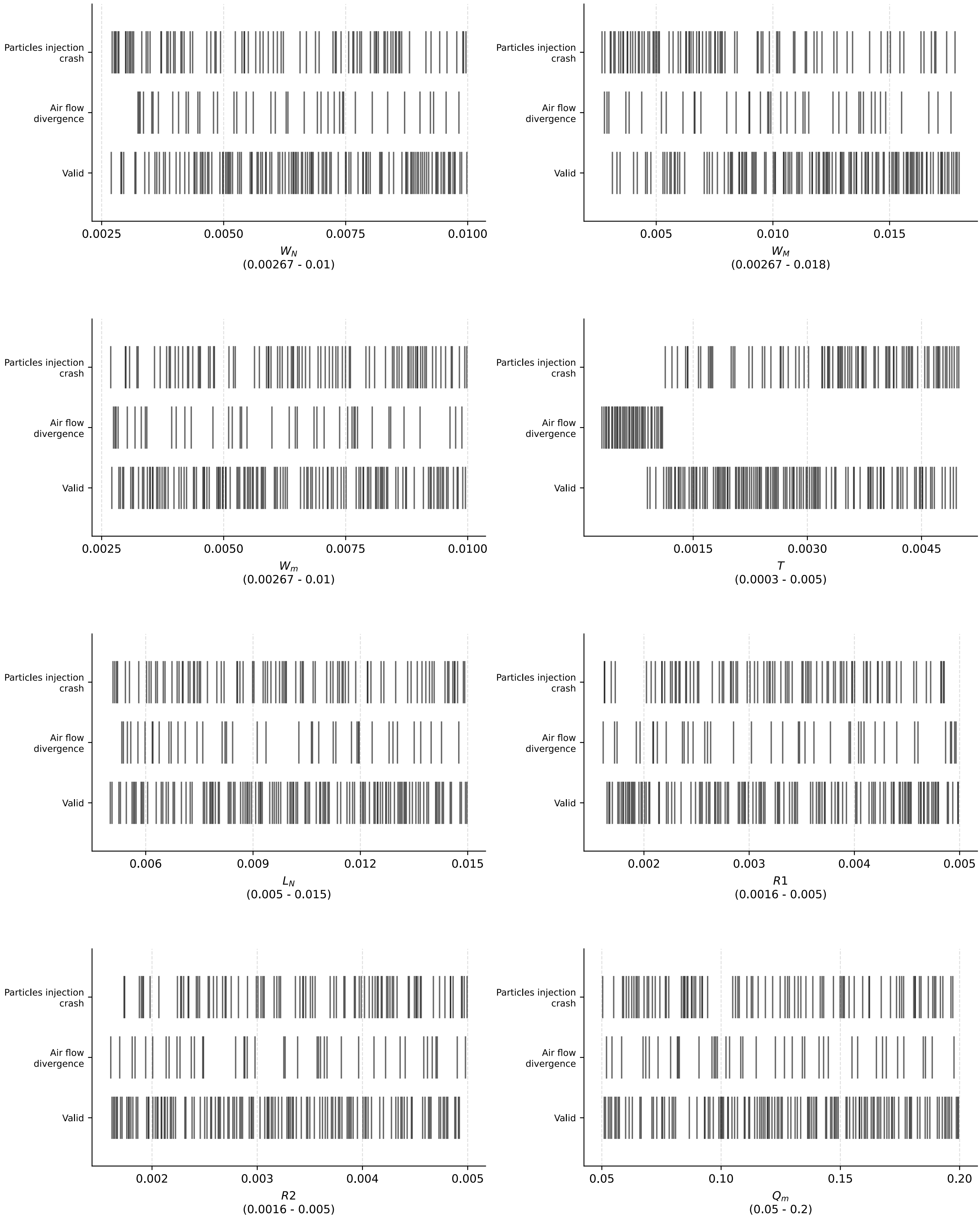


Figure 4.3: Performance analysis of the 250 simulations

4.2 Optimal Virtual Impactor Simulation

4.2.1 Parameters Choices

The parametric exploration allow the observation of trends, but it is interesting to observe how to create an optimal model of VI. The chosen method was to select the parameters value corresponding to minimal deviation according to the weighting of KPIs. However this method might not be optimal, Bayesian optimization for example could probably provide better results. The Table 4.1 details the selection of parameters values based on the analysis of the Figure 4.1 and the Figure 4.4 is the shape of this optimal configuration.

Parameter	Selected value	Unit	Justification
W_N	2.67	mm	Minimal value of the range due to minima of d_{50}, Major contamination and Wall loss regressions.
W_M	8	mm	d_{50} show two minima, as Major contamination regression has positive slope, the left one is selected.
W_m	6.8	mm	Middle value of the range due to minima of d_{50} and Major contamination regressions.
T	1.3	mm	Low converged value of the range due to minima of d_{50}, Major contamination and Wall loss regressions and with Figure 4.3.
L_N	10	mm	Middle value of the range. d_{50} show a minimal deviation at the maximum value of the range but Major contamination and Wall loss shows both minimal deviation at the minimum value of the range.
R_1	1.6	mm	Lower value, according to Hari et al. (2006) in 2.2.4 Optimization Strategies.
R_2	3	mm	Intermediate value corresponding to most KPIs minimal deviation.
Q_m	10	%	d_{50} and Major contamination show two local minima, Wall losses seem to have a quadratic curve centered between those minima (which makes it impossible to make a choice), and Minor contamination has a positive slope. The left value is selected.

Table 4.1: Optimal VI parameters selection

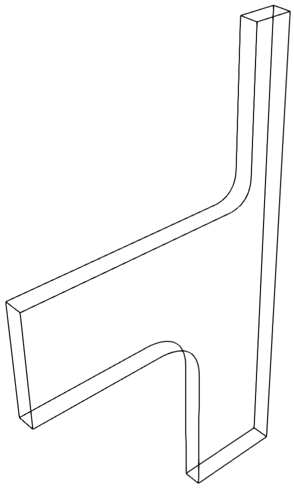


Figure 4.4: 3D Geometry of the optimal configuration

4.2.2 Performance Analysis

Collection efficiency diagram

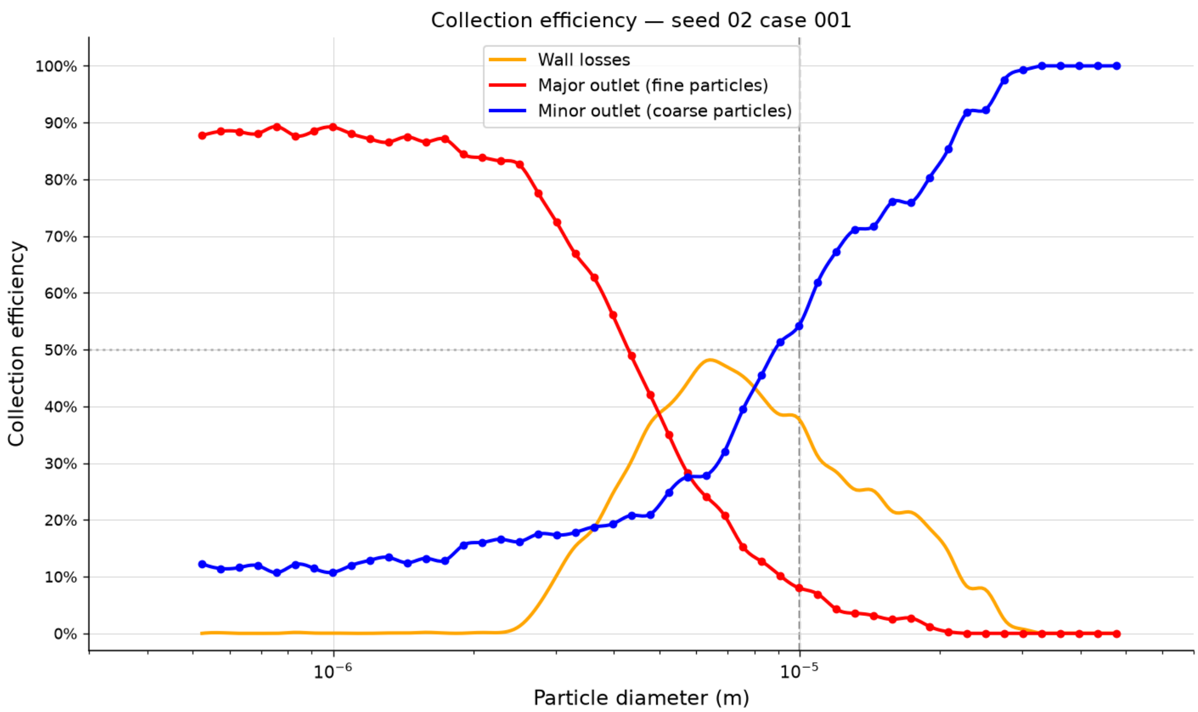


Figure 4.5: Collection efficiency diagram of the optimized set of parameters. The dotted line represents the aimed d_{50}.

The collection efficiency curves obtained with the optimized parameter set is similar to the expected shape of a VI collection efficiency curve. Three collection classes can be defined:

- $d < 2.3$ µm: Fine particles are almost entirely collected in the major flow outlet, a little fraction of $10 - 15$ % is collected by the minor flow outlet, which is similar to Chen et al. (1987) performances and significantly better than Masuda et al. (1988) ones.
- $d \in [2.3 - 10]$ µm: Intermediate particles are mostly lost on walls. Mostly due to the wall adhesion model.

- $d > 10$ µm: Coarse particles are mostly not collected by the major flow outlet, coarser particles are entirely collected by the minor flow outlet but coarse particles < 30 µm are collected by the minor flow outlet and deposited on walls.

In literature, d_{50} is mostly represented as the diameter for which major outlet and minor outlet curves intersect. The definition of d_{50} being the diameter of the particle that has 50 % chances to be collected in the minor outlet is still correct, because wall loss is mostly not considered. In this configuration, particles that have 50 % chances to be collected in the minor outlet, have also 50 % chances to be collected in the major outlet. The two definitions are superposed.

In the configuration proposed here, wall losses are considered, and the diameter corresponding to the intersection between major and minor outlet curves is different than d_{50}. More over, we can observe that the diameter of intersection is slightly similar to the diameter showing maximal wall losses.

Mass collection diagram

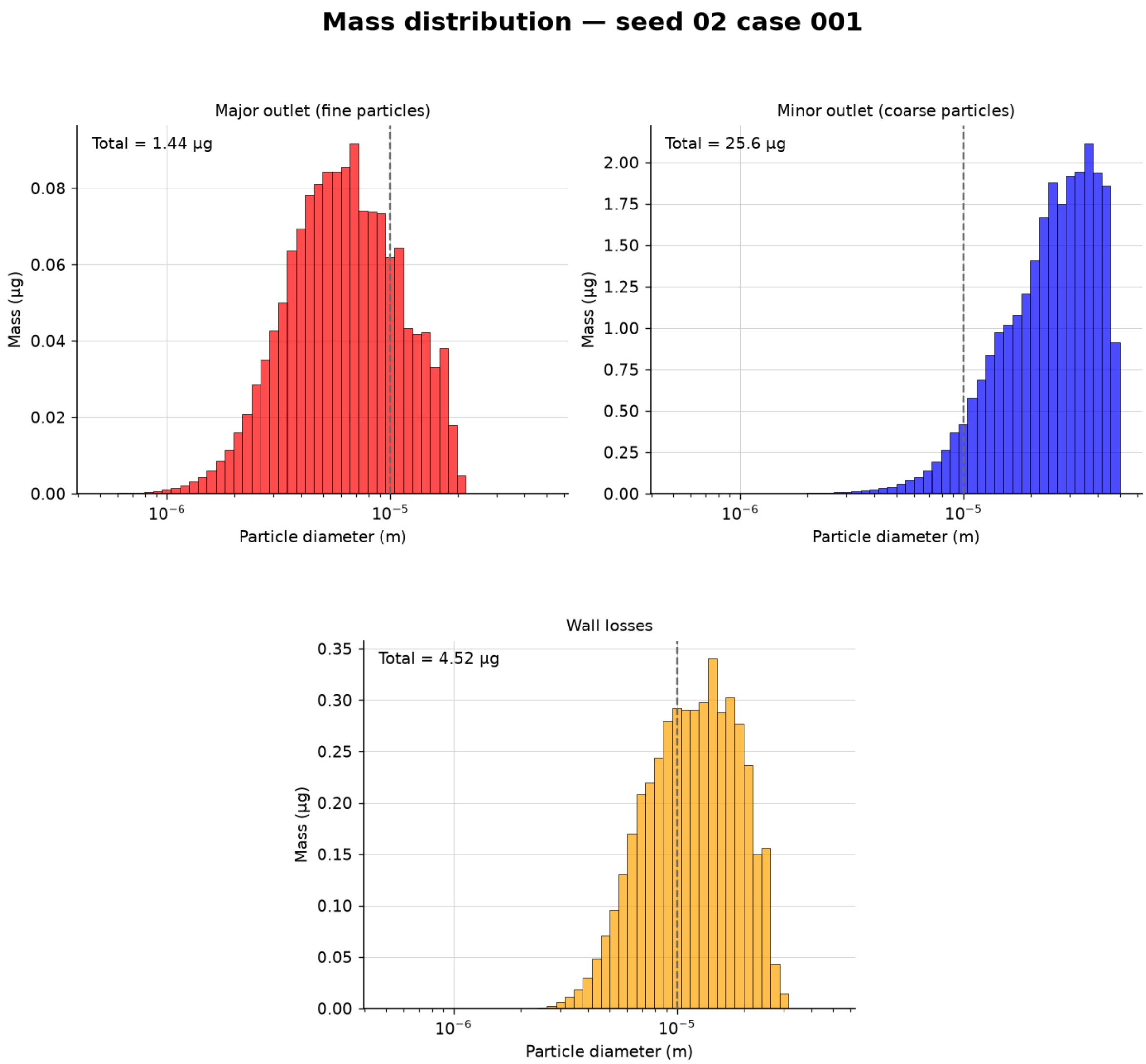


Figure 4.6: Mass distribution diagram of the optimized set of parameters for each outlet. The dotted line represents the aimed d_{50}.

As the minor outlet collects most of coarse particles, collects the majority of the injected mass (25.6 µg), explicitily showing why a small number of coarse particles represent a large proportion of the total mass injected. The major outlet collects 1.44 µg and 4.52 µg of particles are deposited on the walls.

Individual KPIs diagram

The Figure 4.7 shows the deviation of the KPIs of the optimal configuration:

- d_{50} is relatively close to the objective of 10 µm with a 11.4 % deviation and close to Laffolley, Tsubota, et al. (2026) who found 9 µm.
- σ_{VI} is defined, contrarily to the Figure 4.1, Figure 4.4 shows values for d_{16} and d_{84}, however the sharpness is very bad compared to Laffolley, Tsubota, et al. (2026) who found 1.7.
- Contaminations are quite low in terms of mass but they represent $10-15\%$ for the major flow contamination and $0-5\%$ for the minor flow contamination, Laffolley, Tsubota, et al. (2026) found approximatively 20% for the major flow contamination and $0-5\%$ for the minor flow contamination.
- Wall losses were not represented in Laffolley, Tsubota, et al. (2026), that might explain some differences with this study. There is no mention of quantification of wall loss in the literature either. The interpretation of wall loss is detailed in the next paragraph.

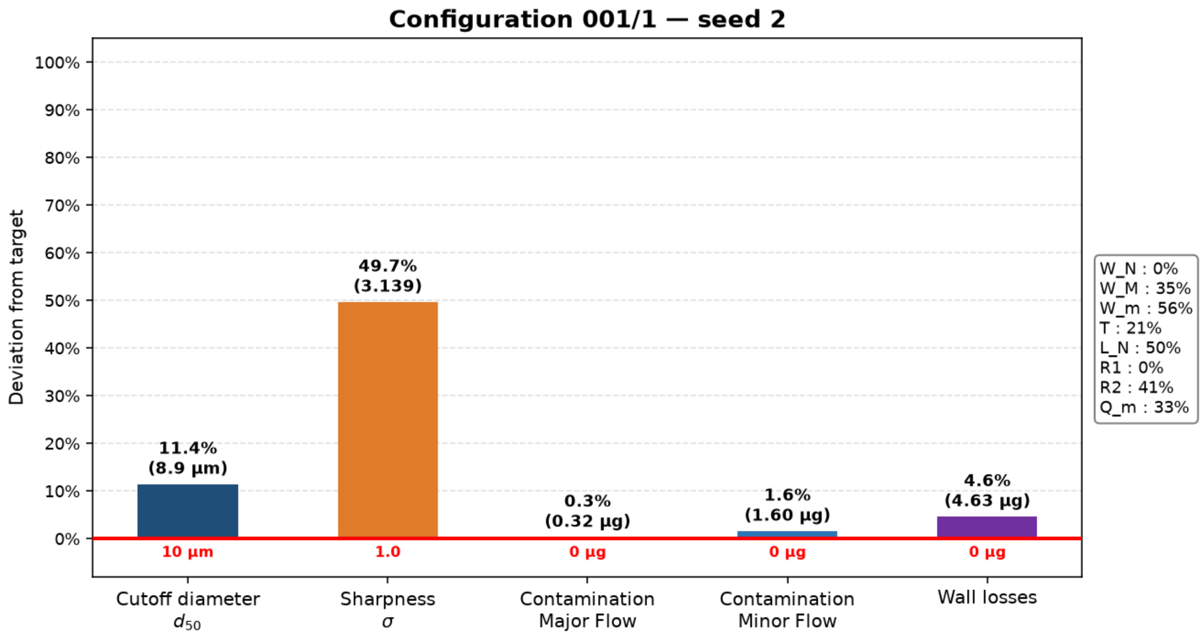


Figure 4.7: Deviation of the optimized set of parameters' KPI to ideal KPI. Parameters values (among the LHS interval) are represented on the right. The ideal KPI are represented by the horizontal red line and the number below.

Radiological Hazard of Wall-Deposited Particles

The core of the µSPLIT project resides in the idea of a low cost VI interchangeable to limit operators exposition to potential ingestion/inhalation, that this separator would never be manipulated by a human. Thus, it is essential to justify that particles deposited on the walls represent a hazard for operators, despite the separation efficiency. For this, the simulation is considered as one cycle of utilization and particles deposition on the walls. The effective dose D_{eff} resulting from the intake of an isotope's activity is (ICRP, 1995):

$$D_{\mathrm{eff}} = A \times e(50) = a_{\mathrm{s}} \times m_{\mathrm{i}} \times e(50) \ , \tag{4.3}$$

where A is the activity of the isotope, $e(50)$ is the dose coefficient for an adult with fast absorption ($e(50) = 4.6 \times 10^{-9}$ Sv $\cdot$ Bq^{-1} for ^{137}Cs), a_{s} is the specific activity of the isotope ($a_{\mathrm{s}} = 3.2 \times 10^{12}$ Bq $\cdot$ g^{-1}) and m_{i} the mass of the isotope (mass collected on the VI walls) (ICRP, 1995).

If the aerosol considered had about 1% of total particles that are ^{137}Cs with equal mass fractions, the collected mass of ^{137}Cs on the VI walls would be 0.0452 µg (Figure 4.5) and if the operator were to inhale every particle deposited on the walls, the effective dose would be $D_{\mathrm{eff}} \approx 0.665$ mSv. For comparison, the effective dose that a civilian can receive is 1 mSv/year and the maximal effective dose that an operator can receive is 20 mSv/year (ICRP, 2007). The hypothesis of ^{137}Cs proportion is only used as an example and might not be reliable, but it shows that the replacement and cleaning of a VI by a human operator is difficult to consider.

5

CONCLUSION

5.1 Summary of Findings

Fukushima Daiichi Nuclear Power Plant dismantling operations represent a severe hazard for on-site operators, by the creation (or resuspension) of radioactive aerosols. Conventional metallic virtual impacters are expensive and require decontamination procedures that expose operators to unnecessary radiation doses. The automated CFD pipeline developed in this work allowed for the optimization of the µSPLIT developed by CLADS (JAEA) through a parametric exploration of the 10 µm separation stage.

This parametric exploration led to an optimal configuration of parameters for the creation of an optimal virtual impactor. This optimal virtual impactor made it possible to obtain a cutoff diameter of 8.9 µm (while µSPLIT aims for 10 µm), a small quantity of contamination in each outlet. Radioactive material (^{137}Cs) can also be deposited on the walls of the virtual impactor, and the radiological risk assessment conducted showed that a single operating cycle could deliver an effective inhalation dose of up to ≈ 0.665 mSv to an operator handling the device. This result justifies the need to eliminate a manual decontamination process, and to create low-cost 3D printed virtual impactors.

This work showed that the fate of a particle (whether it will follow the streamline or deviate from it) is largely decided within the acceleration nozzle itself, well before the flow reaches the major/minor bifurcation. Mostly width, thickness and flow rate influence the good separation of a virtual impactor. The implementation of fillets does not appear as particularly effective, and might be responsible for the poor robustness of the Latin Hypercube Sampling parametric exploration. Other geometrical parameters might also decrease the quality of the simulations, but thickness lower bound, for example, matched the minimal resolution of the 3D printer. This parametric exploration had for objective to find ranges of parameters that improves the model, but another implicit objective was to find ranges of parameters that cannot be exploited for the improvement of the model as well.

5.2 Challenges and Limitations

The project management in research environments was not natural for me. The scope of this internship constantly evolved, while being focused on the development of the µSPLIT. My work ranged from the simple creation of geometries with Fusion 360 to conduct an important parametric exploration (250+ configurations) on a HPC computer, passing through CAD/CFD workflow automation, LHS and OpenFOAM environment learning.

Additionally to this evolution in the project, I started the internship in France, waiting for my certificate of eligibility (CoE) for 1.5 months. And 1 month after my arrival in Japan, my direct supervisor that introduced me to the µSPLIT project, to OpenFOAM and that defined the initial micro-directions, resigned from JAEA and remained only occasionally reachable for online guidance. My academic supervisor at UTokyo was not directly familiar with the technical specifics of the project. This situation shortened the time available with a supervisor directly integrated and competent on the project and limited the technical questions that could be resolved through supervision.

Thus, I was required to, mostly by myself, learn how to use and integrate new tools (CadQuery, cfMesh, OpenFOAM, Latin Hypercube Sampling, HPC infrastructure) to the Python automate, and to make methodological and organizational decisions on my own, sometimes revising the initial directions. I had to prioritize components of the pipeline that had to be more specifically developed, while acquiring the deep scientific knowledge necessary to the project (virtual impactors, aerosol physics, and purpose of µSPLIT project). This constrained the depth of the analysis dedicated to the results and left some aspects of the study less rigorously explored than initially intended.

5.3 Future Work

The work done on µSPLIT reposes on simplifications that deserve further exploration, notably :

→ Compressibility of the flow,

→ Aerodynamic diameter simplification,

→ Mesh quality,

→ Turbulence model selection,

→ Automated particle rebound and deposition/adhesion model,

→ Interparticle interactions (electromagnetism, collisions, adhesions, ...),

→ 3D printing quality and correspondence with the numerical model.

5.4 Concluding Remarks

The µSPLIT project represents a great alternative to industrial virtual impactors. This work allowed the creation of a reusable parametric exploration workflow to observe parameter trends , and create optimal configurations. The Fukushima Daiichi decommissioning process will benefit from this work, and the entire domain of aerosol science education will improve with the realization of µSPLIT by CLADS (JAEA).

Besides the technical part, this internship was a great personal experience. I lived alone in Tōkai, a small town in Ibaraki prefecture far away from big japanese metropolis, with a straight nuclear research environment. I had the opportunity to visit the Fukushima Daiichi site and the surrounding areas affected by the tsunami of March 2011. I also travelled to several regions of Japan during this period, and I particularly enjoyed the Miyajima island.

APPENDICES

A

NUMERICAL TOOLS

A.1 OpenFOAM

A.1.1 Introduction

OpenFOAM (Open Field Operation And Manipulation) is an open-source C++ toolbox dedicated to Computational Fluid Dynamics (CFD). It allows the customization of numerical solvers (some solvers are shown in Figure A.1) and pre/post-processing utilities. It is mainly used in academic institutions and research organizations.

OpenFOAM does not rely on a graphic interface, but it is structured with a specific architecture of directories and dictionaries that governs the physics and numerical resolution of the simulation, main p. This structure, applied to the virtual impactor optimization, is detailed in Figure A.2.

Solver	transient	compressible	turbulence	heat-transfer	buoyancy	combustion	multiphase	particles	dynamic mesh	multi-region	fvOptions
buoyantSimpleFoam		✔	✔	✔	✔	✔					✔
chemFoam	✔			✔		✔					
fireFoam	✔	✔	✔	✔	✔	✔				✔	✔
icoFoam	✔										
interFoam	✔		✔				✔		✔		✔
laplacianFoam	✔										✔
pimpleFoam	✔		✔						✔		✔
pisoFoam	✔		✔								✔
reactingFoam	✔	✔	✔	✔		✔					✔
reactingParcelFoam	✔	✔	✔	✔	✔	✔		✔			✔
rhoCentralFoam	✔	✔	✔	✔							
rhoPimpleFoam	✔	✔	✔	✔					✔		✔
rhoSimpleFoam		✔	✔	✔							✔
simpleFoam			✔								✔
sprayFoam	✔	✔	✔	✔	✔	✔		✔			✔

Figure A.1: Some OpenFOAM solvers and the resolved physics corresponding (The OpenFOAM Foundation, 2024).

A.1.2 General Structure of an OpenFOAM Simulation

- *LHS_parameters.json*
- **Geometry**
  - *geometry.stl*
- **Fluid phase**
  - **0**
    - *k*
    - *mu*
    - *nut*
    - *omega*
    - *p*
    - *U*
  - **constant**
    - **polyMesh**
    - *transportProperties*
    - *turbulenceProperties*
  - **system**
    - *controlDict*
    - *decomposeParDict*
    - *fvSchemes*
    - *fvSolution*
    - *meshDict*
  - *mesh.log*
  - *checkMesh.log*
  - *fluid_simulation.log*
- **Particle phase**
  - **0**
    - *k*
    - *nut*
    - *omega*
    - *p*
    - *phi*
    - *U*
  - **constant**
    - **polyMesh**
    - *g*
    - *kinematicCloudProperties*
    - *particleTrackProperties*
    - *transportProperties*
    - *turbulenceProperties*
  - **system**
    - *controlDict*
    - *fvSchemes*
    - *fvSolution*
  - *particle_simulation.log*

Figure A.2: Representation of an OpenFOAM simulation case directory.

Description of the Case Directories and Dictionaries

At the root of the OpenFOAM simulation case directory, reside a file *LHS_parameters.json* and three directories **Geometry**, **Fluid phase** & **Particle phase**. This file represents the configuration created by the LHS with a particular 3-tuple $(m, n, seed)$. The three directories are essential for organization more than for simulation, it allows the subdivision of the simulation : **Geometry** directory contains the geometry created from the automaton or imported, **Fluid phase** directory contains mesh tool and fluid simulations parameters and **Particle phase** directory contains particles injection simulation parameters and particle tracking parameters. Each one of theses organizational directories contains files *.log* that are simulation reports created by the solver to record the solution history (residuals, CPU time, ...) and three other directories : **0**, **constant** & **system**. Those last three directories can be found in any OpenFOAM simulation.

0

0 directory contains initial and boundary conditions. In this project **0** directory contains multiple dictionaries : *k*, *mu*, *nut*, *omega*, *p*, *U* & *phi*.

- *k*: Specifies the turbulent kinetic energy $(m^2.s^{-2})$.
- *mu*: Specifies the dynamic viscosity of air $(kg.m^{-1}.s^{-1})$.

- *nut*: Represents the turbulent kinematic viscosity ($m^2.s^{-1}$).
- *omega*: Specifies the specific dissipation rate (s^{-1}). It determines the rate of dissipation of turbulent kinetic energy into thermal energy.
- *p*: Specifies the kinematic pressure field ($m^2.s^{-2}$), defined as the static pressure P divided by the fluid density ρ for incompressible flows.
- U: Specifies the fluid velocity vector field ($m.s^{-1}$).
- *phi*: Represents the volumetric flux field ($m^3.s^{-1}$) across cell faces, used by the lagrangian solver to accurately track particle trajectories through the mesh.

constant
constant directory contains the mesh created, and physical quantities that remain constant along the simulation. In this project **constant** contains a **polyMesh** directory that contains the final topology of the mesh (points, faces, cells, and patches) created with *meshDict* dictionary in the **Fluid phase** directory, and multiples dictionaries : *transportProperties*, *turbulenceProperties*, *g*, *kinematicCloudProperties* & *particleTrackProperties*.

- *transportProperties*: Configures the fluid Newtonian rheology model.
- *turbulenceProperties*: Configures the turbulence modeling approach.
- *g*: Specifies the gravitational acceleration vector field of particle phase ($m.s^{-2}$).
- *kinematicCloudProperties*: Configures the transient particles injection and tracking parameters.
- *particleTrackProperties*: Configures the lagrangian particle tracking utility and optimize output file size before generating trajectory files for visualization with ParaView.

system
system directory is the simulation's dashboard. In this project it contains multiple dictionaries : *controlDict*, *decomposeParDict*, *fvSchemes*, *fvSolution* & *meshDict*.

- *controlDict*: Controls the steady-state execution and the transient particle tracking execution.
- *decomposeParDict*: Configures the domain decomposition parameters for parallel computing.
- *fvSchemes*: Configures the numerical discretization schemes (spatial and temporal) for the finite volume formulation.
- *fvSolution*: Configures the linear matrix solvers, tolerances, and convergence algorithms.
- *meshDict*: Configures the *cfMesh* mesh generator.

Code Availability: The complete OpenFOAM case directories, including all input dictionaries, mesh configuration scripts, and post-processing tools, are fully documented and available on: `https://github.com/Pierre-Roger-ns/pierre-roger/tree/main/internship-utokyo-virtual_impactors-2026`

A.1.3 Parallel Computing

This section describes how to run OpenFOAM in parallel on distributed processors. The method of parallel computing used by OpenFOAM is known as domain decomposition, in which the geometry and associated fields are broken into pieces and allocated to separate processors for solution. The process of parallel computation involves: decomposition of mesh and fields; running the application in parallel; and, post-processing the decomposed case as described in the following sections. The parallel running uses the public domain `openMPI` implementation of the standard message passing interface (MPI)(OpenCFD, 2026).

The mesh and fields are decomposed using the `decomposePar` utility: the geometry and fields are broken up according to a set of parameters specified in a dictionary named *decomposeParDict* that must be located in the **system** directory of the case of interest.

The user has a choice of four methods of decomposition:

- `simple`: Simple geometric decomposition in which the domain is split into pieces by direction.
- `hierarchical`: Hierarchical geometric decomposition which is the same as simple except the user specifies the order in which the directional split is done.
- `scotch`: Scotch decomposition requires no geometric input from the user and attempts to minimise the number of processor boundaries.
- `manual`: Manual decomposition, where the user directly specifies the allocation of each cell to a particular processor.

The run is then made using `mpirun` utility, in which the set of parameters used in *DecomposeParDict* and *ControlDict* dictionaries is specified, after each timestep of calculation `mpirun` aggregates the information in each subdomain to a common domain to process next timestep calculation. Finally, the mesh and fields are merged back together with the utility `reconstructPar`. Then the case is disposable as an usual case (OpenCFD, 2026).

However, it is important to care about the process of parallel computing, because it would be easy to think that simulations must be divided in the maximum processors that the calcul station disposes to increase the calculation speed. If the number of subdomains is superior or equal to the numbers of available cores, the computer will shutdown because the calcul will deallocate cores dedicated to the OS. Usually, the number of cores used in a simulation depends of the number of cells : a core can optimally process about 10 000 cells. If a core is allocated for the calculation of too little number of cells, it will be counterproductive, as the writing in each time step will take a lot of time.

A.2 High-Perfomance Computing (HPC)

High-Performance Computing enables complex simulations by leveraging clusters of interconnected nodes to perform large-scale parallel processing. This architecture works by dividing computational tasks and data across multiple processors, significantly reducing execution time compared to traditional machines. The integration of high-speed networks and specialized accelerators, such as GPUs, further optimizes these systems for intensive tasks like CFD, allowing for high-fidelity modeling that would be infeasible on a single computer (NVIDIA, 2026).

A.3 Numerical Model Validation

A.3.1 Sampling Uniformity Study

For the sampling uniformity study, the LHS distribution was compared to a uniform random distribution within the same characteristics (250 points) in the function `LHS_uniformity.py`. The distances between the points of 8 coordinates $\in [0,1]$ (corresponding to normalized LHS parameters) were evaluated as :

$$\forall i,j \quad distance(i,j) = \sqrt{\begin{array}{l}(W_{N\ i} - W_{N\ j})^2 + (W_{M\ i} - W_{M\ j})^2 + (W_{m\ i} - W_{m\ j})^2 + (T_i - T_j)^2 \\ +(L_{N\ i} - L_{N\ j})^2 + (R_{1\ i} - R_{1\ j})^2 + (R_{2\ i} - R_{2\ j})^2 + (q_{m\ i} - q_{m\ j})^2\end{array}}$$

where i,j represents two points.

The results obtained are :

```
===========================
RANDOM UNIFORM
===========================
Mean Distance : 1.1430
Standard Deviation : 0.2503

===========================
LHS
===========================
Mean Distance : 1.1311
Standard Deviation : 0.2438

========================================================================
COMPARISON
========================================================================

Quantity              Random Uniform    LHS           Relative error (%)
------------------------------------------------------------------------
Mean Distance         1.1430            1.1311        1.0417
Standard Deviation    0.2503            0.2438        2.5968

========================================================================
```

The LHS distribution have distances statistics nearly identical to the uniform random reference. LHS shows effective and uniform coverage of the 8-dimensional parameter space.

A.3.2 Mesh Independence Study

The Mesh Independence study is essential to ensure the independence between the mesh refinement and the result obtained. This independance study has been conducted looking at the relative pressure loss. This choice is motivated because it characterizes the overall behaviour of the flow rather than a single localised value. The pressure loss encapsulates the cumulative dissipation occurring throughout the geometry (viscous friction along the walls, losses through the bends, and turbulent dissipation in the recirculation zone) that are affected by the spatial resolution of the mesh. Moreover, point measurements' values may differ according to the position of flow structures from one mesh to another, where pressure loss is unaffected by any sampling location and converges in a stable, reliable manner.

Under 2% of relative error in the pressure loss measurement between mesh refinement levels, the variation of pressure is considered small compared to other sources of uncertainty such as the turbulence model, the boundary conditions and the physical assumptions. A further refinement of the mesh would increase too much the computational cost for a campaign of 250 simulations, without improving considerably the reliability fo the results.

As an adaptive mesh is used to cover LHS parameters, the mesh independence study has been made with multiplicative refinement factor applied to the mesh caracteristics.

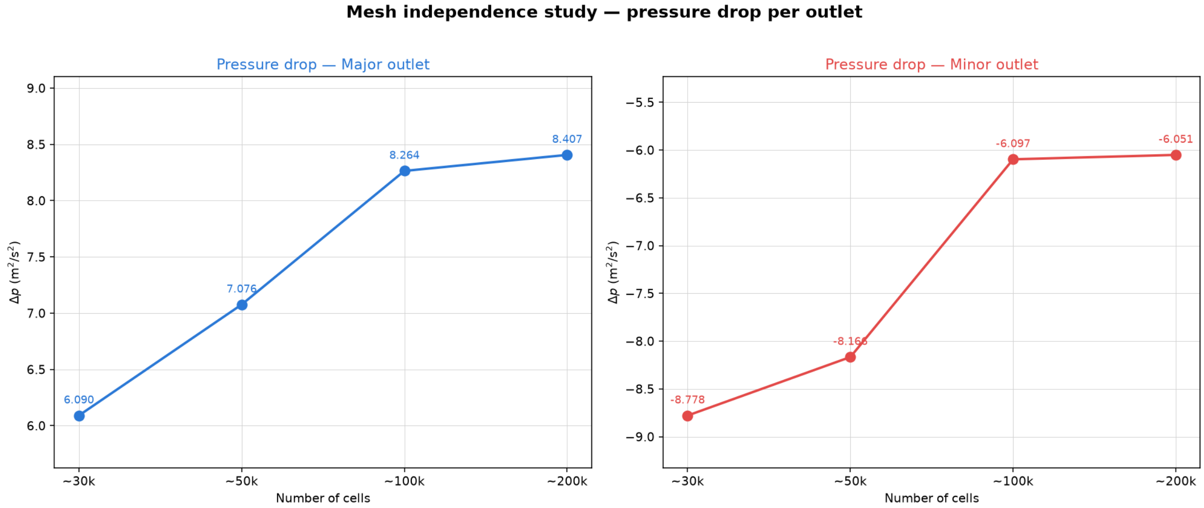


Figure A.3: Caption

Refinement factor	Number of cells	Δp Major outlet	Relative error from finest
$\times$1.0	30764	6.090316	27.5%
$\times$0.7	51816	7.076219	15.8%
$\times$0.5	100929	8.264179	1.7%
$\times$0.35	196531	8.406526	0%

Refinement factor	Number of cells	Δp Minor outlet	Relative error from finest
$\times$1.0	30764	-8.778459	45%
$\times$0.7	51816	-8.166390	35%
$\times$0.5	100929	-6.097125	0.8%
$\times$0.35	196531	-6.051001	0%

Table A.1: Mesh independence study

As we can see the coarser mesh refinement that fits the requirement of 2% of relative error is the one with a refinement factor of $\times$0.5. This is the refinement factor that is taken for the complete campaign.

A.3.3 Statistical Independence Study

The objective of this study is to know the minimal number of parcels that ensures independence of the number of parcels on the different KPIs' deviation. A parcel is a group of particles that has the exact same characteristics, for this study the number of particles considered in each parcel is 1. The increasement of the number of particles in the parcels only has an effect on the total weight collected.

Parcel number	d_{50}	Sharpness	Major contamination	Minor contamination	Wall losses
1e5	182.45	68.135	7.9	1.25	32.75
2e5	172.3	73.66	8.3	1.45	38.45
3e5	152.1	83.3	7.9	1.25	35.45
4e5	157	97.35	7.85	1.3	36.6
5e5	153.3	75.1	7.95	1.3	35.7
6e5	155.05	75.76	8.1	1.3	35.8
7e5	152.2	80.88	7.9	1.25	34.35
8e5	153.8	75.4	7.9	1.3	36.7
9e5	153.35	89.2	8.25	1.3	36.65
10e5	152.5	84.8	8.3	1.3	35.85

Table A.2: Deviation from KPIs in function of the diameter. The green slots corresponds to the KPIs deviation that has an error $<2\%$ with 10e5. The green slots corresponds to the KPIs deviation that has an error $<5\%$ with 10e5. The red slots corresponds to the KPIs deviation that has an error $<10\%$ with 10e5.

Beyond the injection of 10^5 parcel the available RAM is maxed out, preventing higher resolution simulations execution. The statistic independence study show a progressive stabilization of the KPIs near this value. 10^5 parcel is considered as a good compromise between model precision and computational cost.

B

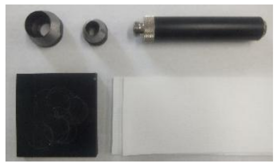
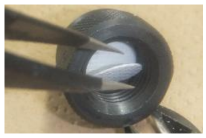
EXPERIMENTAL PROTOCOL

This appendix documents the experimental protocol used during a preliminary test in JAEA, it is intended mainly as a reference for future work rather than as a experimental validation of the numerical model

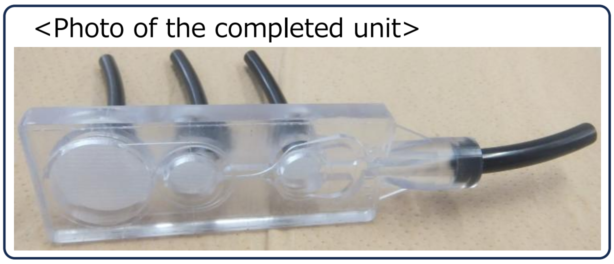
B.1 Virtual Impactor Manufacturing Process

Author	Nobuyoshi Konakawa	Department	CLADS, Decommissioning Assessment Group	Date	July 15, 2026
Title	Component Installation for the Aerosol Classifier (μSPLIT)				
Purpose of this Manual	After the μSPLIT is fabricated with a 3D printer, the individual parts (dust-monitor filter paper, Mylar film, and tubing) must be attached using various tools and adhesives. Because this work is delicate, must be carried out quickly, and involves a number of precautions, this manual has been established.				

Stage	Procedure / Key Points	Items to Prepare	Work Photo
1	(1) Remove the support material from the 3D-printed μSPLIT using long-nose pliers or a similar tool. (2) Inspect the surrounding area carefully to make sure no support material remains (especially in the exhaust port hole). (3) Confirm that the surface is smooth. (4) If burrs are present, cut them off with nippers or a similar tool.	• Long-nose pliers • Nippers	

2	(1) Using scissors, cut off the amount of filter paper needed to cover the exhaust port of the µSPLIT. (2) Prepare the size 13 and size 25 punches from the gasket punch set. (2) Fit the punch into the punch holder, and prepare the punch mat, the hammer, and the filter paper. (4) Place the punch mat on a sturdy part of the workbench, lay the filter paper on it with the fiber side facing up, set the punch holder so that the blade contacts the paper evenly, and strike the head firmly with the hammer about three times to cut out the filter paper.	• Filter paper (ADVANTEC, HE-40T) • Scissors • Gasket punch set (HHH, 3H-P17) • Punch mat (PVC, ESCO, EA576HR-1) • Hammer	 
3	(1) Using tweezers, lift the cut filter paper out of the punch and place it on the intake / exhaust port of the µSPLIT with the fiber side facing down. Align the grid pattern of the filter paper with the orientation of the µSPLIT. (2) Press gradually around the outer edge of the µSPLIT exhaust port with the tweezers; when it stops moving, the paper is seated.	• Tweezers	 
4	(1) Cut the Mylar film to be applied to the µSPLIT surface slightly larger than needed beforehand. (2) Stick Rega tape on the cutting mat, dispense about 1 cm each of the epoxy adhesive base resin and hardener (1:1 ratio), close the caps, and mix them quickly with a cotton swab. (3) The epoxy adhesive cures in 5 minutes, so within 2 minutes apply it evenly and thinly to the µSPLIT surface with a cotton swab, avoiding the corners of the exhaust port. (4) Slowly lower the film onto the µSPLIT. (5) Smooth it outward with the pad of your index finger.	• Scissors • Mylar film • Rega tape • Epoxy adhesive (Cemedine, Hi-Super 5) • Cotton swabs	

5	(1) Once the film applied to the μSPLIT surface has fully bonded (approx. 1 hour), place the μSPLIT in the drawer of the dedicated cutting jig set in order to trim off the excess film, insert the drawer into the cutting base, and turn the handle to lock it in position (there are two insertion slots, one for vertical cuts and one for horizontal cuts). (2) Insert a utility knife into the slit of the cutting base and cut the film.	• Utility knife • Dedicated cutting jig set (drawer, cutting base)	
6	(1) Cut the φ6 × φ4 tube to approx. 5.5 cm using the scale on the cutting mat, taking care that the cut end is not angled. (2) Stick Rega tape on the cutting mat, dispense about 1 cm of the modified silicone elastic adhesive, and close the cap. (3) The modified silicone elastic adhesive begins to set in 4 minutes, so within 2 minutes apply it evenly around the end of the tube to a thickness of about 1 mm with a cotton swab, taking care that it does not get inside the tube. (4) Insert the tube into the exhaust port and confirm that the adhesive seals the gap.	• Scissors • φ6 × φ4 tube (PISCO Tube, UE0640, COLOR: B) • Rega tape • Modified silicone elastic adhesive (Cemedine, SUPER XG, black) • Cotton swabs	 
7	(1) Cut the φ8 × φ5 tube to approx. 5.5 cm using the scale on the cutting mat, taking care that the cut end is not angled. (2) Stick Rega tape on the cutting mat, dispense about 1 cm of the modified silicone elastic adhesive, and close the cap. (3) The modified silicone elastic adhesive begins to set in 4 minutes, so within 2 minutes apply it evenly around the end of the tube to a thickness of about 1 mm with a cotton swab, taking care that it does not get inside the tube. (4) Insert the tube into the intake port and confirm that the adhesive seals the gap.	• Scissors • φ8 × φ5 tube (PISCO Tube, UE0850, COLOR: B) • Rega tape • Modified silicone elastic adhesive (Cemedine, SUPER XG, black) • Cotton swabs	 



B.1.1 Experimental Protocol

The powder is inserted in the mixing chamber to be aired out, then the connection between the mixing chamber and the VI (µSPLIT) is open. The vacuum pump and the flow controllers are running so the flow inside the µSPLIT can be considered in a steady state. The particles are carried by the flow and are filtered in different outlets by HEPA filters. The high-speed camera records the motion of particles in the 10 µm separation region, using a variable intensity light, then this record is transferred to the post-processing software PIVLAB. Also, the particles are injected in a multi-stage inertial impactor, to observe the mass or size distribution of the powder, and the information are transferred to the post-processing software. The list of facilities used is detailed below.

B.1.2 Experimental Results

The conducted experiences were using the repartition 8/1/1, 4/0.5/0.5. 2/0.25/0.25 $\mathrm{L \cdot min^{-1}}$, of the flow controllers. Recirculation were observed in the minor and major outlets, and wall particles deposition, neglected in the simulation is clearly observed. This deposition of particles on walls seems to affect PIV recognition, thus performance. It modifies also the interior topology of the impactor, which modify the physics, this creates a permanent transient regime.

B.1.3 Post-processing

The recorded video is analyzed with a Particle Image Velocimetry (PIV) algorithm, which separates background from moving particles, and determines size and velocity of the particles identified. It allows to represent the collection efficiency diagram of one separation stage. It allows as well to observe different recirculation and particles flow behavior.

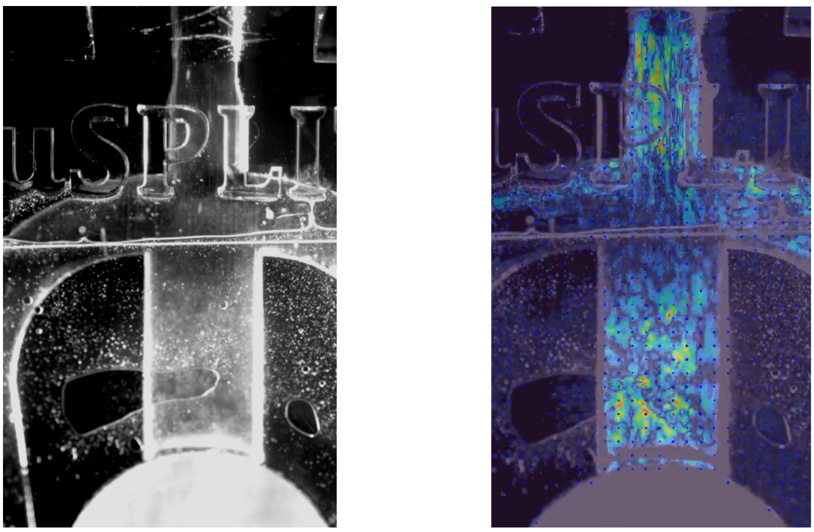


Figure B.1: Comparison between the video and the PIV analysis, the measurement is about particles speed here, with the software PIVLAB

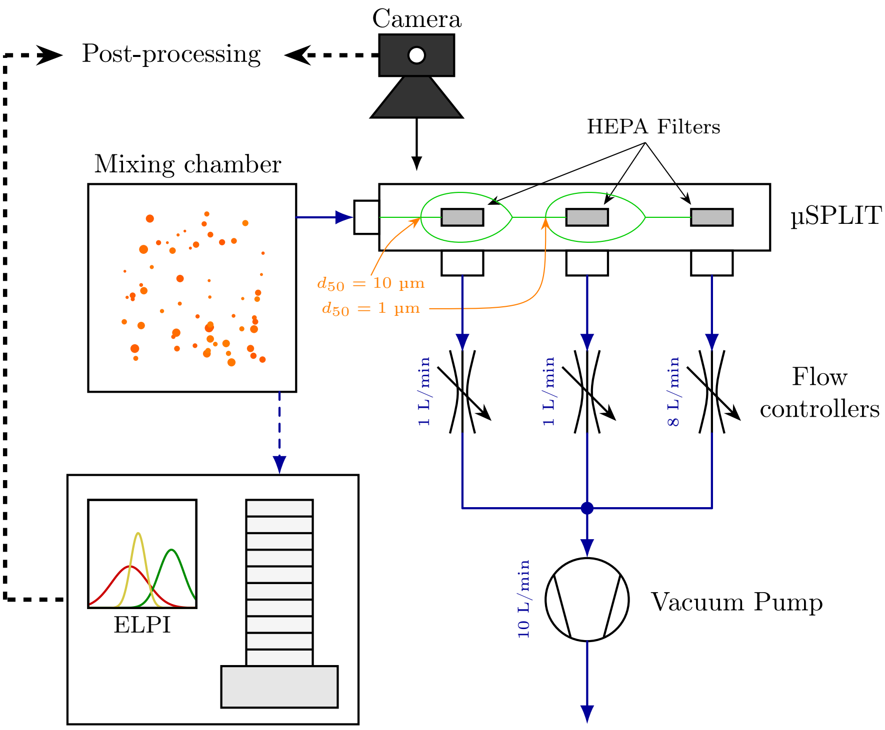


Figure B.2: Schematic of the experimental setup.
Plain arrows corresponds to airflow, dashed arrow corresponds to data transport.

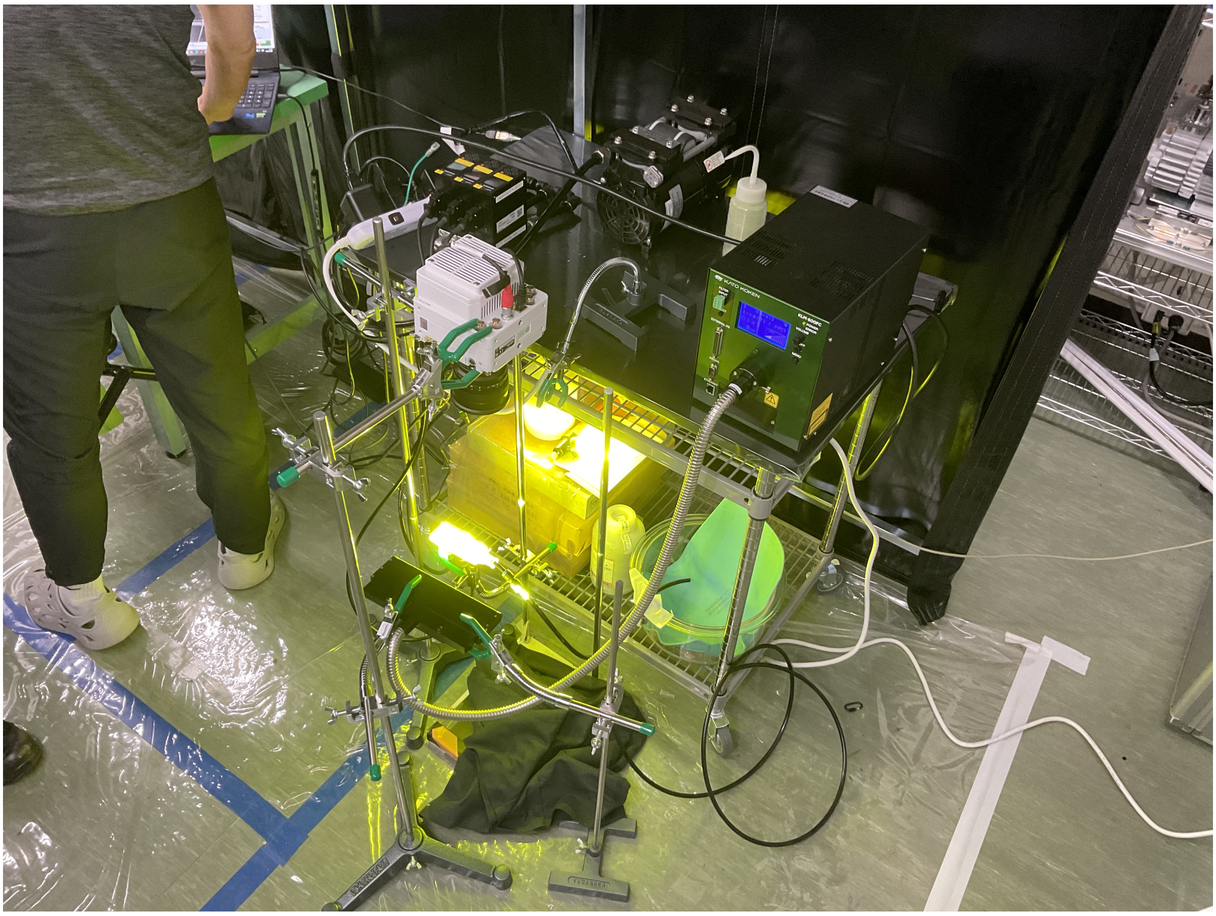


Figure B.3: Photography of the experimental setup

B.2 3D printing

B.2.1 Technical Specifications of the 3D Printer

Parameter	Specification
Printer Model	FORMLABS Form 4
Technology	Masked Stereolithography (MSLA)
Print System	Low Force Display (LFD)
Minimum Access Dimensions ($L \times W \times H$)	40.7 cm × 47.8 cm × 84.4 cm
Printer Dimensions ($L \times W \times H$)	39.8 cm × 36.7 cm × 55.4 cm
Printer Weight	18.3 kg
Build Volume ($L \times W \times H$)	20.0 cm × 12.5 cm × 21.0 cm
Layer Thickness (Vertical Resolution)	25 µm to 300 µm
XY Resolution	50 µm pixel size with pre-set anti-aliasing
Optical Power Intensity	16 mW cm^{-2} on the print plane
Optical Wavelength	405 nm
Resin Cartridges	1
Resin Distribution System	Automatic
Biocompatible Materials	No (available on the Form 4B model)
Supports	Automatically generated, light-touch style
Operating Conditions	18 °C to 28 °C, low ambient humidity
Operating Temperature	Automatic resin heating to 25 °C to 45 °C$^{(2)}$
Temperature Control	Direct resin heating
Power Requirements	100 V to 240 V AC, 4.8 A max, 50 Hz to 60 Hz, 480 W
Connectivity	Wi-Fi (2.4 GHz and 5 GHz), Ethernet (1000 Mbit), USB 2.0
USB Connectivity	USB-C Port with USB C-to-A cable
Printer Control Panel	Interactive touchscreen
Integrated Camera	2592 × 1944 (5 MP) resolution
Print Preparation Software	PreForm desktop software
Compatible File Types	STL, OBJ, or 3MF

Table B.1: Technical Specifications of the Formlabs Form 4 3D Printer

B.2.2 Technical Specifications of the Printing Material

Property	Symbol	Value
Tensile Modulus	E	2750.0 MPa
Poisson coefficient	ν	0.35
Flexural Modulus	E_f	2700.0 MPa
Ultimate Tensile Strength	σ_u	60.0 MPa
Elongation at Break	ϵ_b	8.0 %
Flexural Strength	σ_f	105.0 MPa
Notched Izod Impact Strength	a_{kn}	29.0 J m^{-1}
Heat Deflection Temperature @ 0.45 MPa	$T_{hdt,1}$	74.0 °C
Heat Deflection Temperature @ 1.8 MPa	$T_{hdt,2}$	59.0 °C

Table B.2: Mechanical and Thermal Properties of the Printed Material

Multi-stage Inertial Impactor DEKATI ELPI+

Dekati®
High-Resolution ELPI®+

Description

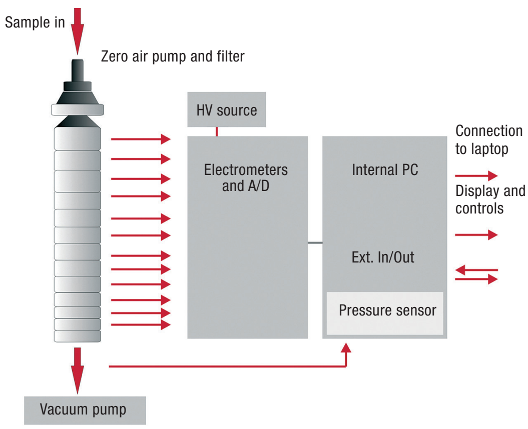
The Dekati® High-Resolution ELPI®+ (HR-ELPI®+) is an advanced version of the widely used ELPI®+ (Electrical Low Pressure Impactor). It is designed for accurate real-time aerosol particle size distribution measurements over a size range of 6 nm to 10 µm. The HR-ELPI®+ uses a dedicated data inversion algorithm to provide real-time particle size distributions with up to 500 size classes. The instrument supports a wide particle concentration range and its robust design makes it suitable for both laboratory and field use. In addition to real-time measurements, the HR-ELPI®+ allows collection of size-segregated particles for subsequent laboratory analysis. The included Data Analysis Tool enables efficient data processing and visualization, providing particle number, LDSA, and mass concentration (6 nm–2.5 µm) outputs.

.

Operating principle

First, the particles pass through a unipolar corona charger where they are charged up to a known positive charge level. Second, the particles are size classified in a cascade impactor into 14 size classes depending on their aerodynamic size.

Charger-Impactor assembly of the High-Resolution ELPI®+ (left), collection plates (middle) and the impactor stack showing the 14 stages (right)

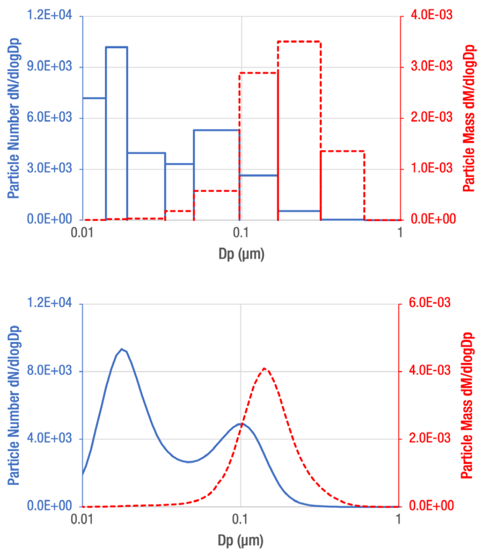


Operating principle of the High-Resolution ELPI®+

The classification of particles is based on the inertia of the particles, with larger particles getting collected on the upper impactor stages and smaller particles getting collected on the lower stages.

All the impactor stages are electrically insulated, and each impactor stage is connected to an electrometer. The primary particle collection efficiencies have been determined for each impactor stage allowing determination of impactor Kernel functions.

The data inversion calculation method used in the HR-ELPI®+ is based on these Kernel functions and iterative calculation routines, resulting in an accurate and reliable determination of particle size distribution with high size resolution. The inversion calculation runs in real-time and does not use any initial assumptions on the size distribution.



Particle number and mass size distributions for the same sample, measured in standard ELPI®+ mode (graph above) and when using the HR-ELPI®+ data inversion algorithm (graph below).

Stage [µm]	D50% [µm]	Di [µm]	Size range of Collected Particles [µm]	Number Concentration Min [#/cm³]	Number Concentration Max [#/cm³]	Mass Concentration Min [mg/m³]	Mass Concentration Max [mg/m³]
15	9,88		> 10				
14	5,4	7,3	5,4 < 10	0,03	2,4E+04	1,0E-02	4,9E+03
13	3,7	4,4	3,7 < 5,4	0,06	4,1E+04	4,8E-03	1,8E+03
12	2,5	3,0	2,5 < 3,7	0,09	7,3E+04	2,8E-03	1,0E+03
11	1,6	2,0	1,6 < 2,5	0,18	1,4E+05	1,5E-03	5,7E+02
10	0,91	1,2	0,91 < 1,6	0,31	2,9E+05	6,0E-04	2,7E+02
9	0,59	0,73	0,59 < 0,91	0,58	5,6E+05	3,0E-04	1,2E+02
8	0,32	0,43	0,32 < 0,59	0,98	1,1E+06	1,3E-04	4,5E+01
7	0,17	0,23	0,17 < 0,32	1,6	2,3E+06	7,0E-05	1,5E+01
6	0,099	0,13	0,099 < 0,17	3,3	4,7E+06	3,5E-05	5,3E+00
5	0,051	0,071	0,051 < 0,099	3,6	9,7E+06	1,5E-05	1,9E+00
4	0,034	0,042	0,034 < 0,051	19	1,9E+07	7,7E-06	7,1E-01
3	0,019	0,025	0,019 < 0,034	39	3,5E+07	3,1E-06	2,9E-01
2	0,014	0,016	0,014 < 0,019	180	6,1E+07	1,7E-06	1,3E-01
1	0,006	0,009	0,006 < 0,014	1800	1,2E+08	6,0E-07	4,7E-02

Main characteristics of the HR-ELPI®+ impactor stages. Each HR-ELPI®+ unit is individually calibrated before delivery; the calibration includes detailed determination of the exact sample flow rate and D50% values. The values presented in this table are nominal values.

Measurement Applications

The robust structure of the HR-ELPI®+, and its ability to measure in real-time over a wide size range, make the HR-ELPI®+ the ideal choice for various measurement applications. In combination with Dekati® Sample Conditioning Instruments, Dekati can provide complete HR-ELPI®+ measurement solutions for a very broad range of applications and even demanding measurement conditions. Typical applications for the HR-ELPI®+ include:

- Brake and tyre wear measurements
- Nanoparticle measurements
- Nuclear safety research
- Outdoor and indoor air quality measurements
- Occupational Health & Safety studies
- Inhalation toxicology
- Engine exhaust measurements
- Blow-by gas measurements
- Combustion process studies
- Emission monitoring
- Carbon Capture research and process monitoring
- Filtration efficiency measurements
- ENDS measurements

Features and benefits

- Wide particle size range 6 nm - 10 µm with one instrument
- Up to 6 months continuous use without maintenance
- Real-time (10 Hz) particle size distribution in up to 500 size classes
- Particle number, LDSA and mass concentration calculation
- Long-term, maintenance free operation
- Wide operational concentration range
- Sampling from up to 180 °C with the High-Temperature ELPI®+ upgrade
- ELPI®+ Data Analysis Tool software for easy data processing and visualization
- Simple and robust construction
- Insensitivity to variations in sample pressure and humidity
- Possibility of post-measurement chemical characterization of size classified impactor samples
- Each unit individually characterized
- Integrated flow control and pressure adjustment
- ISO16000-34 compliant measurement method for real-time determination of PM indoors

Vacuum Pump: Fuji Medical Instruments Co. - Elite OSP-90W

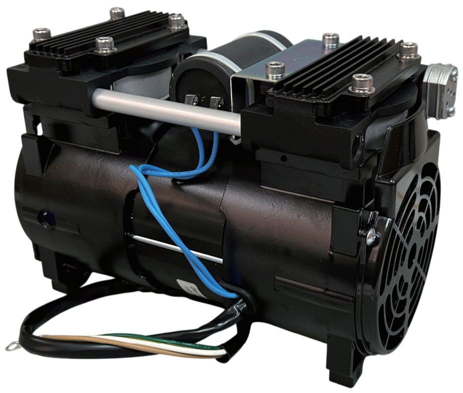
Elite Fuji

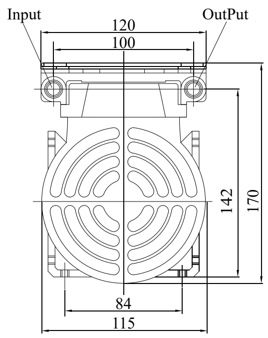
Since 1964
Made in Japan

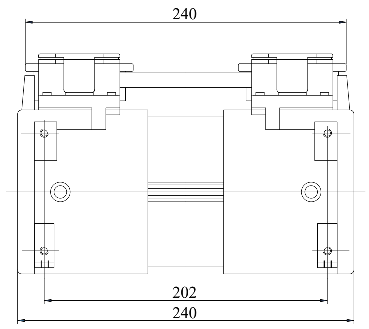
Oil-less Piston Air Compressors and Vacuum Pumps

MODEL: ***OSP-90W***

Oil-less Double Piston Pump





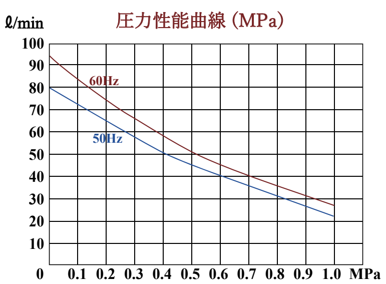


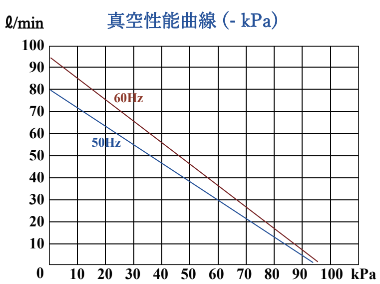
■ 主用途

・医療機器　・包装機械
・分析装置　・産業機器

■ 特長

・オイルレス仕様　・最大真空度 -95 kPa(g)
・連続運転対応＊1　・最大圧力 0.9 MPa(g)





OSP-90W 製品仕様

電源 / 消費電力	単相 AC100V, 50 / 60 Hz / 250 W	寸法 (W×D×H)	120 ×240 ×170 mm
開放流量 (50 / 60 Hz)	80 / 94 L / min	重量	7.3 Kg
最大圧力	0.9 MPa(g)	接続タイプ	PT1/4　本体側 目ねじ
常用圧力	0.5 MPa(g)	動作周囲温度	0 ~ 40 ℃ ＊1
最大真空圧	- 95.0 KPa(g) / - 710 mmHg	メーカー希望小売価格	98,000 (税別)

＊1 連続運転を行う際はモーターコアの温度を90℃以下に保ってください。

Fuji Medical Instruments Co., Ltd.
1-32-18 Hokima, Adachi-ku, Tokyo 121-0064 Japan
TEL:+81-3-884-6414
https://elite-fuji.com/

Digital Mass Flow Controller
Model F4Q

Overview
This device is a high-performance digital mass flow controller with advanced functions for the general industrial market. It features high accuracy, high-speed response, and a wide control range.

Features
- High-speed controllability
  High-speed response of 300 ms (typ.).* (700 ms for F4Q0050(J,K)/0200)
  * When control begins from the fully closed state or when the setting is changed during control, the controlled flow rate reaches the set value ± 2 % within 300 ms.
- Low differential pressure operation
  This device can be used even at a low differential pressures. (The minimum operating differential pressure differs depending on the model)
- Wide control range
  Provides a wide control range of 1–100 % full scale (FS).
- A lineup of products suited to customers' applications
  The lineup consists of integrated-display models and separate-display models.
  With a separate-display model, a special 2 m cable (included) can be used for remote display and operation of the controller.
- Ease of use
  The product runs on 24 V DC from a single general-purpose power supply. In addition, the power supply circuit and I/O circuit inside the device are isolated.
  When multiple F4Qs are driven through analog input/output using a programmable logic controller (PLC) or the like, even if the analog module channels on the PLC side are not isolated, a common power supply can be used to power the units.
  Even without using an individual power supply for each device, there is no need for concern regarding effects from one circuit on adjacent ones.
  In addition, for easy use in a laboratory, a convenient AC adapter (optional) is available.
- Ease of setup
  This device can be set up easily by connecting it and a PC with a general-purpose USB 2.0 cable and using the dedicated PC loader software.
  There is no need for an external power supply during setup because power is supplied from the PC through the USB cable.
  (Note that USB power cannot be used for control or for powering the display.)
- Changeable display orientation
  The orientation of the display unit can be changed according to the gas flow direction to make it easy to read.
- Better design
  A large indicator for better visibility and design.
  The use of an LCD on the display makes it easy to check settings and numerical values.
- A variety of functions
  A variety of functions are included as standard features.

■ F4Q0050 (B,C)

Item		F4Q0050 (B, C)
Valve type		Proportional solenoid valve
Valve operation		Normally closed (N.C.) when not powered
Standard full scale flow rate (air/nitrogen)*1		50 L/min
Standard compatible gas type*2	Fluoroelastomer gasket	Air/nitrogen, oxygen, argon, carbon dioxide, fuel gas 13A (45 MJ/m^3), 100 % methane, 100 % propane, 100 % butane
	EPDM gasket	Air/nitrogen, argon, carbon dioxide
Control	Control range	1 to 100 % FS
	Response time*3	0.3 s (typ.) to the setting ±2 % FS (when control begins with the valve fully closed, or when the setting is changed during control)
	Accuracy (under standard conditions. Q: flow rate)*4	±1 % SP (15 ≤ Q ≤ 100 %) ±0.15 % FS (1 ≤ Q < 15 %)
	Repeatability (Q: Flow rate)	±0.25 % SP (15 ≤ Q ≤ 100 %) ±0.0375 % FS (1 ≤ Q < 15 %)
	Effect of temperature (Q: Flow rate)*3 *5	±0.1 % SP/°C (15 ≤ Q ≤ 100 %) ±0.015 % FS/°C (1 ≤ Q < 15 %)
	Effect of pressure*5	0.3 % FS / 100 kPa
Pressure	Standard differential pressure	200 kPa (inlet pressure: 200 kPa (gauge); outlet pressure: 0 kPa (gauge))
	Operating differential pressure*6 *7	100 to 300 kPa
	Allowable inlet pressure	0.5 MPa (gauge)
	Pressure resistance	1 MPa (gauge)
Operating conditions	Ambient temperature	–10 to + 60 °C
	Ambient humidity	10 to 90 % RH (without condensation)
	Vibration condition	No vibration
Transport/storage conditions	Ambient temperature	−20 to +70 °C
	Drop	0.8 m (when packed)
	Vibration resistance	4.9 m/s^2 (10 to 60 Hz, in the X, Y, and Z directions for two hours when mounted with the bracket)
	Shock resistance	X direction: 147 m/s^2, Y and Z directions: 490 m/s^2 (three times in each direction when mounted with the bracket. See ■ External dimensions (for directions))
External leakage		1×10^{-8} Pa•m^3/s (He) (not including O-ring permeability)
Flow rate setting	Method	(1) Use of the keys, (2) External analog input, (3) Loader communication, (4) RS-485 communication (3-wire system)
Analog input	Input type	0–5 V DC, 1–5 V DC, 4–20 mA (switchable)
	Input impedance	1 MΩ ±10 % for DC voltage input, 250 Ω ±10 % for current input
Analog output	Output type	Instantaneous flow rate (PV) output or flow rate set point (SP) output (switchable)
	Output scale	0 to full-scale flow rate (scale can be changed)
	Output type	0–5 V DC, 1–5 V DC, 4–20 mA (switchable)
	Maximum output	7 V DC or less / 28 mA or less
	Accuracy	±0.3 % FS (overall output accuracy: indication accuracy ±0.3 % FS)
	External load resistance	250 kΩ min. for voltage output, 300 Ω max. for current output
Digital output	Number of outputs	3
	Output rating	External power supply: 30 V DC max., maximum load current: 30 mA max.
	Totalization pulse width	20–100 ms (when totalizer pulse output is selected, output at multiples of 20 ms)
	Totalization pulse weight	200–1,000,000 mL/pulse
Digital input	Number of inputs	3
	Required circuit type	Non-voltage contacts or open collector
	Terminal voltage with contacts OFF	5 ±0.5 V
	Terminal current with contacts ON	Approx. 0.5 mA (current to contacts)
	Allowable ON contact resistance	250 Ω max.
	Allowable OFF contact resistance	100 kΩ min.
	Allowable ON residual voltage	1.0 V max. (with open collector)
	Allowable OFF leakage current	100 μA max. (with open collector)
Communication	Method*8	(1) USB 2.0 (2) RS-485 communication (3-wire system, CPL or Modbus RTU)
	Transfer speed selection	4800, 9600, 19200, 38400 bps (with RS-485 communication)
Power	Rating	24 V DC, current consumption: 300 mA max.
	Allowable supply voltage range	21.6–26.4 V DC (ripple: 5 % max.)
	Isolation	The power supply circuit and I/O circuit are isolated.

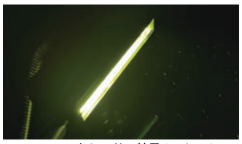
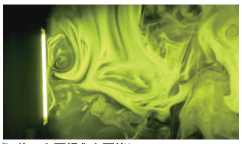
KATO KOKEN

可視化用ルミネッセンス光源

KLM-600FC

Light Beyond Lasers: The Alternative Illumination.

なぜ黄色い光なのか？（黄色い光の特性）

トレーサー粒子のハレーションを抑え、均一な可視化を可能に

黄色い光は、LEDの白色光と比較すると奥行方向へ強く浸透します。そのため、トレーサー粒子が高濃度にシーディングされた空間でも、 均一な輝度を保ちます。霧の中で使用するフォグランプが黄色いのは、その浸透性から視界を維持しやすいという理由があります（一方、白色光は水蒸気で反射してまぶしさを引き起こします）トレーサー粒子のハレーションを抑え、均一な視認性を可能にする光です。

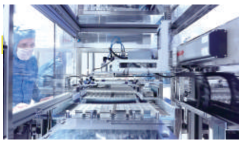
ルミネッセンス光源とは？

青色レーザーダイオードと蛍光体を組み合わせることで、特定の光を発生させる光源です。 レーザーダイオードが放出する短い波長の光は、蛍光体ターゲットに照射され、励起状態となります。この時、蛍光体から発振する特定の非干渉性の光が生成されます。この技術のメリットは、発散角の制御が可能で、光をシート状に取り出すことができる点です。高輝度で高品質なシート光を生成し、クラス4の高出力レーザーシートに匹敵する明瞭な可視化を実現します。

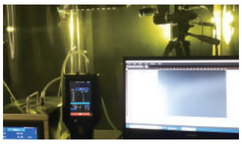
レーザークラス1に相当

可視化用ルミネッセンス光源KLM-600FCは、レーザー光が外部に漏れないよう堅牢に封入されているため、 レーザープロジェクターのような組込型レーザー製品の扱いになります。これは”レーザー製品としては合理的に予見可能な条件下で安全” に相当し、レーザークラス1の製品として運用が可能です。

クリーンルームで異物を可視化



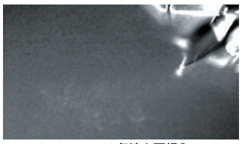
クリーンルーム内の作業



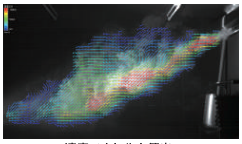
粒子を可視化

半導体製造プロセスでは異物対策が非常に重要です。しかし、クリーンルームの管理基準となる0.3 μm の異物を可視化する場合、 高出力レーザーを使用した可視化性能が求められます。 高出力レーザー（レーザークラス3B以上）の運用は、安全上多くの制約がかかり、可視化による異物検証はハードルが高くなることが現状です。**KLM-600FCはレーザークラス1相当の光源となるため、レーザー安全対策は必要はありません。**様々な環境で浮遊する異物や発塵源の特定の検証に使用できます。

LED光源では難しかったPIVにも対応



エアコンの気流を可視化



速度ベクトルを算出

可視化用ルミネッセンスKLM-600FCはPIV に対応します。従来のLED光源では難しかった一つ一つの粒子像を鮮明に撮影できるため、 高出力レーザーシートを用いた実験に限りなく近い計測が可能です。 また、KLM-600FCのシート光の厚みはステレオPIVにも最適です。

三脚に取付可能

ファイバー光学系・本体は、それぞれ三脚取付に対応。撮影条件や被写体に合わせて、高さ・角度・距離をスムーズに調整でき、安定した照明配置が可能です。 現場での再現性と作業効率を高めます。

可視化用ルミネッセンスKLM-600FC仕様

発光色	黄色
駆動方式	定電流方式
調光方式	電流可変制御
チャネル数	1チャネル
入力電源	AC100～240V（±10%）、50/60
消費電力（typ.）	410VA（100V入力）、420VA（240V入力）
突入電流（typ.）	40A コールドスタート時
使用環境（屋内限定）	温度：0～40℃、湿度：20～80%RH（結露なきこと） 高度：最高2,200m AC過電圧：カテゴリーII 汚染度：2
保存温湿度	温度：-15～60℃、湿度：20～80%RH（結露なきこと）
冷却方式	強制空冷方式
本体寸法／重量	寸法：高さ215mm×幅175mm×奥行270mm 重量：約7.8kg
シート光学系寸法（ファイバー光学系）	寸法：縦60mm×横170mm×高さ18mm ファイバー長：3m
付属品	取扱説明書×1、3Pアース極付きACコード（2m）×1

製品のデザイン、仕様は予告なく変更する場合がございますのでご了承ください

流れの可視化　カトウ光研株式会社　PIV・画像解析

本　社　〒259-1116 神奈川県伊勢原市石田737-2
TEL：0463-91-1281　FAX：0463-91-1291

名古屋営業所　〒451-0045 愛知県名古屋市西区名駅2-29-3 Sur303
TEL：052-485-8691　FAX：052-485-8692

大阪営業所　〒577-0022 大阪府東大阪市荒本新町8-37-102
TEL：06-7492-6658　FAX：06-7492-6489

お問合せ・技術相談
https://www.kk-co.jp
E-mail:info@kk-co.jp

ホームページ上でカタログのダウンロードも受け付けております

Anti-static Tube

PISCO

>>> https://www.pisco.co.jp/en/

Anti-static conductive Polyurethane Tube

⚠ Safety instructions for this product

Safety instructions, Common safety instructions for each product category and Detailed safety instructions for each product are in the end of this catalog and our website.

Tube

Model Designation (Example)

UE (1) 0640 (2) - 20 (3) - B (4)

(1) Anti-static Tube

(2) Tube dia. (O.D. x I.D.)

	mm size (mm)					
Code	0320	0425	0640	0850	1065	1280
O.D. (mm)	ø3	ø4	ø6	ø8	ø10	ø12
I.D. (mm)	ø2	ø2.5	ø4	ø5	ø6.5	ø8

(3) Tube length

Code	20	100
Length (m)	20	100

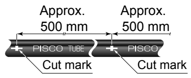
(4) Tube color

Code	B
Tube color	Black

Characteristics

- Best suited for use on the manufacturing or assembly line that requires dissipation of static electricity and prevention of electrification.
- The cut marks at 500 mm interval are printed on the tube as a guide.

*The marks are just as a guide and the accuracy is not guaranteed.



Specifications

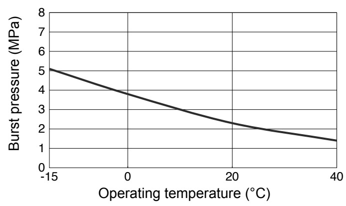
	Model code	UE0320	UE0425	UE0640	UE0850	UE1065	UE1280
Size	O.D. x I.D. (mm)	3×2	4×2.5	6×4	8×5	10×6.5	12×8
Size	Tube length (m)	20 · 100					
	Tube color: 1 color	Black (B)					
Specifications	Fluid medium	Air					
Specifications	Max. operating pressure (MPa)	0.6 (65% RH at 20ºC*1)					
Specifications	Max. vacuum (kPa)	-100					
Specifications	Operating temp. range (°C)	-15 to 40 (No freezing)					
Specifications	Min. bending radius (JIS) (mm)	8	10	15		20	30
Specifications	Min. installation radius (Vice) (mm)	12	15	23		30	45
Specifications	Volume resistance (Ω·cm)*2	1.4×10^3					

⚠ Warning

*1. The value of max. operating pressure is measured at 20ºC and 65% RH. For other temperatures, the Burst Pressure Curve should be used with a sufficient safety margin. In addition, if the tube is in a moving part, such as a swinging or bending part, the temperature may rise due to self-heating caused by intermolecular heat generation, which may cause the tube to break.

*2. The value of volume resistivity is from the data of the material supplier, which is just for reference but not guaranteed.

Burst Pressure Curve (Reference Value)



RoHS2 (2011/65/EU+EU2015/863) compliant

Tube type	Model Code
Anti-static Tube UE 	UE0320-3-B
	UE0425-3-B
	UE0640-3-B
	UE0850-3-B
	UE1065-3-B
	UE1280-3-B

Notes

*1. Tubes with desired length are available as make-to-order items. Contact us for the price.

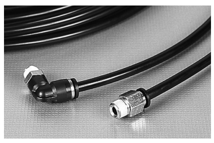
*2. For 3 in model code, please select a code of tube length.

Packaging specifications

20 M lengths / bag: Applicable to all types

100 M lengths / box: Applicable to all types

Related product



Tube Fitting Anti-static

► P.58

JIS TEST POWDERS 1 (JIS Z 8901) (Class 8)

Chemical component	% **(weight)**
SiO_2	34 - 40
Fe_2O_3	17 - 23
Al_2O_3	26 - 32
CaO	0 - 3
MgO	0 - 7
TiO_2	0 - 4
Ignition Loss	0 - 4
Particle size µm	**Oversize (weight basis)** %
1	
2	
4	
5	61±5
6	
8	
10	43±3
20	27±3
30	15±3
40	9±3
75	≤ 3
Material	KANTO (Japanese) loam
Particle density	$2.9\ g \cdot cm^{-3}$ – $3.1\ g \cdot cm^{-3}$

Table B.3: Technical Specifications of the JIS TEST POWDERS 1 (JIS Z 8901) (Class 8).

Mixing Chamber

PHANTOM® / AMETEK® VISION RESEARCH

VEO-SERIES

DATASHEET

PHANTOM VEO E-310L VEO E-340L

HIGH-SPEED CAMERA

1280 × 800 up-to 3,260 fps (E-310L)
2560 × 1600 up-to 800 fps (E-340L)

PHANTOM HIGH-SPEED / FORZA SILICON AMETEK

FEATURES & BENEFITS

PHANTOM VEO PRODUCT FAMILY

Phantom high-speed cameras are utilized every day in demanding test and measurement applications around the world. The VEO platform is known for high quality and dependable image capture due to proprietary sensor design, rugged and compact housings, unique workflow features and overall system versatility.

VEO-E models leverage this platform, offer many of the same features and are:

- 20% smaller and lighter than the core VEO models
- Designed for an efficient and easy set-up with industry standard connections
- Cost-effective for laboratories and academic institutions

Now available: VEO-E E225 models with a maximum frame rate of 225,000 fps at reduced resolutions.

IMAGE & SENSITIVITY

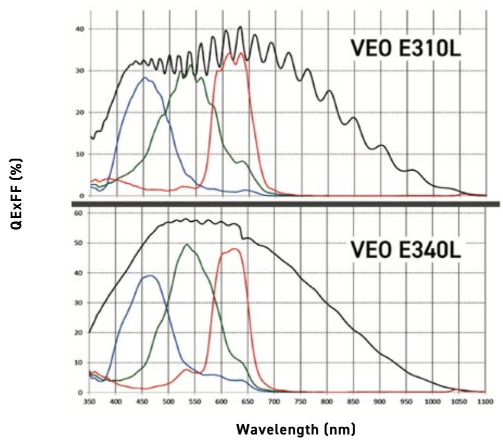
Sensor Type	CMOS, with Global Shutter	
Maximum Resolution	E-310L: 1280 × 800	E-340L: 2560 × 1600
CAR Increments	E-310L: 64 × 8	E-340L: 128 × 4
Pixel Size	E-310L: 20 µm	E-340L: 10 µm
Sensor Size	25.6 × 16 mm	
Bit Depth	12 bit	
	EMVA 1288 Measurements (at 532 nm)	
	E-310L	E-340L
Quantum Efficiency (%)	33.9% mono 29.5% color	60.2% mono 43.6% color
Max. SNR (dB)	44.1	41.6
Absolute Sensitivity Threshold (p)	99.4 mono 115.9 color	40.4 mono 54.1 color
Saturation Capacity (e-)	25,918 mono 27,880 color	14,512 mono 14,335 color
Temporal Dark Noise (e-)	33.1	23.8
Dynamic Range (dB)	57.7	55.5

- *Reported measurements were taken at 532 nm with both monochrome and color cameras, using the EMVA 1288 3.1 standard*
- *Visit: www.phantomhighspeed.com/emva for more information on EMVA 1288*

VEO L-model rear view

SPECTRAL RESPONSE

Quantum Efficiency Monochrome and Color



CONNECTIVITY & SIGNALS

Ethernet	Gigabit Ethernet
Timecode	IRIG-B Modulated and Un-modulated
Port Descriptions	Ethernet: Standard RJ45 port Power: Fischer 6-pin Range Data: N/A USB: N/A Video output: 3G-SDI (1 port), HDMI Dedicated BNC: 2 ports for Trigger, Timecode-in Programmable I/O BNC: 2 ports
I/O Signals	Programmable I/O (2 ports) for Fsync, Strobe, Ready, Timecode-out, Event, Pretrigger. Assign and define signals in PCC
Hardware Trigger	Dedicated BNC
Software Trigger	via Ethernet; via Image-based auto trigger (IBAT)
Synchronization	External Sync via FSync or IRIG Timecode
Recording Features	Burst mode; Image-based auto trigger, Continuous
Video Output	3G-SDI via Din and Micro HDMI type D port *Cameras prior to 2021 had HDMI type A port*
Accessory Power	4-pin Hirose (front) for 12V monitors up to 1 Amp

MEMORY & STORAGE

RAM Buffer	18GB, 36GB RAM options
Multi-Cine	Up to 64 Partitions
Non-Volatile Media	N/A

FRAME RATES & EXPOSURE

Top FPS at Max Resolution	E-310L: 3,260	E-340L: 800
1 Megapixel FPS	E-310L: 3,260	E-340L: 2,950
Maximum FPS	E-310L: 650,000 E-310L-E225: 225,000	E-340L: 287,000 E-340L-E225: 225,000
Minimum FPS	24	
Minimum Exposure	1 µs	
PIV Features	Shutter-off mode with straddle time of 480 ns (310) and 1.7 µs (340); Supports Burst mode	
Exposure Features	Extreme Dynamic Range (EDR), Auto-Exposure, Overexposure indication over video and in PCC	

FRAME RATE CHART

Table provides examples of common resolutions and frame rates. The record times shown are for 36GB RAM at the frame rate shown. Duration will be 1/2 the time for 18GB RAM.

MAXIMUM FRAME RATE - FPS; (36GB RECORD TIME - SEC)				
Resolution (H × V)	**E-310L**	**E-310L-E225**	**E-340L**	**E-340L-E225**
2560 × 1600	N/A	N/A	800 (3.9)	800 (3.9)
2560 × 1440	N/A	N/A	890 (3.9)	890 (3.9)
1536 × 1536	N/A	N/A	1,320 (4.1)	1,320 (4.1)
1920 × 1080	N/A	N/A	1,540 (4)	1,540 (4)
1280 × 1280	N/A	N/A	1,850 (4.2)	1,850 (4.2)
1280 × 800	3,260 (7.7)	3,260 (7.7)	2,950 (4.2)	2,950 (4.2)
1280 × 720	3,630 (7.7)	3,630 (7.7)	3,270 (4.2)	3,270 (4.2)
640 × 480	10,100 (8.3)	10,100 (8.3)	8,430 (4.9)	8,430 (4.9)
512 × 512	11,500 (8.5)	11,500 (8.5)	9,250 (5.2)	9,250 (5.2)
256 × 256	39,700 (9.9)	39,700 (9.9)	26,800 (7.3)	26,800 (7.3)
128 × 128	120,400 (13)	120,400 (13)	64,500 (12)	64,500 (12)
128 × 64	224,900 (13)	224,900 (13)	108,700 (14)	108,700 (14)
128 × 32	397,100 (15)	225,000 (25)	165,100 (19)	165,100 (19)
128 × 8	650,000 (38)	225,000 (100)	270,000 (46)	225,000 (50)
128 × 4	N/A	N/A	287,000 (87)	225,000 (100)

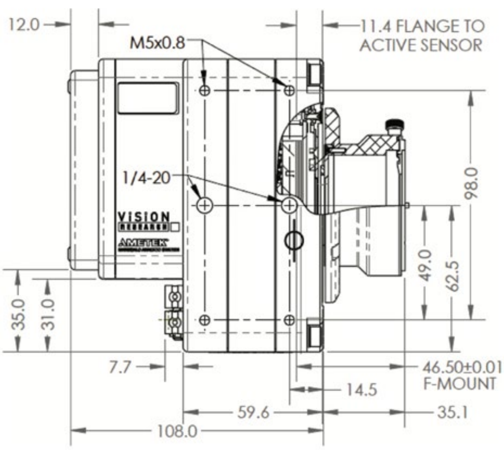
**Certain Phantom cameras are held to export licensing standards. Details available at: www.phantomhighspeed.com/export*

CONTROL

Software & OS	Phantom PCC (Windows x64); SDK available for C/C++, C#, Python, MatLab and LabView
On-Camera Controls	N/A
Primary File Format	Phantom Cine RAW (.cine)
Alternative File Formats	Easily convert to formats including .mp4, Apple ProRes .mov, .avi, Tiff, JPG, DNG and many more using PCC. Cine files are directly compatible with many major video editing and motion analysis programs
Software Features	Continuous recording can eliminate downtime between shots, Integrated Data Acquisition (NI-DAQ), Support for DIC Calibration with Sync-Snapshot menu, Image Processing

MECHANICAL

Housing Variants	L-model only
Size	5 × 5 × 4.2" (12.7 × 12.7 × 10.8 cm)
Weight	4 lbs (1.8 kg)
Lens Mounts	F-Mount standard (aperture support for Nikon G-style lenses). Also available: Canon EF (with electronic focus and iris control), PL, C-mount
Mounting Points	Standard 1/4 × 20" mounting points on bottom, top and side of camera
Internal Shutter	Standard, for remote black references
Cooling	Active cooling. Quiet mode disables fans during capture



POWER

AC Power	100-240 VAC, 80W power supply included
Voltage Range	16-32VDC Primary
Power Consumption	40W typical
Battery Options	Works with 16-32V battery sources only

ENVIRONMENTAL

Operating Temperature	-10 to +50°C
Storage Temperature	-20 to +70°C
Relative Humidity	≤85% non condensing
Operational Shock	30G, 11msec sawtooth, 3 axes, 2 directions per axis, 10 shocks per direction (60 pulses total)
Operational Vibration	MIL-STD-202G Method 214-A. Rated 12Grms; Figure 2A-1, Test Condition D, 15 min per axis
Regulatory	Made in the USA **Emissions** – CE & UKCA Compliant EN 61326-1 **Immunity** – CE & UKCA Compliant EN 61326-1 **FCC** – CFR 47, Part 15, Subpart B & ICES-0003, Class A**KC** **Emissions** – KC Compliant KN32 **KC Immunity** – KC Compliant KN35 **Safety** – IEC 60950-1 (2012)

GLOBAL SUPPORT NETWORK

Phantom cameras are supported by Vision Research's Global Service and Support network, providing PhantomCare services from multiple sites around the globe.

ABOUT PHANTOM HIGH-SPEED

Focused. Since 1950, Phantom has been designing, and manufacturing high-speed cameras. Our single focus is to invent, build, and support the most advanced cameras possible.

Phantomhighspeed.com
+1 973.696.4500
100 Dey Rd., Wayne,
NJ 07470, USA
@PhantomHighSpeed
@Phantom High-Speed Cameras

WWW.PHANTOMHIGHSPEED.COM

PIVLAB

PIVlab (`https://www.pivlab.de/`) is a free and open-source particle image velocimetry (PIV) software and is currently the most frequently cited PIV tool. It can be used to calculate the velocity distribution within imported (or captured) images. It can also control lasers, cameras and synchronizers, and derive, display and export multiple parameters of the flow pattern. The simple graphical user interface makes PIV data acquisition and data post-processing fast and efficient.

C

SUSTAINABLE DEVELOPMENT & SOCIETAL RESPONSIBILITIES

The 2030 Agenda for Sustainable Development, adopted by all United Nations Member States in 2015, defines 17 Sustainable Development Goals (SDGs) as a framework shared for global action (Figure C.1). This internship may have contributed to that framework: a brief reflection on this contribution is developed below.

Figure C.1: 17 Sustainable Development Goals (SDGs) (UN Organization, 2015).

This work supports **SDG 3 (Good Health and Well-Being)** by considering radiation exposure risks for Fukushima Daiichi decommissioning operators, and **SDG 4 (Quality Education)**, **SDG 9 (Industry, Innovation and Infrastructure)** by developing a low-cost, 3D-printable alternative to expensive industrial virtual impactors, potentially granting access to aerosol science instrumentation for educative institutions. It also contributes to **SDG 17 (Partnerships for the Goals)** through the international partnership between Polytech Nancy, UTokyo, and JAEA that made this work possible.

REFERENCES

— Marple, Virgil A et al. (2011). “History of Virtual Impactors”. In: *Aerosol science and technology: history and reviews*, p. 509.

— Laffolley, Hugo and Youichi Tsubota (2026). “Development of a low-cost and disposable virtual impactor for particle-size-dependent radioactivity assessment at Fukushima Daiichi NPS”. In: *AESJ 2026 Annual Meeting*.

— Colbeck, Ian et al. (2014). *Aerosol science: Technology and applications*. John Wiley & Sons.

— NEA (2009). “State-of-the-art report on Nuclear Aerosols”. In: *Committee on the safety of nuclear installations*.

— Hari, Sridhar et al. (2006). “Optimization studies on a slit virtual impactor”. In: *Particulate science and technology* 24.2, pp. 105–136.

— Crowe, Clayton et al. (1998). “Multiphase flows with droplets and particles”. In: *Taylor Francis Group*.

— Hinds, William C et al. (1999). *Aerosol technology: properties, behavior, and measurement of airborne particles*. John Wiley & Sons.

— Zhuang, Shuhan et al. (2026). “Wet deposition modeling and behavior analysis of 137Cs aerosols using ten observation-based continuous size distributions following the Fukushima accident”. In: *Environmental Pollution*, p. 127773.

— Högberg, Lars (2013). “Root causes and impacts of severe accidents at large nuclear power plants”. In: *Ambio* 42.3, pp. 267–284.

— Laffolley, Hugo, Christophe Journeau, et al. (2024). “Simulant molten core–concrete interaction experiments in view of understanding Fukushima Daiichi Nuclear Power Station Cs-bearing particles generation mechanism”. In: *Scientific Reports* 14.1, p. 6611.

— IRSN (2012). “Fukushima, one year later: initial analyses of the accident and its consequences.” In: *Report n°IRSN/DG/2012-003*. URL: `https://recherche-expertise.asnr.fr/sites/default/files/documents/expertise/rapports_expertise/surete/IRSN_Rapport_Fukushima-1-an-apres_032012.pdf`.

— Mészáros, Róbert et al. (2016). “Predictability of the dispersion of Fukushima-derived radionuclides and their homogenization in the atmosphere”. In: *Nature, Scientific Reports*.

— Katata, Genki et al. (2012). “Atmospheric discharge and dispersion of radionuclides during the Fukushima Dai-ichi Nuclear Power Plant accident. Part I: Source term estimation and local-scale atmospheric dispersion in early phase of the accident”. In: *Journal of Environmental Radioactivity* 109, pp. 103–113.

— TEPCO (2026). “Fukushima Daiichi Nuclear Power Station Unit 3 : Commencement of PCV Internal Investigation (non-submerged area) using Micro-drones”. In: URL: `https://www.tepco.co.jp/en/hd/decommission/information/newsrelease/reference/pdf/2026/reference_20260305_02-e.pdf`.

— Porcheron, Emmanuel et al. (2021). “Fukushima Daiichi fuel debris retrieval: results of aerosol characterization during laser cutting of non-radioactive corium simulants”. In: *JOURNAL OF NUCLEAR SCIENCE AND TECHNOLOGY, VOL. 58, NO. 1, 87–99*. URL: `https://doi.org/10.1080/00223131.2020.1806135`.

— Steinhauser, Georg et al. (2014). “Comparison of the Chernobyl and Fukushima nuclear accidents: A review of the environmental impacts”. In: *Science of The Total Environment* 470-471. ISSN: 00489697. DOI: 10.1016/j.scitotenv.2013.10.029. URL: https://linkinghub.elsevier.com/retrieve/pii/S004896971301173X.

— Li, Chunsheng et al. (2022). “Public health response and medical management of internal contamination in past radiological or nuclear incidents: A narrative review”. In: *Environment international* 163, p. 107222.

— Długosz-Lisiecka, Magdalena et al. (2025). “Fate and health implications of the radioactive ultrafine aerosols in the air”. In: *Sci Rep* 15. ISSN: 2045-2322. DOI: 10.1038/s41598-025-10923-0. URL: https://www.nature.com/articles/s41598-025-10923-0.

— Darquenne, Chantal (2020). “Deposition Mechanisms”. In: *Journal of Aerosol Medicine and Pulmonary Drug Delivery* 33.4, pp. 181–185. DOI: 10.1089/jamp.2020.29029.cd. URL: https://doi.org/10.1089/jamp.2020.29029.cd.

— Hinds, William C et al. (2022). *Aerosol technology: properties, behavior, and measurement of airborne particles*. John Wiley & Sons.

— Le, Thi-Cuc et al. (2021). “Inertial Impaction Technique for the Classification of Particulate Matters and Nanoparticles: A Review”. In: *KONA* 38. DOI: 10.14356/kona.2021004. URL: https://www.jstage.jst.go.jp/article/kona/38/0/38_2021004/_article.

— Romay, FJ et al. (2023). *Design of Round-Nozzle Inertial Impactors Review with Updated Design Parameters, Aerosol Air Qual. Res., 23, 220436*.

— Chen, B. T. et al. (1987). “An Improved Virtual Impactor : Design And Performance”. In: *Journal of Aerosol Science, Vol. 18, No. 2, pp. 203-214*.

— Marple, Virgil A. et al. (1980). “Virtual impactors: a theoretical study.” In: *Environmental Science & Technology*.

— Masuda, H. et al. (1988). “Classification performance of a rectangular jet virtual impactor—effect of nozzle width ratio of collection nozzle to acceleration jet”. In: *Journal of Aerosol Science*. DOI: https://doi.org/10.1016/0021-8502(88)90226-1. URL: https://www.sciencedirect.com/science/article/pii/0021850288902261.

— McFarland, Andrew R. et al. (1978). “Particle collection characteristics of a single-stage dichotomous sampler”. In: *Environmental Science & Technology*.

— Zhao, Heng et al. (2026). “Low-Temperature Carbon Dioxide-Enabled Virtual Impactor: Improved Cutoff Performance for Fine Particle Sorting”. In: *Atmosphere* 17.3, p. 248.

— Novick, Andrea Lee (1987). *Traveling light: lithic inference, site structure and hunter-gatherer mobility strategies at Gatecliff Shelter, Nevada*. Washington State University.

— Marple, Virgil et al. (2013). “A PM1. 0/2.5/10 trichotomous virtual impactor based sampler: design and applied to arid southwest aerosols part I: design”. In: *KONA Powder and Particle Journal* 30, pp. 15–21.

— Virgil, Marple et al. (2014). “A PM1.0/2.5/10 Trichotomous Virtual Impactor Based Sampler: Design and Applied to Arid Southwest Aerosols (Part 2) Application to Arid Southwest Aerosols”. In: *KONA Powder and Particle Journal* 31, pp. 2–9.

— Zahir, Muhammad Zeeshan et al. (2021). “Development of electric virtual impactor with variable sampling particle size range”. In: *Advanced Powder Technology* 32.10, pp. 3456–3464.

— Wang, Pan et al. (2025). “Innovative microseparator design for enhanced aerosol separation”. In: *Analytica Chimica Acta*.

— Ferziger, Joel H et al. (2020). *Computational methods for fluid dynamics*. Vol. 4. Springer.

— Cao, Yang et al. (2023). “Advances in Coupling Computational Fluid Dynamics and Discrete Element Method in Geotechnical Problems”. In: *Geotechnics* 3, pp. 1162–1179. ISSN: 2673-7094. DOI: 10.3390/geotechnics3040063. URL: https://www.mdpi.com/2673-7094/3/4/63.

— Elghobashi, Said (1994). “On predicting particle-laden turbulent flows”. In: *Applied scientific research* 52.4, pp. 309–329.

— Thornton, Colin et al. (1998). “A theoretical model for the stick/bounce behaviour of adhesive, elastic-plastic spheres”. In: *Powder technology* 99.2, pp. 154–162.

— Shukla, Usha et al. (Aug. 2022). “A Brief Note on Turbulent flow”. In: *Research & Reviews Journal of Pure and Applied Physics* 10, pp. 9–10.

— Moin, Parviz et al. (1998). “Direct numerical simulation: a tool in turbulence research”. In: *Annual review of fluid mechanics* 30.1, pp. 539–578.

— Sagaut, Pierre (2006). *Large eddy simulation for incompressible flows: an introduction*. Springer.

— Germano, Massimo et al. (1991). “A dynamic subgrid-scale eddy viscosity model”. In: *Physics of fluids a: Fluid dynamics* 3.7, pp. 1760–1765.

— Reynolds, Osborne (1895). “IV. On the dynamical theory of incompressible viscous fluids and the determination of the criterion”. In: *Philosophical transactions of the royal society of london.(a.)* 186, pp. 123–164.

— Versteeg, Henk Kaarle (2007). *An introduction to computational fluid dynamics the finite volume method, 2/E*. Pearson Education India.

— Launder, Brian Edward et al. (1983). “The numerical computation of turbulent flows”. In: *Numerical prediction of flow, heat transfer, turbulence and combustion*. Elsevier, pp. 96–116.

— Wilcox, David C (1988). “Reassessment of the scale-determining equation for advanced turbulence models”. In: *AIAA journal* 26.11, pp. 1299–1310.

— Menter, Florian R (1994). “Two-equation eddy-viscosity turbulence models for engineering applications”. In: *AIAA journal* 32.8, pp. 1598–1605.

— Spalart, Philippe et al. (1992). “A one-equation turbulence model for aerodynamic flows”. In: *30th aerospace sciences meeting and exhibit*, p. 439.

— Pant, Pranshu et al. (2021). “Deep learning for reduced order modelling and efficient temporal evolution of fluid simulations”. In: *Physics of Fluids*.

— Waldbaur, Ansgar et al. (2011). “Let there be chip—towards rapid prototyping of microfluidic devices: one-step manufacturing processes”. en. In: *Anal. Methods* 3.12, p. 2681. ISSN: 1759-9660, 1759-9679. DOI: 10.1039/c1ay05253e. URL: https://xlink.rsc.org/?DOI=c1ay05253e.

— Wang, Yong et al. (2024). “System integration of an optimally designed virtual impactor with a QCM sensor as a one-stop pm2. 5 classification and detection platform”. In: *ACS omega* 9.5, pp. 5751–5760.

— Bazan, Anna et al. (2025). “The Influence of Model Orientation on the Surface Roughness of Polymeric Models Produced by FFF, mSLA, PJ, and SLS Methods”. In: *Materials* 18.24, p. 5600.

— Altmeppen, Johannes et al. (2020). “An analytical approach to estimate the effect of surface roughness on particle rebound”. In: *Journal of the Global Power and Propulsion Society* 4, pp. 27–37.

— Altmeppen, Johannes et al. (2022). “Experimental and numerical investigation into the effect of surface roughness on particle rebound”. In: *Frontiers in Mechanical Engineering* 8, p. 918708.

— Laffolley, Hugo, Youichi Tsubota, et al. (2026). “Development of a 3D-Printed Two-Stage Virtual Impactor for Radioactive Aerosol Size Classification and Direct Analysis”. In: *ACS OMEGA*.

— Lilienfeld, Pedro et al. (1990). “METHOD AND APPARATUS FOR REAL TIME ASBESTOS AEROSOL MONITORING”. In: *United States Patent*.

— Blevins, Robert D (1984). “Applied fluid dynamics handbook”. In: *New York*.

— Mohammadi, Samira et al. (2022). “Efficiency of uncertainty propagation methods for moment estimation of uncertain model outputs”. In: *Computers & Chemical Engineering* 166, p. 107954.

— Loeppky, Jason L et al. (2009). “Choosing the sample size of a computer experiment: A practical guide”. In: *Technometrics* 51.4, pp. 366–376.

— ICRP (1995). “Age-dependent Doses to Members of the Public from Intake of Radionuclides - Part 5 Compilation of Ingestion and Inhalation Coefficients.” In: *ICRP Publication 72*.

— ICRP (2007). “Recommendations of the International Commission on Radiological Protection”. In: *ICRP Publication 103*.

— The OpenFOAM Foundation (2024). *OpenFOAM Standard Solvers Documentation*. URL: `https://doc.openfoam.com/2312/tools/processing/solvers/`.

— OpenCFD (2026). *OpenFOAM: Running Applications in Parallel*. URL: `https://www.openfoam.com/documentation/user-guide/3-running-applications/3.2-running-applications-in-parallel`.

— NVIDIA (2026). *What is High-Performance Computing (HPC)?* URL: `https://www.nvidia.com/en-us/glossary/high-performance-computing/`.

— UN Organization (2015). *17 Sustainable Development Goals*. URL: `https://sdgs.un.org/goals`.